T3 general group factors: UV

Original RGE-structure / prefactor comparison layout

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Field representations

Model	$S_1 : (d, Y)$	$S_2 : (d, Y)$	$F : (d, Y)$
A_+0	$(1, 0)$	$(3, 1)$	$(2, \frac{1}{2})$
A_-2	$(1, -1)$	$(3, 0)$	$(2, -\frac{1}{2})$
A_-4	$(1, -2)$	$(3, -1)$	$(2, -\frac{3}{2})$
B_+1	$(2, \frac{1}{2})$	$(2, \frac{3}{2})$	$(1, 1)$
B_-1	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(1, 0)$

B_-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(1,-1)$
C_+1	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	$(3,1)$
C_-1	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$
C_-3	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	$(3,-1)$
D_+0	$(3,0)$	$(1,1)$	$(2,\frac{1}{2})$
D_+2	$(3,1)$	$(1,2)$	$(2,\frac{3}{2})$
D_-2	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$
E_+0	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$
E_+2	$(3,1)$	$(3,2)$	$(2,\frac{3}{2})$
E_-2	$(3,-1)$	$(3,0)$	$(2,-\frac{1}{2})$
E_-4	$(3,-2)$	$(3,-1)$	$(2,-\frac{3}{2})$

One-loop renormalisable RGE structure prefactors

$\beta^{(1)}_{M_F}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$-\frac{9}{2}$g2^{2}Matrix(MF)(heavy(\$i), heavy(\$j))	1	1	1
$-\frac{3}{2}$gY^{2}Matrix(MF)(heavy(\$i), heavy(\$j))	1	1	0
$\frac{3}{4}$Matrix(MF, Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))	1	1	1
$\frac{1}{2}$Matrix(Trans(y1), Bar(y1), MF)(heavy(\$i), heavy(\$j))	1	1	1
$\frac{1}{2}$Matrix(Trans(y1Conj), Bar(y1Conj), MF)(heavy(\$i), heavy(\$j))	1	0	0
$\frac{3}{4}$Matrix(MF, Trans(Bar(y2Conj)), y2Conj)(heavy(\$i), heavy(\$j))	0	1	0
$-\frac{27}{2}$gY^{2}Matrix(MF)(heavy(\$i), heavy(\$j))	0	0	1
$\frac{1}{2}$Matrix(MF, Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{3}{4}$Matrix(Trans(y1), Bar(y1), MF)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{3}{4}$Matrix(Trans(y1Conj), Bar(y1Conj), MF)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{1}{2}$Matrix(MF, Trans(Bar(y2Conj)), y2Conj)(heavy(\$i), heavy(\$j))	0	0	0
gY^{2}Matrix(MF)(heavy(\$i), heavy(\$j))	0	0	0
g2^{2}Matrix(MF)(heavy(\$i), heavy(\$j))	0	0	0
g2^{2}Matrix(Trans(MF))(heavy(\$i), heavy(\$j))	0	0	0
$\frac{1}{2}$Matrix(Trans(MF), Trans(Bar(y1)), y1)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{1}{2}$Matrix(Trans(MF), Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{1}{2}$Matrix(Trans(y2), Bar(y2), MF)(heavy(\$i), heavy(\$j))	0	0	0
Matrix(MF, Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))	0	0	0
Matrix(Trans(y1), Bar(y1), MF)(heavy(\$i), heavy(\$j))	0	0	0
Matrix(Trans(MF), Trans(Bar(y1)), y1)(heavy(\$i), heavy(\$j))	0	0	0
Matrix(Trans(MF), Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))	0	0	0
Matrix(Trans(y2), Bar(y2), MF)(heavy(\$i), heavy(\$j))	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$-\frac{9}{2}$g2^{2}Matrix(MF)(heavy(\$i), heavy(\$j))	0	0	0
$-\frac{3}{2}$gY^{2}Matrix(MF)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{3}{4}$Matrix(MF, Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{1}{2}$Matrix(Trans(y1), Bar(y1), MF)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{1}{2}$Matrix(Trans(y1Conj), Bar(y1Conj), MF)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{3}{4}$Matrix(MF, Trans(Bar(y2Conj)), y2Conj)(heavy(\$i), heavy(\$j))	0	0	0
$-\frac{27}{2}$gY^{2}Matrix(MF)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{1}{2}$Matrix(MF, Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{3}{4}$Matrix(Trans(y1), Bar(y1), MF)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{3}{4}$Matrix(Trans(y1Conj), Bar(y1Conj), MF)(heavy(\$i), heavy(\$j))	0	0	0
$\frac{1}{2}$Matrix(MF, Trans(Bar(y2Conj)), y2Conj)(heavy(\$i), heavy(\$j))	0	0	0
gY^{2}Matrix(MF)(heavy(\$i), heavy(\$j))	-6	0	-6
g2^{2}Matrix(MF)(heavy(\$i), heavy(\$j))	0	0	0
g2^{2}Matrix(Trans(MF))(heavy(\$i), heavy(\$j))	0	0	0
$\frac{1}{2}$Matrix(Trans(MF), Trans(Bar(y1)), y1)(heavy(\$i), heavy(\$j))	0	0	0

$\frac{1}{2}$Matrix(Trans(MF), Trans(Bar(y2)), y2)(heavy($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(Trans(y2), Bar(y2), MF)(heavy($i), heavy($j))	0	0	0
Matrix(MF, Trans(Bar(y2)), y2)(heavy($i), heavy($j))	1	0	1
Matrix(Trans(y1), Bar(y1), MF)(heavy($i), heavy($j))	1	1	1
Matrix(Trans(MF), Trans(Bar(y1)), y1)(heavy($i), heavy($j))	0	1	0
Matrix(Trans(MF), Trans(Bar(y2)), y2)(heavy($i), heavy($j))	0	1	0
Matrix(Trans(y2), Bar(y2), MF)(heavy($i), heavy($j))	0	1	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$-\frac{9}{2}$g2^{2}Matrix(MF)(heavy($i), heavy($j))	0	0	0
$-\frac{3}{2}$gY^{2}Matrix(MF)(heavy($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(MF, Trans(Bar(y2)), y2)(heavy($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(Trans(y1), Bar(y1), MF)(heavy($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(Trans(y1Conj), Bar(y1Conj), MF)(heavy($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(MF, Trans(Bar(y2Conj)), y2Conj)(heavy($i), heavy($j))	0	0	0
$-\frac{27}{2}$gY^{2}Matrix(MF)(heavy($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(MF, Trans(Bar(y2)), y2)(heavy($i), heavy($j))	1	0	1
$\frac{3}{4}$Matrix(Trans(y1), Bar(y1), MF)(heavy($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(Trans(y1Conj), Bar(y1Conj), MF)(heavy($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(MF, Trans(Bar(y2Conj)), y2Conj)(heavy($i), heavy($j))	0	0	0
gY^{2}Matrix(MF)(heavy($i), heavy($j))	-6	0	-6
g2^{2}Matrix(MF)(heavy($i), heavy($j))	-12	-6	-12
g2^{2}Matrix(Trans(MF))(heavy($i), heavy($j))	0	-6	0
$\frac{1}{2}$Matrix(Trans(MF), Trans(Bar(y1)), y1)(heavy($i), heavy($j))	0	1	0
$\frac{1}{2}$Matrix(Trans(MF), Trans(Bar(y2)), y2)(heavy($i), heavy($j))	0	1	0
$\frac{1}{2}$Matrix(Trans(y2), Bar(y2), MF)(heavy($i), heavy($j))	0	1	0
Matrix(MF, Trans(Bar(y2)), y2)(heavy($i), heavy($j))	0	0	0
Matrix(Trans(y1), Bar(y1), MF)(heavy($i), heavy($j))	0	0	0
Matrix(Trans(MF), Trans(Bar(y1)), y1)(heavy($i), heavy($j))	0	0	0
Matrix(Trans(MF), Trans(Bar(y2)), y2)(heavy($i), heavy($j))	0	0	0
Matrix(Trans(y2), Bar(y2), MF)(heavy($i), heavy($j))	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$-\frac{9}{2}$g2^{2}Matrix(MF)(heavy($i), heavy($j))	1	1	1
$-\frac{3}{2}$gY^{2}Matrix(MF)(heavy($i), heavy($j))	1	0	1
$\frac{3}{4}$Matrix(MF, Trans(Bar(y2)), y2)(heavy($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(Trans(y1), Bar(y1), MF)(heavy($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(Trans(y1Conj), Bar(y1Conj), MF)(heavy($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(MF, Trans(Bar(y2Conj)), y2Conj)(heavy($i), heavy($j))	0	0	0
$-\frac{27}{2}$gY^{2}Matrix(MF)(heavy($i), heavy($j))	0	1	0
$\frac{1}{2}$Matrix(MF, Trans(Bar(y2)), y2)(heavy($i), heavy($j))	1	1	1
$\frac{3}{4}$Matrix(Trans(y1), Bar(y1), MF)(heavy($i), heavy($j))	1	1	1
$\frac{3}{4}$Matrix(Trans(y1Conj), Bar(y1Conj), MF)(heavy($i), heavy($j))	1	0	0
$\frac{1}{2}$Matrix(MF, Trans(Bar(y2Conj)), y2Conj)(heavy($i), heavy($j))	0	0	1
gY^{2}Matrix(MF)(heavy($i), heavy($j))	0	0	0
g2^{2}Matrix(MF)(heavy($i), heavy($j))	0	0	0
g2^{2}Matrix(Trans(MF))(heavy($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(Trans(MF), Trans(Bar(y1)), y1)(heavy($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(Trans(MF), Trans(Bar(y2)), y2)(heavy($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(Trans(y2), Bar(y2), MF)(heavy($i), heavy($j))	0	0	0
Matrix(MF, Trans(Bar(y2)), y2)(heavy($i), heavy($j))	0	0	0
Matrix(Trans(y1), Bar(y1), MF)(heavy($i), heavy($j))	0	0	0
Matrix(Trans(MF), Trans(Bar(y1)), y1)(heavy($i), heavy($j))	0	0	0
Matrix(Trans(MF), Trans(Bar(y2)), y2)(heavy($i), heavy($j))	0	0	0
Matrix(Trans(y2), Bar(y2), MF)(heavy($i), heavy($j))	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$-\frac{9}{2}g2^2\text{Matrix(MF)(heavy(\$i), heavy(\$j))}$	1	1	1
$-\frac{3}{2}gY^2\text{Matrix(MF)(heavy(\$i), heavy(\$j))}$	1	0	1
$\frac{3}{4}\text{Matrix(MF, Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))}$	1	1	1
$\frac{1}{2}\text{Matrix(Trans(y1), Bar(y1), MF)(heavy(\$i), heavy(\$j))}$	0	0	0
$\frac{1}{2}\text{Matrix(Trans(y1Conj), Bar(y1Conj), MF)(heavy(\$i), heavy(\$j))}$	0	0	0
$\frac{3}{4}\text{Matrix(MF, Trans(Bar(y2Conj)), y2Conj)(heavy(\$i), heavy(\$j))}$	0	0	1
$-\frac{27}{2}gY^2\text{Matrix(MF)(heavy(\$i), heavy(\$j))}$	0	1	0
$\frac{1}{2}\text{Matrix(MF, Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))}$	0	0	0
$\frac{3}{4}\text{Matrix(Trans(y1), Bar(y1), MF)(heavy(\$i), heavy(\$j))}$	1	1	1
$\frac{3}{4}\text{Matrix(Trans(y1Conj), Bar(y1Conj), MF)(heavy(\$i), heavy(\$j))}$	1	0	0
$\frac{1}{2}\text{Matrix(MF, Trans(Bar(y2Conj)), y2Conj)(heavy(\$i), heavy(\$j))}$	0	0	0
$gY^2\text{Matrix(MF)(heavy(\$i), heavy(\$j))}$	0	0	0
$g2^2\text{Matrix(MF)(heavy(\$i), heavy(\$j))}$	0	0	0
$g2^2\text{Matrix(Trans(MF))(heavy(\$i), heavy(\$j))}$	0	0	0
$\frac{1}{2}\text{Matrix(Trans(MF), Trans(Bar(y1)), y1)(heavy(\$i), heavy(\$j))}$	0	0	0
$\frac{1}{2}\text{Matrix(Trans(MF), Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))}$	0	0	0
$\frac{1}{2}\text{Matrix(Trans(y2), Bar(y2), MF)(heavy(\$i), heavy(\$j))}$	0	0	0
$\text{Matrix(MF, Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))}$	0	0	0
$\text{Matrix(Trans(y1), Bar(y1), MF)(heavy(\$i), heavy(\$j))}$	0	0	0
$\text{Matrix(Trans(MF), Trans(Bar(y1)), y1)(heavy(\$i), heavy(\$j))}$	0	0	0
$\text{Matrix(Trans(MF), Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))}$	0	0	0
$\text{Matrix(Trans(y2), Bar(y2), MF)(heavy(\$i), heavy(\$j))}$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$-\frac{9}{2}g2^2\text{Matrix(MF)(heavy(\$i), heavy(\$j))}$	1
$-\frac{3}{2}gY^2\text{Matrix(MF)(heavy(\$i), heavy(\$j))}$	0
$\frac{3}{4}\text{Matrix(MF, Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))}$	1
$\frac{1}{2}\text{Matrix(Trans(y1), Bar(y1), MF)(heavy(\$i), heavy(\$j))}$	0
$\frac{1}{2}\text{Matrix(Trans(y1Conj), Bar(y1Conj), MF)(heavy(\$i), heavy(\$j))}$	0
$\frac{3}{4}\text{Matrix(MF, Trans(Bar(y2Conj)), y2Conj)(heavy(\$i), heavy(\$j))}$	0
$-\frac{27}{2}gY^2\text{Matrix(MF)(heavy(\$i), heavy(\$j))}$	1
$\frac{1}{2}\text{Matrix(MF, Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))}$	0
$\frac{3}{4}\text{Matrix(Trans(y1), Bar(y1), MF)(heavy(\$i), heavy(\$j))}$	1
$\frac{3}{4}\text{Matrix(Trans(y1Conj), Bar(y1Conj), MF)(heavy(\$i), heavy(\$j))}$	0
$\frac{1}{2}\text{Matrix(MF, Trans(Bar(y2Conj)), y2Conj)(heavy(\$i), heavy(\$j))}$	0
$gY^2\text{Matrix(MF)(heavy(\$i), heavy(\$j))}$	0
$g2^2\text{Matrix(MF)(heavy(\$i), heavy(\$j))}$	0
$g2^2\text{Matrix(Trans(MF))(heavy(\$i), heavy(\$j))}$	0
$\frac{1}{2}\text{Matrix(Trans(MF), Trans(Bar(y1)), y1)(heavy(\$i), heavy(\$j))}$	0
$\frac{1}{2}\text{Matrix(Trans(MF), Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))}$	0
$\frac{1}{2}\text{Matrix(Trans(y2), Bar(y2), MF)(heavy(\$i), heavy(\$j))}$	0
$\text{Matrix(MF, Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))}$	0
$\text{Matrix(Trans(y1), Bar(y1), MF)(heavy(\$i), heavy(\$j))}$	0
$\text{Matrix(Trans(MF), Trans(Bar(y1)), y1)(heavy(\$i), heavy(\$j))}$	0
$\text{Matrix(Trans(MF), Trans(Bar(y2)), y2)(heavy(\$i), heavy(\$j))}$	0
$\text{Matrix(Trans(y2), Bar(y2), MF)(heavy(\$i), heavy(\$j))}$	0

$\beta_{g_2}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$-\frac{g2^3}{2}$	1	1	1
$-\frac{17g2^3}{6}$	0	0	0
$\frac{31g2^3}{6}$	0	0	0

$\frac{7g2^3}{6}$	0	0	0
$\frac{g2^3}{6}$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$-\frac{g2^3}{2}$	0	0	0
$-\frac{17g2^3}{6}$	1	1	1
$\frac{31g2^3}{6}$	0	0	0
$\frac{7g2^3}{6}$	0	0	0
$\frac{g2^3}{6}$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$-\frac{g2^3}{2}$	0	0	0
$-\frac{17g2^3}{6}$	0	0	0
$\frac{31g2^3}{6}$	1	0	1
$\frac{7g2^3}{6}$	0	1	0
$\frac{g2^3}{6}$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$-\frac{g2^3}{2}$	1	1	1
$-\frac{17g2^3}{6}$	0	0	0
$\frac{31g2^3}{6}$	0	0	0
$\frac{7g2^3}{6}$	0	0	0
$\frac{g2^3}{6}$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$-\frac{g2^3}{2}$	0	0	0
$-\frac{17g2^3}{6}$	0	0	0
$\frac{31g2^3}{6}$	0	0	0
$\frac{7g2^3}{6}$	0	0	0
$\frac{g2^3}{6}$	1	1	1

$\mathcal{S}$	E ($\alpha = -4$)
$-\frac{g2^3}{2}$	0
$-\frac{17g2^3}{6}$	0
$\frac{31g2^3}{6}$	0
$\frac{7g2^3}{6}$	0
$\frac{g2^3}{6}$	1

$\beta_{g_3}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$g3^3$	-7	-7	-7

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$g3^3$	-7	-7	-7

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$g3^3$	-7	-7	-7

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$g3^3$	-7	-7	-7

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$g3^3$	-7	-7	-7

$\mathcal{S}$	E ($\alpha = -4$)
$g3^3$	-7

$\beta_{g_1}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\frac{59gY^3}{6}$	1	0	0
$\frac{55gY^3}{6}$	0	1	0
$\frac{163gY^3}{6}$	0	0	1
$\frac{25gY^3}{2}$	0	0	0
$\frac{43gY^3}{6}$	0	0	0
$\frac{41gY^3}{2}$	0	0	0
$\frac{179gY^3}{6}$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\frac{59gY^3}{6}$	0	0	0
$\frac{55gY^3}{6}$	0	0	0
$\frac{163gY^3}{6}$	0	0	0
$\frac{25gY^3}{2}$	1	0	1
$\frac{43gY^3}{6}$	0	1	0
$\frac{41gY^3}{2}$	0	0	0
$\frac{179gY^3}{6}$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\frac{59gY^3}{6}$	0	0	0
$\frac{55gY^3}{6}$	0	0	0
$\frac{163gY^3}{6}$	0	0	0
$\frac{25gY^3}{2}$	0	0	0
$\frac{43gY^3}{6}$	0	1	0
$\frac{41gY^3}{2}$	1	0	1

$\frac{179 g Y^3}{6}$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\frac{59 g Y^3}{6}$	0	0	1
$\frac{55 g Y^3}{6}$	1	0	0
$\frac{163 g Y^3}{6}$	0	1	0
$\frac{25 g Y^3}{2}$	0	0	0
$\frac{43 g Y^3}{6}$	0	0	0
$\frac{41 g Y^3}{2}$	0	0	0
$\frac{179 g Y^3}{6}$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\frac{59 g Y^3}{6}$	1	0	1
$\frac{55 g Y^3}{6}$	0	0	0
$\frac{163 g Y^3}{6}$	0	0	0
$\frac{25 g Y^3}{2}$	0	0	0
$\frac{43 g Y^3}{6}$	0	0	0
$\frac{41 g Y^3}{2}$	0	0	0
$\frac{179 g Y^3}{6}$	0	1	0

$\mathcal{S}$	E ($\alpha = -4$)
$\frac{59 g Y^3}{6}$	0
$\frac{55 g Y^3}{6}$	0
$\frac{163 g Y^3}{6}$	0
$\frac{25 g Y^3}{2}$	0
$\frac{43 g Y^3}{6}$	0
$\frac{41 g Y^3}{2}$	0
$\frac{179 g Y^3}{6}$	1

$\beta^{(1)}_{\lambda^{(1)}_{12}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
g2^{2}lambda12	-12	-12	-12
gY^{2}lambda12	-6	-6	-30
lambda12^2	4	4	4
lambdaH1lambdaH2	4	4	4
lambda12lambdaS1	4	4	4
lambda12lambdaS2	8	8	8
lambda12lambdaS2Adj	4	4	4
lambdaT3Bar(lambdaT3)	4	4	4
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	16	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I1)	12	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ001300I1)	12	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambda12Tr[y1.Trans(Bar(y1))]	4	4	4
lambdaT3GeneralQ000211I1Tr[y1.Trans(Bar(y1Conj))]	8	0	0
Bar(lambdaT3GeneralQ000211I1)Tr[y1Conj.Trans(Bar(y1))]	8	0	0
lambda12Tr[y1Conj.Trans(Bar(y1Conj))]	4	0	0

lambda12Tr[y2.Trans(Bar(y2))]	2	2	2
Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	-4	-4	-4
Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	-4	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ000013I1)	0	20	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I1)	0	20	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	16	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	4	0
Bar(lambdaT3GeneralQ001102I1)Tr[y2.Trans(Bar(y2Conj))]	0	4	0
lambdaT3GeneralQ001102I1Tr[y2Conj.Trans(Bar(y2))]	0	4	0
lambda12Tr[y2Conj.Trans(Bar(y2Conj))]	0	2	0
Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	-4	0
gY^4	0	0	48
$\frac{9g2^4}{4}$	0	0	0
$\frac{27gY^4}{4}$	0	0	0
$\frac{3lambda12Adj^2}{4}$	0	0	0
$\frac{3gY^4}{4}$	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
Tr[y1.Trans(Bar(y2)).y2.Trans(Bar(y1))]	0	0	0
$-\frac{9}{2}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
lambda12lambdaS1Adj	0	0	0
$\frac{63g2^4}{5}$	0	0	0
$lambda12Adj^2$	0	0	0
$lambda12Cross^2$	0	0	0
lambda12CrosslambdaS1	0	0	0
lambda12CrosslambdaS1Adj	0	0	0
lambda12CrosslambdaS2	0	0	0
lambda12CrosslambdaS2Adj	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ001300I1)	0	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{22}{15}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$g2^2$lambda12	-9	-9	-9
gY^2lambda12	-15	-3	-15
$lambda12^2$	4	4	4
lambdaH1lambdaH2	4	4	4
lambda12lambdaS1	6	6	6
lambda12lambdaS2	6	6	6
lambda12lambdaS2Adj	0	0	0
lambdaT3Bar(lambdaT3)	3	3	3
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ001300I1)	0	0	0

lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambda12Tr[y1.Trans(Bar(y1))]	2	2	2
lambdaT3GeneralQ000211I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(lambdaT3GeneralQ000211I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambda12Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambda12Tr[y2.Trans(Bar(y2))]	2	2	2
Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	-2	-2	-2
Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
Bar(lambdaT3GeneralQ001102I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambda12Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
gY^4	0	0	0
$\frac{9g2^4}{4}$	1	1	1
$\frac{27gY^4}{4}$	1	0	1
$\frac{3lambda12Adj^2}{4}$	1	1	1
$\frac{3gY^4}{4}$	0	1	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	2	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	1	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	24	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	9	0
lambdaS1S1bar2S2barBar(lambdaS1barS2S2bar2)	0	9	0
lambdaS1barS2S2bar2Bar(lambdaS1S1bar2S2bar)	0	9	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	9	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	1	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	2	0
Tr[y1.Trans(Bar(y2)).y2.Trans(Bar(y1))]	0	-4	0
$-\frac{9}{2}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
lambda12lambdaS1Adj	0	0	0
$\frac{63g2^4}{5}$	0	0	0
$lambda12Adj^2$	0	0	0
$lambda12Cross^2$	0	0	0
lambda12CrosslambdaS1	0	0	0
lambda12CrosslambdaS1Adj	0	0	0
lambda12CrosslambdaS2	0	0	0
lambda12CrosslambdaS2Adj	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ001300I1)	0	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{22}{15}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$g2^2$lambda12	-9	-9	-9
gY^2lambda12	-15	-3	-15
$lambda12^2$	4	4	4
lambdaH1lambdaH2	4	4	4
lambda12lambdaS1	6	6	6
lambda12lambdaS2	6	6	6

lambda12lambdaS2Adj	0	0	0
lambdaT3Bar(lambdaT3)	3	3	3
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambda12Tr[y1.Trans(Bar(y1))]	3	3	3
lambdaT3GeneralQ000211I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(lambdaT3GeneralQ000211I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambda12Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambda12Tr[y2.Trans(Bar(y2))]	3	3	3
Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
Bar(lambdaT3GeneralQ001102I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambda12Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
gY^4	0	0	0
$\frac{9g2^4}{4}$	1	1	1
$\frac{27gY^4}{4}$	1	0	1
$\frac{3lambda12Adj^2}{4}$	1	1	1
$\frac{3gY^4}{4}$	0	1	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	2	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	1	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	24	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	9	0
lambdaS1S1bar2S2barBar(lambdaS1barS2S2bar2)	0	9	0
lambdaS1barS2S2bar2Bar(lambdaS1S1bar2S2bar)	0	9	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	9	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	1	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	2	0
Tr[y1.Trans(Bar(y2)).y2.Trans(Bar(y1))]	0	-3	0
$-\frac{9}{2}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	1	1	1
lambda12lambdaS1Adj	0	0	0
$\frac{63g2^4}{5}$	0	0	0
$lambda12Adj^2$	0	0	0
$lambda12Cross^2$	0	0	0
lambda12CrosslambdaS1	0	0	0
lambda12CrosslambdaS1Adj	0	0	0
lambda12CrosslambdaS2	0	0	0
lambda12CrosslambdaS2Adj	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ001300I1)	0	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{22}{15}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$g2^2$lambda12	-12	-12	-12
gY^2lambda12	-6	-30	-6

lambda12^2	4	4	4
lambdaH1lambdaH2	4	4	4
lambda12lambdaS1	8	8	8
lambda12lambdaS2	4	4	4
lambda12lambdaS2Adj	0	0	0
lambdaT3Bar(lambdaT3)	4	4	4
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	16	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I1)	20	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ001300I1)	20	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambda12Tr[y1.Trans(Bar(y1))]	2	2	2
lambdaT3GeneralQ000211I1Tr[y1.Trans(Bar(y1Conj))]	4	0	0
Bar(lambdaT3GeneralQ000211I1)Tr[y1Conj.Trans(Bar(y1))]	4	0	0
lambda12Tr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
lambda12Tr[y2.Trans(Bar(y2))]	4	4	4
Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	-4	-4	-4
Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	-4	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ000013I1)	0	0	12
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I1)	0	0	12
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	16
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	4
Bar(lambdaT3GeneralQ001102I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	8
lambdaT3GeneralQ001102I1Tr[y2Conj.Trans(Bar(y2))]	0	0	8
lambda12Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	4
Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	-4
gY^4	0	48	0
$\frac{9\text{g2}^4}{4}$	0	0	0
$\frac{27\text{gY}^4}{4}$	0	0	0
$\frac{3\text{lambda12Adj}^2}{4}$	0	0	0
$\frac{3\text{gY}^4}{4}$	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
Tr[y1.Trans(Bar(y2)).y2.Trans(Bar(y1))]	0	0	0
$-\frac{9}{2}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
lambda12lambdaS1Adj	4	4	4
$\frac{63\text{g2}^4}{5}$	0	0	0
lambda12Adj^2	0	0	0
lambda12Cross^2	0	0	0
lambda12CrosslambdaS1	0	0	0
lambda12CrosslambdaS1Adj	0	0	0
lambda12CrosslambdaS2	0	0	0
lambda12CrosslambdaS2Adj	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ001300I1)	0	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{22}{15}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$g2^2$lambda12	-24	-24	-24
gY^2lambda12	-6	-30	-6
lambda12^2	4	4	4
lambdaH1lambdaH2	4	4	4
lambda12lambdaS1	8	8	8
lambda12lambdaS2	8	8	8
lambda12lambdaS2Adj	4	4	4
lambdaT3Bar(lambdaT3)	2	2	2
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	16	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I1)	20	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ001300I1)	20	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	2	0	0
lambda12Tr[y1.Trans(Bar(y1))]	2	2	2
lambdaT3GeneralQ000211I1Tr[y1.Trans(Bar(y1Conj))]	4	0	0
Bar(lambdaT3GeneralQ000211I1)Tr[y1Conj.Trans(Bar(y1))]	4	0	0
lambda12Tr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
lambda12Tr[y2.Trans(Bar(y2))]	2	2	2
Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ000013I1)	0	0	20
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I1)	0	0	20
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	16
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	2
Bar(lambdaT3GeneralQ001102I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	4
lambdaT3GeneralQ001102I1Tr[y2Conj.Trans(Bar(y2))]	0	0	4
lambda12Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	2
Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
gY^4	0	48	0
$\frac{9g2^4}{4}$	0	0	0
$\frac{27gY^4}{4}$	0	0	0
$\frac{3\text{lambda12Adj}^2}{4}$	0	0	0
$\frac{3gY^4}{4}$	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
Tr[y1.Trans(Bar(y2)).y2.Trans(Bar(y1))]	0	0	0
$-\frac{9}{2}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
lambda12lambdaS1Adj	4	4	4
$\frac{63g2^4}{5}$	1	1	1
lambda12Adj^2	16	16	16
lambda12Cross^2	4	4	4
lambda12CrosslambdaS1	4	4	4
lambda12CrosslambdaS1Adj	4	4	4
lambda12CrosslambdaS2	4	4	4
lambda12CrosslambdaS2Adj	4	4	4
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	4	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I2)	4	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ001300I1)	4	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	1	1	1
$-\frac{22}{15}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	1	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ000013I1)	0	0	4

lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I2)	0	0	4
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	4
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	1

$\mathcal{S}$	E ($\alpha = -4$)
$g2^2$lambda12	-24
gY^2lambda12	-30
lambda12^2	4
lambdaH1lambdaH2	4
lambda12lambdaS1	8
lambda12lambdaS2	8
lambda12lambdaS2Adj	4
lambdaT3Bar(lambdaT3)	2
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I1)	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ001300I1)	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0
lambda12Tr[y1.Trans(Bar(y1))]	2
lambdaT3GeneralQ000211I1Tr[y1.Trans(Bar(y1Conj))]	0
Bar(lambdaT3GeneralQ000211I1)Tr[y1Conj.Trans(Bar(y1))]	0
lambda12Tr[y1Conj.Trans(Bar(y1Conj))]	0
lambda12Tr[y2.Trans(Bar(y2))]	2
Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0
Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ000013I1)	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I1)	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0
Bar(lambdaT3GeneralQ001102I1)Tr[y2.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ001102I1Tr[y2Conj.Trans(Bar(y2))]	0
lambda12Tr[y2Conj.Trans(Bar(y2Conj))]	0
Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0
gY^4	48
$\frac{9g2^4}{4}$	0
$\frac{27gY^4}{4}$	0
$\frac{3\text{lambda12Adj}^2}{4}$	0
$\frac{3gY^4}{4}$	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0
lambdaS1S1bar2S2barBar(lambdaS1barS2S2bar2)	0
lambdaS1barS2S2bar2Bar(lambdaS1S1bar2S2bar)	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0
Tr[y1.Trans(Bar(y2)).y2.Trans(Bar(y1))]	0
$-\frac{9}{2}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0
lambda12lambdaS1Adj	4
$\frac{63g2^4}{5}$	1
lambda12Adj^2	16
lambda12Cross^2	4
lambda12CrosslambdaS1	4
lambda12CrosslambdaS1Adj	4
lambda12CrosslambdaS2	4
lambda12CrosslambdaS2Adj	4
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0

lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I2)	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ001300I1)	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	1
$-\frac{22}{15}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ000013I1)	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I2)	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0

$\beta^{(1)}_{\lambda_{12}^{(A)}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$g2^2gY^2$	0	0	0
$g2^2$lambda12Adj	0	0	0
gY^2lambda12Adj	0	0	0
lambda12lambda12Adj	0	0	0
lambdaH1AdjlambdaH2Adj	0	0	0
lambda12AdjlambdaS1	0	0	0
lambda12AdjlambdaS2	0	0	0
$\frac{4}{3}$lambdaT3Bar(lambdaT3)	0	0	0
lambda12AdjTr[y1.Trans(Bar(y1))]	0	0	0
lambda12AdjTr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	0	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y2)).y2.Trans(Bar(y1))]	0	0	0
$\frac{16\text{lambda12Adjlambda12Cross}}{3}$	0	0	0
lambda12AdjlambdaS1Adj	0	0	0
lambda12AdjlambdaS2Adj	0	0	0
$-\frac{2}{9}$lambdaT3Bar(lambdaT3)	0	0	0
$\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{1}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambda12AdjTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
$-\frac{2}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{1}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambda12AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$g2^2gY^2$	18	-6	18
$g2^2$lambda12Adj	-9	-9	-9
gY^2lambda12Adj	-15	-3	-15
lambda12lambda12Adj	8	8	8
lambdaH1AdjlambdaH2Adj	1	1	1
lambda12AdjlambdaS1	2	2	2
lambda12AdjlambdaS2	2	2	2
$\frac{4}{3}$lambdaT3Bar(lambdaT3)	1	1	1
lambda12AdjTr[y1.Trans(Bar(y1))]	2	2	2
lambda12AdjTr[y2.Trans(Bar(y2))]	2	2	2
$-\frac{8}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	1	1	1

lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	4	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	-64	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	-20	0
lambdaS1S1bar2S2barBar(lambdaS1barS2S2bar2)	0	-4	0
lambdaS1barS2S2bar2Bar(lambdaS1S1bar2S2bar)	0	-4	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	-20	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	4	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y2)).y2.Trans(Bar(y1))]	0	0	0
$\frac{16\text{lambda12Adjlambda12Cross}}{3}$	0	0	0
lambda12AdjlambdaS1Adj	0	0	0
lambda12AdjlambdaS2Adj	0	0	0
$-\frac{2}{9}$lambdaT3Bar(lambdaT3)	0	0	0
$\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{1}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambda12AdjTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
$-\frac{2}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{1}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambda12AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$g2^2gY^2$	18	-6	18
$g2^2$lambda12Adj	-9	-9	-9
gY^2lambda12Adj	-15	-3	-15
lambda12lambda12Adj	8	8	8
lambdaH1AdjlambdaH2Adj	1	1	1
lambda12AdjlambdaS1	2	2	2
lambda12AdjlambdaS2	2	2	2
$\frac{4}{3}$lambdaT3Bar(lambdaT3)	1	1	1
lambda12AdjTr[y1.Trans(Bar(y1))]	3	3	3
lambda12AdjTr[y2.Trans(Bar(y2))]	3	3	3
$-\frac{8}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	4	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	-64	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	-20	0
lambdaS1S1bar2S2barBar(lambdaS1barS2S2bar2)	0	-4	0
lambdaS1barS2S2bar2Bar(lambdaS1S1bar2S2bar)	0	-4	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	-20	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	4	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	1	1	1
$-\frac{8}{3}$Tr[y1.Trans(Bar(y2)).y2.Trans(Bar(y1))]	0	1	0
$\frac{16\text{lambda12Adjlambda12Cross}}{3}$	0	0	0
lambda12AdjlambdaS1Adj	0	0	0
lambda12AdjlambdaS2Adj	0	0	0
$-\frac{2}{9}$lambdaT3Bar(lambdaT3)	0	0	0
$\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{1}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambda12AdjTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
$-\frac{2}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{1}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambda12AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$g2^2gY^2$	0	0	0

$g2^2$lambda12Adj	0	0	0
gY^2lambda12Adj	0	0	0
lambda12lambda12Adj	0	0	0
lambdaH1AdjlambdaH2Adj	0	0	0
lambda12AdjlambdaS1	0	0	0
lambda12AdjlambdaS2	0	0	0
$\frac{4}{3}$lambdaT3Bar(lambdaT3)	0	0	0
lambda12AdjTr[y1.Trans(Bar(y1))]	0	0	0
lambda12AdjTr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	0	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y2)).y2.Trans(Bar(y1))]	0	0	0
$\frac{16\text{lambda12Adjlambda12Cross}}{3}$	0	0	0
lambda12AdjlambdaS1Adj	0	0	0
lambda12AdjlambdaS2Adj	0	0	0
$-\frac{2}{9}$lambdaT3Bar(lambdaT3)	0	0	0
$\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{1}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambda12AdjTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
$-\frac{2}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{1}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambda12AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$g2^2gY^2$	0	16	0
$g2^2$lambda12Adj	-24	-24	-24
gY^2lambda12Adj	-6	-30	-6
lambda12lambda12Adj	8	8	8
lambdaH1AdjlambdaH2Adj	3	3	3
lambda12AdjlambdaS1	2	2	2
lambda12AdjlambdaS2	2	2	2
$\frac{4}{3}$lambdaT3Bar(lambdaT3)	0	0	0
lambda12AdjTr[y1.Trans(Bar(y1))]	2	2	2
lambda12AdjTr[y2.Trans(Bar(y2))]	2	2	2
$-\frac{8}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	0	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y2)).y2.Trans(Bar(y1))]	0	0	0
$\frac{16\text{lambda12Adjlambda12Cross}}{3}$	1	1	1
lambda12AdjlambdaS1Adj	16	16	16
lambda12AdjlambdaS2Adj	16	16	16
$-\frac{2}{9}$lambdaT3Bar(lambdaT3)	1	1	1
$\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	1	1	1
$-\frac{1}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	1	0	0

lambda12AdjTr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
$-\frac{2}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	1	0	0
$-\frac{1}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	1
lambda12AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	2
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	1

$\mathcal{S}$	E ($\alpha=-4$)
$g2^2gY^2$	16
$g2^2$lambda12Adj	−24
gY^2lambda12Adj	−30
lambda12lambda12Adj	8
lambdaH1AdjlambdaH2Adj	3
lambda12AdjlambdaS1	2
lambda12AdjlambdaS2	2
$\frac{4}{3}$lambdaT3Bar(lambdaT3)	0
lambda12AdjTr[y1.Trans(Bar(y1))]	2
lambda12AdjTr[y2.Trans(Bar(y2))]	2
$-\frac{8}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0
lambdaS1S1bar2S2barBar(lambdaS1barS2S2bar2)	0
lambdaS1barS2S2bar2Bar(lambdaS1S1bar2S2bar)	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y2)).y2.Trans(Bar(y1))]	0
$\frac{16\text{lambda12Adjlambda12Cross}}{3}$	1
lambda12AdjlambdaS1Adj	16
lambda12AdjlambdaS2Adj	16
$-\frac{2}{9}$lambdaT3Bar(lambdaT3)	1
$\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	1
$-\frac{1}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0
lambda12AdjTr[y1Conj.Trans(Bar(y1Conj))]	0
$-\frac{2}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0
$-\frac{1}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0
lambda12AdjTr[y2Conj.Trans(Bar(y2Conj))]	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0

$\beta^{(1)}_{\lambda_{12}^{(\times)}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\frac{51g2^4}{10}$	0	0	0
$g2^2$lambda12Cross	0	0	0
gY^2lambda12Cross	0	0	0
lambda12lambda12Cross	0	0	0
lambda12Cross^2	0	0	0
lambda12CrosslambdaS1	0	0	0
lambda12CrosslambdaS1Adj	0	0	0
lambda12CrosslambdaS2	0	0	0
lambda12CrosslambdaS2Adj	0	0	0
lambdaT3Bar(lambdaT3)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I2)	0	0	0

lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambda12CrossTr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000211I2Tr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(lambdaT3GeneralQ000211I2)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambda12CrossTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambda12CrossTr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{4}{5}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{4}{5}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
Bar(lambdaT3GeneralQ001102I2)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I2Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambda12CrossTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{4}{5}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\frac{51 g2^4}{10}$	0	0	0
$g2^2$lambda12Cross	0	0	0
gY^2lambda12Cross	0	0	0
lambda12lambda12Cross	0	0	0
lambda12Cross^2	0	0	0
lambda12CrosslambdaS1	0	0	0
lambda12CrosslambdaS1Adj	0	0	0
lambda12CrosslambdaS2	0	0	0
lambda12CrosslambdaS2Adj	0	0	0
lambdaT3Bar(lambdaT3)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambda12CrossTr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000211I2Tr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(lambdaT3GeneralQ000211I2)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambda12CrossTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambda12CrossTr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{4}{5}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{4}{5}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
Bar(lambdaT3GeneralQ001102I2)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I2Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambda12CrossTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{4}{5}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)

$\frac{51g2^4}{10}$	0	0	0
$g2^2$lambda12Cross	0	0	0
gY^2lambda12Cross	0	0	0
lambda12lambda12Cross	0	0	0
lambda12Cross^2	0	0	0
lambda12CrosslambdaS1	0	0	0
lambda12CrosslambdaS1Adj	0	0	0
lambda12CrosslambdaS2	0	0	0
lambda12CrosslambdaS2Adj	0	0	0
lambdaT3Bar(lambdaT3)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambda12CrossTr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000211I2Tr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(lambdaT3GeneralQ000211I2)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambda12CrossTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambda12CrossTr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{4}{5}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{4}{5}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
Bar(lambdaT3GeneralQ001102I2)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I2Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambda12CrossTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{4}{5}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\frac{51g2^4}{10}$	0	0	0
$g2^2$lambda12Cross	0	0	0
gY^2lambda12Cross	0	0	0
lambda12lambda12Cross	0	0	0
lambda12Cross^2	0	0	0
lambda12CrosslambdaS1	0	0	0
lambda12CrosslambdaS1Adj	0	0	0
lambda12CrosslambdaS2	0	0	0
lambda12CrosslambdaS2Adj	0	0	0
lambdaT3Bar(lambdaT3)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambda12CrossTr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000211I2Tr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(lambdaT3GeneralQ000211I2)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambda12CrossTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambda12CrossTr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{4}{5}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{4}{5}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0	0	0

lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
Bar(lambdaT3GeneralQ001102I2)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I2Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambda12CrossTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{4}{5}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\frac{51g2^4}{10}$	1	1	1
g2^{2}lambda12Cross	-24	-24	-24
gY^{2}lambda12Cross	-6	-30	-6
lambda12lambda12Cross	8	8	8
lambda12Cross^2	10	10	10
lambda12CrosslambdaS1	2	2	2
lambda12CrosslambdaS1Adj	-2	-2	-2
lambda12CrosslambdaS2	2	2	2
lambda12CrosslambdaS2Adj	-2	-2	-2
lambdaT3Bar(lambdaT3)	-1	-1	-1
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	8	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	8	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	10	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I2)	4	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ001300I1)	4	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	-1	0	0
lambda12CrossTr[y1.Trans(Bar(y1))]	2	2	2
lambdaT3GeneralQ000211I2Tr[y1.Trans(Bar(y1Conj))]	2	0	0
Bar(lambdaT3GeneralQ000211I2)Tr[y1Conj.Trans(Bar(y1))]	2	0	0
lambda12CrossTr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
lambda12CrossTr[y2.Trans(Bar(y2))]	2	2	2
$-\frac{4}{5}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	1	1	1
$-\frac{4}{5}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	1	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ000013I1)	0	0	4
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	8
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I2)	0	0	4
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	8
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	10
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	-1
Bar(lambdaT3GeneralQ001102I2)Tr[y2.Trans(Bar(y2Conj))]	0	0	2
lambdaT3GeneralQ001102I2Tr[y2Conj.Trans(Bar(y2))]	0	0	2
lambda12CrossTr[y2Conj.Trans(Bar(y2Conj))]	0	0	2
$-\frac{4}{5}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0	0	1

$\mathcal{S}$	E ($\alpha = -4$)
$\frac{51g2^4}{10}$	1
g2^{2}lambda12Cross	-24
gY^{2}lambda12Cross	-30
lambda12lambda12Cross	8
lambda12Cross^2	10
lambda12CrosslambdaS1	2
lambda12CrosslambdaS1Adj	-2
lambda12CrosslambdaS2	2
lambda12CrosslambdaS2Adj	-2

lambdaT3Bar(lambdaT3)	-1
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ000211I2)	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ001300I1)	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0
lambda12CrossTr[y1.Trans(Bar(y1))]	2
lambdaT3GeneralQ000211I2Tr[y1.Trans(Bar(y1Conj))]	0
Bar(lambdaT3GeneralQ000211I2)Tr[y1Conj.Trans(Bar(y1))]	0
lambda12CrossTr[y1Conj.Trans(Bar(y1Conj))]	0
lambda12CrossTr[y2.Trans(Bar(y2))]	2
$-\frac{4}{5}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2))]	1
$-\frac{4}{5}$Tr[y1Conj.Trans(Bar(y1Conj)).y2.Trans(Bar(y2))]	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ000013I1)	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ001102I2)	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0
Bar(lambdaT3GeneralQ001102I2)Tr[y2.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ001102I2Tr[y2Conj.Trans(Bar(y2))]	0
lambda12CrossTr[y2Conj.Trans(Bar(y2Conj))]	0
$-\frac{4}{5}$Tr[y1.Trans(Bar(y1)).y2Conj.Trans(Bar(y2Conj))]	0

$\beta^{(1)}_{\lambda_1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\frac{9g2^4}{4}$	1	1	1
$\frac{3g2^2gY^2}{2}$	1	1	1
$\frac{3gY^4}{4}$	1	1	1
$g2^2$lambdaH	-9	-9	-9
gY^2lambdaH	-3	-3	-3
lambdaH^2	12	12	12
lambdaH1^2	2	2	2
lambdaH2^2	6	6	6
lambdaH2Adj^2	3	3	3
lambdaT3Bar(lambdaT3)	4	4	4
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	8	0	0
lambdaHTr[yd.Trans(Bar(yd))]	12	12	12
lambdaHTr[ye.Trans(Bar(ye))]	4	4	4
lambdaHTr[yu.Trans(Bar(yu))]	12	12	12
Tr[yd.Trans(Bar(yd)).yd.Trans(Bar(yd))]	-12	-12	-12
Tr[ye.Trans(Bar(ye)).ye.Trans(Bar(ye))]	-4	-4	-4
Tr[yu.Trans(Bar(yu)).yu.Trans(Bar(yu))]	-12	-12	-12
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	4	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	24	0
$\frac{\text{lambdaH1Adj}^2}{4}$	0	0	0
$\frac{\text{lambdaH2Adj}^2}{4}$	0	0	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	0
lambdaH1Adj^2	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\frac{9g2^4}{4}$	1	1	1
$\frac{3g2^2gY^2}{2}$	1	1	1
$\frac{3gY^4}{4}$	1	1	1
$g2^2$lambdaH	−9	−9	−9
gY^2lambdaH	−3	−3	−3
lambdaH^2	12	12	12
lambdaH1^2	4	4	4
lambdaH2^2	4	4	4
lambdaH2Adj^2	0	0	0
lambdaT3Bar(lambdaT3)	4	4	4
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaHTr[yd.Trans(Bar(yd))]	12	12	12
lambdaHTr[ye.Trans(Bar(ye))]	4	4	4
lambdaHTr[yu.Trans(Bar(yu))]	12	12	12
Tr[yd.Trans(Bar(yd)).yd.Trans(Bar(yd))]	−12	−12	−12
Tr[ye.Trans(Bar(ye)).ye.Trans(Bar(ye))]	−4	−4	−4
Tr[yu.Trans(Bar(yu)).yu.Trans(Bar(yu))]	−12	−12	−12
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
$\frac{\text{lambdaH1Adj}^2}{4}$	1	1	1
$\frac{\text{lambdaH2Adj}^2}{4}$	1	1	1
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	8	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	8	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	4	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	8	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	4	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	8
lambdaH1Adj^2	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\frac{9g2^4}{4}$	1	1	1
$\frac{3g2^2gY^2}{2}$	1	1	1
$\frac{3gY^4}{4}$	1	1	1
$g2^2$lambdaH	−9	−9	−9
gY^2lambdaH	−3	−3	−3
lambdaH^2	12	12	12
lambdaH1^2	4	4	4
lambdaH2^2	4	4	4
lambdaH2Adj^2	0	0	0
lambdaT3Bar(lambdaT3)	4	4	4
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaHTr[yd.Trans(Bar(yd))]	12	12	12
lambdaHTr[ye.Trans(Bar(ye))]	4	4	4
lambdaHTr[yu.Trans(Bar(yu))]	12	12	12
Tr[yd.Trans(Bar(yd)).yd.Trans(Bar(yd))]	−12	−12	−12
Tr[ye.Trans(Bar(ye)).ye.Trans(Bar(ye))]	−4	−4	−4
Tr[yu.Trans(Bar(yu)).yu.Trans(Bar(yu))]	−12	−12	−12
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
$\frac{\text{lambdaH1Adj}^2}{4}$	1	1	1
$\frac{\text{lambdaH2Adj}^2}{4}$	1	1	1
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	8	0	0

lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	8	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	4	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	8	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	4	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	8
lambdaH1Adj^2	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\frac{9\text{g2}^4}{4}$	1	1	1
$\frac{3\text{g2}^2\text{gY}^2}{2}$	1	1	1
$\frac{3\text{gY}^4}{4}$	1	1	1
$\text{g2}^2\text{lambdaH}$	-9	-9	-9
$\text{gY}^2\text{lambdaH}$	-3	-3	-3
lambdaH^2	12	12	12
lambdaH1^2	6	6	6
lambdaH2^2	2	2	2
lambdaH2Adj^2	0	0	0
lambdaT3Bar(lambdaT3)	4	4	4
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	24	0	0
lambdaHTr[yd.Trans(Bar(yd))]	12	12	12
lambdaHTr[ye.Trans(Bar(ye))]	4	4	4
lambdaHTr[yu.Trans(Bar(yu))]	12	12	12
Tr[yd.Trans(Bar(yd)).yd.Trans(Bar(yd))]	-12	-12	-12
Tr[ye.Trans(Bar(ye)).ye.Trans(Bar(ye))]	-4	-4	-4
Tr[yu.Trans(Bar(yu)).yu.Trans(Bar(yu))]	-12	-12	-12
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	4
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	8
$\frac{\text{lambdaH1Adj}^2}{4}$	0	0	0
$\frac{\text{lambdaH2Adj}^2}{4}$	0	0	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	0
lambdaH1Adj^2	3	3	3

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\frac{9\text{g2}^4}{4}$	1	1	1
$\frac{3\text{g2}^2\text{gY}^2}{2}$	1	1	1
$\frac{3\text{gY}^4}{4}$	1	1	1
$\text{g2}^2\text{lambdaH}$	-9	-9	-9
$\text{gY}^2\text{lambdaH}$	-3	-3	-3
lambdaH^2	12	12	12
lambdaH1^2	6	6	6
lambdaH2^2	6	6	6
lambdaH2Adj^2	3	3	3
lambdaT3Bar(lambdaT3)	4	4	4
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	24	0	0
lambdaHTr[yd.Trans(Bar(yd))]	12	12	12
lambdaHTr[ye.Trans(Bar(ye))]	4	4	4
lambdaHTr[yu.Trans(Bar(yu))]	12	12	12
Tr[yd.Trans(Bar(yd)).yd.Trans(Bar(yd))]	-12	-12	-12

Tr[ye.Trans(Bar(ye)).ye.Trans(Bar(ye))]	−4	−4	−4
Tr[yu.Trans(Bar(yu)).yu.Trans(Bar(yu))]	−12	−12	−12
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	4
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	24
$\frac{\text{lambdaH1Adj}^2}{4}$	0	0	0
$\frac{\text{lambdaH2Adj}^2}{4}$	0	0	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	0
lambdaH1Adj^2	3	3	3

$\mathcal{S}$	E ($\alpha = -4$)
$\frac{9\text{g2}^4}{4}$	1
$\frac{3\text{g2}^2\text{gY}^2}{2}$	1
$\frac{3\text{gY}^4}{4}$	1
$\text{g2}^2\text{lambdaH}$	−9
$\text{gY}^2\text{lambdaH}$	−3
lambdaH^2	12
lambdaH1^2	6
lambdaH2^2	6
lambdaH2Adj^2	3
lambdaT3Bar(lambdaT3)	4
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0
lambdaHTr[yd.Trans(Bar(yd))]	12
lambdaHTr[ye.Trans(Bar(ye))]	4
lambdaHTr[yu.Trans(Bar(yu))]	12
Tr[yd.Trans(Bar(yd)).yd.Trans(Bar(yd))]	−12
Tr[ye.Trans(Bar(ye)).ye.Trans(Bar(ye))]	−4
Tr[yu.Trans(Bar(yu)).yu.Trans(Bar(yu))]	−12
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0
$\frac{\text{lambdaH1Adj}^2}{4}$	0
$\frac{\text{lambdaH2Adj}^2}{4}$	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0
lambdaH1Adj^2	3

$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$-\frac{9\text{g2}^2\text{lambdaH1}}{2}$	1	1	1
$-\frac{3\text{gY}^2\text{lambdaH1}}{2}$	1	0	0
lambdaHlambdaH1	6	6	6
lambdaH1^2	4	4	4
lambda12lambdaH2	6	6	6
lambdaH1lambdaS1	4	4	4

lambdaT3Bar(lambdaT3)	12	12	12
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ001300I1)	12	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	12	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ110200I1)	12	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	16	0	0
lambdaH1Tr[y1.Trans(Bar(y1))]	4	4	4
lambdaT3GeneralQ110200I1Tr[y1.Trans(Bar(y1Conj))]	8	0	0
Bar(lambdaT3GeneralQ110200I1)Tr[y1Conj.Trans(Bar(y1))]	8	0	0
lambdaH1Tr[y1Conj.Trans(Bar(y1Conj))]	4	0	0
lambdaH1Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaH1Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaH1Tr[yu.Trans(Bar(yu))]	6	6	6
Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	-4	-4	-4
Tr[y1Conj.Trans(Bar(y1Conj)).ye.Trans(Bar(ye))]	-4	0	0
gY^4	0	3	12
$-\frac{15gY^2lambdaH1}{2}$	0	1	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ001102I1)	0	12	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	12	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ110002I1)	0	12	0
$-\frac{51gY^2lambdaH1}{2}$	0	0	1
$g2^4$	0	0	0
$-\frac{33g2^2lambdaH1}{2}$	0	0	0
$lambdaH1Adj^2$	0	0	0
lambdaH1lambdaS1Adj	0	0	0
lambda12CrosslambdaH2	0	0	0
$\frac{9g2^4}{4}$	0	0	0
$\frac{3gY^4}{4}$	0	0	0
$g2^2lambdaH1$	0	0	0
$gY^2lambdaH1$	0	0	0
$\frac{3lambdaH1Adj^2}{4}$	0	0	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1S1bar2S2barBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaS1S1bar2S2bar)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaS1S1bar2S2barBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
$\frac{27gY^4}{4}$	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ110002I1)	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$-\frac{9g2^2lambdaH1}{2}$	0	0	0
$-\frac{3gY^2lambdaH1}{2}$	0	0	0
lambdaHlambdaH1	6	6	6
$lambdaH1^2$	4	4	4
lambda12lambdaH2	4	4	4
lambdaH1lambdaS1	6	6	6
lambdaT3Bar(lambdaT3)	6	6	6
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaH1Tr[y1.Trans(Bar(y1))]	2	2	2

lambdaT3GeneralQ110200I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(lambdaT3GeneralQ110200I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaH1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaH1Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaH1Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaH1Tr[yu.Trans(Bar(yu))]	6	6	6
Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	-2	-2	-2
Tr[y1Conj.Trans(Bar(y1Conj)).ye.Trans(Bar(ye))]	0	0	0
gY^4	0	0	0
$-\frac{15gY^2lambdaH1}{2}$	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
$-\frac{51gY^2lambdaH1}{2}$	0	0	0
$g2^4$	0	0	0
$-\frac{33g2^2lambdaH1}{2}$	0	0	0
$lambdaH1Adj^2$	0	0	0
lambdaH1lambdaS1Adj	0	0	0
lambda12CrosslambdaH2	0	0	0
$\frac{9g2^4}{4}$	1	1	1
$\frac{3gY^4}{4}$	1	1	0
$g2^2lambdaH1$	-9	-9	-9
$gY^2lambdaH1$	-3	-3	-15
$\frac{3lambdaH1Adj^2}{4}$	1	1	1
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	24	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	24	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	4	0
lambdaS1S1bar2S2barBar(lambdaHHdagS1barS2barCross)	0	3	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	2	0
lambdaHHdagS1barS2barCrossBar(lambdaS1S1bar2S2bar)	0	3	0
lambdaT3GeneralQ110101I1Bar(lambdaS1S1bar2S2bar)	0	-3	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	2	0
lambdaS1S1bar2S2barBar(lambdaT3GeneralQ110101I1)	0	-3	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	4	0
$\frac{27gY^4}{4}$	0	0	1
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ110002I1)	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$-\frac{9g2^2lambdaH1}{2}$	0	0	0
$-\frac{3gY^2lambdaH1}{2}$	0	0	0
lambdaHlambdaH1	6	6	6
$lambdaH1^2$	4	4	4
lambda12lambdaH2	4	4	4
lambdaH1lambdaS1	6	6	6
lambdaT3Bar(lambdaT3)	6	6	6
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaH1Tr[y1.Trans(Bar(y1))]	3	3	3
lambdaT3GeneralQ110200I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(lambdaT3GeneralQ110200I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaH1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaH1Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaH1Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaH1Tr[yu.Trans(Bar(yu))]	6	6	6

Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	-3	-3	-3
Tr[y1Conj.Trans(Bar(y1Conj)).ye.Trans(Bar(ye))]	0	0	0
gY^4	0	0	0
$-\frac{15gY^2lambdaH1}{2}$	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
$-\frac{51gY^2lambdaH1}{2}$	0	0	0
$g2^4$	0	0	0
$-\frac{33g2^2lambdaH1}{2}$	0	0	0
$lambdaH1Adj^2$	0	0	0
lambdaH1lambdaS1Adj	0	0	0
lambda12CrosslambdaH2	0	0	0
$\frac{9g2^4}{4}$	1	1	1
$\frac{3gY^4}{4}$	1	1	0
$g2^2lambdaH1$	-9	-9	-9
$gY^2lambdaH1$	-3	-3	-15
$\frac{3lambdaH1Adj^2}{4}$	1	1	1
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	24	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	24	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	4	0
lambdaS1S1bar2S2barBar(lambdaHHdagS1barS2barCross)	0	3	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	2	0
lambdaHHdagS1barS2barCrossBar(lambdaS1S1bar2S2bar)	0	3	0
lambdaT3GeneralQ110101I1Bar(lambdaS1S1bar2S2bar)	0	-3	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	2	0
lambdaS1S1bar2S2barBar(lambdaT3GeneralQ110101I1)	0	-3	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	4	0
$\frac{27gY^4}{4}$	0	0	1
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ110002I1)	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$-\frac{9g2^2lambdaH1}{2}$	0	0	0
$-\frac{3gY^2lambdaH1}{2}$	1	0	0
lambdaHlambdaH1	6	6	6
$lambdaH1^2$	4	4	4
lambda12lambdaH2	2	2	2
lambdaH1lambdaS1	8	8	8
lambdaT3Bar(lambdaT3)	4	4	4
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ001300I1)	20	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ110200I1)	20	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	16	0	0
lambdaH1Tr[y1.Trans(Bar(y1))]	2	2	2
lambdaT3GeneralQ110200I1Tr[y1.Trans(Bar(y1Conj))]	4	0	0
Bar(lambdaT3GeneralQ110200I1)Tr[y1Conj.Trans(Bar(y1))]	4	0	0
lambdaH1Tr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
lambdaH1Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaH1Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaH1Tr[yu.Trans(Bar(yu))]	6	6	6
Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	-2	-2	-2
Tr[y1Conj.Trans(Bar(y1Conj)).ye.Trans(Bar(ye))]	-2	0	0
gY^4	0	3	3
$-\frac{15gY^2lambdaH1}{2}$	0	1	1
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ001102I1)	0	0	4
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	4

lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ110002I1)	0	0	4
$-\frac{51 g_Y^2 \text{lambdaH1}}{2}$	0	0	0
g_2^4	6	6	6
$-\frac{33 g_2^2 \text{lambdaH1}}{2}$	1	1	1
lambdaH1Adj^2	6	6	6
lambdaH1lambdaS1Adj	4	4	4
lambda12CrosslambdaH2	0	0	0
$\frac{9 g_2^4}{4}$	0	0	0
$\frac{3 g_Y^4}{4}$	0	0	0
$g_2^2 \text{lambdaH1}$	0	0	0
$g_Y^2 \text{lambdaH1}$	0	0	0
$\frac{3 \text{lambdaH1Adj}^2}{4}$	0	0	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1S1bar2S2barBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaS1S1bar2S2bar)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaS1S1bar2S2barBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
$\frac{27 g_Y^4}{4}$	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ110002I1)	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$-\frac{9 g_2^2 \text{lambdaH1}}{2}$	0	0	0
$-\frac{3 g_Y^2 \text{lambdaH1}}{2}$	1	0	0
lambdaHlambdaH1	6	6	6
lambdaH1^2	4	4	4
lambda12lambdaH2	6	6	6
lambdaH1lambdaS1	8	8	8
lambdaT3Bar(lambdaT3)	4	4	4
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ001300I1)	20	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ110200I1)	20	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	16	0	0
lambdaH1Tr[y1.Trans(Bar(y1))]	2	2	2
lambdaT3GeneralQ110200I1Tr[y1.Trans(Bar(y1Conj))]	4	0	0
Bar(lambdaT3GeneralQ110200I1)Tr[y1Conj.Trans(Bar(y1))]	4	0	0
lambdaH1Tr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
lambdaH1Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaH1Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaH1Tr[yu.Trans(Bar(yu))]	6	6	6
Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	-2	-2	-2
Tr[y1Conj.Trans(Bar(y1Conj)).ye.Trans(Bar(ye))]	-2	0	0
g_Y^4	0	3	3
$-\frac{15 g_Y^2 \text{lambdaH1}}{2}$	0	1	1
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ001102I1)	0	0	12
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	4
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ110002I1)	0	0	12
$-\frac{51 g_Y^2 \text{lambdaH1}}{2}$	0	0	0
g_2^4	6	6	6
$-\frac{33 g_2^2 \text{lambdaH1}}{2}$	1	1	1
lambdaH1Adj^2	6	6	6

lambdaH1lambdaS1Adj	4	4	4
lambda12CrosslambdaH2	4	4	4
$\frac{9g2^4}{4}$	0	0	0
$\frac{3gY^4}{4}$	0	0	0
$g2^2$lambdaH1	0	0	0
gY^2lambdaH1	0	0	0
$\frac{3\text{lambdaH1Adj}^2}{4}$	0	0	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1S1bar2S2barBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaS1S1bar2S2bar)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaS1S1bar2S2barBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
$\frac{27gY^4}{4}$	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ001102I2)	0	0	4
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ110002I1)	0	0	4

$\mathcal{S}$	E ($\alpha = -4$)
$-\frac{9g2^2\text{lambdaH1}}{2}$	0
$-\frac{3gY^2\text{lambdaH1}}{2}$	0
lambdaHlambdaH1	6
lambdaH1^2	4
lambda12lambdaH2	6
lambdaH1lambdaS1	8
lambdaT3Bar(lambdaT3)	4
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ001300I1)	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ110200I1)	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0
lambdaH1Tr[y1.Trans(Bar(y1))]	2
lambdaT3GeneralQ110200I1Tr[y1.Trans(Bar(y1Conj))]	0
Bar(lambdaT3GeneralQ110200I1)Tr[y1Conj.Trans(Bar(y1))]	0
lambdaH1Tr[y1Conj.Trans(Bar(y1Conj))]	0
lambdaH1Tr[yd.Trans(Bar(yd))]	6
lambdaH1Tr[ye.Trans(Bar(ye))]	2
lambdaH1Tr[yu.Trans(Bar(yu))]	6
Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	-2
Tr[y1Conj.Trans(Bar(y1Conj)).ye.Trans(Bar(ye))]	0
gY^4	12
$-\frac{15gY^2\text{lambdaH1}}{2}$	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ001102I1)	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ110002I1)	0
$-\frac{51gY^2\text{lambdaH1}}{2}$	1
$g2^4$	6
$-\frac{33g2^2\text{lambdaH1}}{2}$	1
lambdaH1Adj^2	6
lambdaH1lambdaS1Adj	4
lambda12CrosslambdaH2	4
$\frac{9g2^4}{4}$	0
$\frac{3gY^4}{4}$	0
$g2^2$lambdaH1	0

$gY^2 lambdaH1$	0
$\frac{3 lambdaH1Adj^2}{4}$	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0
lambdaS1S1bar2S2barBar(lambdaHHdagS1barS2barCross)	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0
lambdaHHdagS1barS2barCrossBar(lambdaS1S1bar2S2bar)	0
lambdaT3GeneralQ110101I1Bar(lambdaS1S1bar2S2bar)	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0
lambdaS1S1bar2S2barBar(lambdaT3GeneralQ110101I1)	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0
$\frac{27 gY^4}{4}$	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ001102I2)	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ110002I1)	0

$\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$g2^2 gY^2$	0	0	0
$g2^2 lambdaH1Adj$	0	0	0
$gY^2 lambdaH1Adj$	0	0	0
lambdaHlambdaH1Adj	0	0	0
lambdaH1lambdaH1Adj	0	0	0
lambda12AdjlambdaH2Adj	0	0	0
lambdaH1AdjlambdaS1	0	0	0
$\frac{16}{3}$lambdaT3Bar(lambdaT3)	0	0	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0	0	0
lambdaH1AdjTr[y1.Trans(Bar(y1))]	0	0	0
lambdaH1AdjTr[yd.Trans(Bar(yd))]	0	0	0
lambdaH1AdjTr[ye.Trans(Bar(ye))]	0	0	0
lambdaH1AdjTr[yu.Trans(Bar(yu))]	0	0	0
$\frac{8}{3}$Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1S1bar2S2barBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaS1S1bar2S2bar)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
$-\frac{4}{3}$Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	0	0	0
$-\frac{33 g2^2 lambdaH1Adj}{2}$	0	0	0
lambdaH1AdjlambdaS1Adj	0	0	0
$-\frac{8}{3}$lambdaT3Bar(lambdaT3)	0	0	0
$-\frac{15 gY^2 lambdaH1Adj}{2}$	0	0	0
$-\frac{8}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
$-\frac{4}{3}$lambdaT3Bar(lambdaT3)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
$-\frac{51 gY^2 lambdaH1Adj}{2}$	0	0	0
$-\frac{3 gY^2 lambdaH1Adj}{2}$	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaH1AdjTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
$\frac{4}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).ye.Trans(Bar(ye))]	0	0	0
$-\frac{8}{9}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$g2^2gY^2$	6	-6	-18
$g2^2$lambdaH1Adj	-9	-9	-9
gY^2lambdaH1Adj	-3	-3	-15
lambdaHlambdaH1Adj	2	2	2
lambdaH1lambdaH1Adj	8	8	8
lambda12AdjlambdaH2Adj	1	1	1
lambdaH1AdjlambdaS1	2	2	2
$\frac{16}{3}$lambdaT3Bar(lambdaT3)	1	1	1
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	64	0	0
lambdaH1AdjTr[y1.Trans(Bar(y1))]	2	2	2
lambdaH1AdjTr[yd.Trans(Bar(yd))]	6	6	6
lambdaH1AdjTr[ye.Trans(Bar(ye))]	2	2	2
lambdaH1AdjTr[yu.Trans(Bar(yu))]	6	6	6
$\frac{8}{3}$Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	1	1	1
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	-64	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	8	0
lambdaS1S1bar2S2barBar(lambdaHHdagS1barS2barCross)	0	4	0
lambdaHHdagS1barS2barCrossBar(lambdaS1S1bar2S2bar)	0	4	0
lambdaT3GeneralQ110101I1Bar(lambdaS1S1bar2S2bar)	0	4	0
lambdaS1S1bar2S2barBar(lambdaT3GeneralQ110101I1)	0	4	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	-8	0
$-\frac{4}{3}$Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	0	0	0
$-\frac{33g2^2\text{lambdaH1Adj}}{2}$	0	0	0
lambdaH1AdjlambdaS1Adj	0	0	0
$-\frac{8}{3}$lambdaT3Bar(lambdaT3)	0	0	0
$-\frac{15gY^2\text{lambdaH1Adj}}{2}$	0	0	0
$-\frac{8}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
$-\frac{4}{3}$lambdaT3Bar(lambdaT3)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
$-\frac{51gY^2\text{lambdaH1Adj}}{2}$	0	0	0
$-\frac{3gY^2\text{lambdaH1Adj}}{2}$	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaH1AdjTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
$\frac{4}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).ye.Trans(Bar(ye))]	0	0	0
$-\frac{8}{9}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$g2^2gY^2$	6	-6	-18
$g2^2$lambdaH1Adj	-9	-9	-9
gY^2lambdaH1Adj	-3	-3	-15
lambdaHlambdaH1Adj	2	2	2
lambdaH1lambdaH1Adj	8	8	8
lambda12AdjlambdaH2Adj	1	1	1
lambdaH1AdjlambdaS1	2	2	2
$\frac{16}{3}$lambdaT3Bar(lambdaT3)	1	1	1
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	64	0	0
lambdaH1AdjTr[y1.Trans(Bar(y1))]	3	3	3
lambdaH1AdjTr[yd.Trans(Bar(yd))]	6	6	6
lambdaH1AdjTr[ye.Trans(Bar(ye))]	2	2	2
lambdaH1AdjTr[yu.Trans(Bar(yu))]	6	6	6
$\frac{8}{3}$Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	-64	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	8	0
lambdaS1S1bar2S2barBar(lambdaHHdagS1barS2barCross)	0	4	0
lambdaHHdagS1barS2barCrossBar(lambdaS1S1bar2S2bar)	0	4	0

lambdaT3GeneralQ110101I1Bar(lambdaS1S1bar2S2bar)	0	4	0
lambdaS1S1bar2S2barBar(lambdaT3GeneralQ110101I1)	0	4	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	−8	0
$-\frac{4}{3}$Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	1	1	1
$-\frac{33g2^2\text{lambdaH1Adj}}{2}$	0	0	0
lambdaH1AdjlambdaS1Adj	0	0	0
$-\frac{8}{3}$lambdaT3Bar(lambdaT3)	0	0	0
$-\frac{15gY^2\text{lambdaH1Adj}}{2}$	0	0	0
$-\frac{8}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
$-\frac{4}{3}$lambdaT3Bar(lambdaT3)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
$-\frac{51gY^2\text{lambdaH1Adj}}{2}$	0	0	0
$-\frac{3gY^2\text{lambdaH1Adj}}{2}$	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaH1AdjTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
$\frac{4}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).ye.Trans(Bar(ye))]	0	0	0
$-\frac{8}{9}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$g2^2gY^2$	0	4	−4
$g2^2$lambdaH1Adj	0	0	0
gY^2lambdaH1Adj	0	0	0
lambdaHlambdaH1Adj	2	2	2
lambdaH1lambdaH1Adj	8	8	8
lambda12AdjlambdaH2Adj	0	0	0
lambdaH1AdjlambdaS1	2	2	2
$\frac{16}{3}$lambdaT3Bar(lambdaT3)	0	0	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0	0	0
lambdaH1AdjTr[y1.Trans(Bar(y1))]	2	2	2
lambdaH1AdjTr[yd.Trans(Bar(yd))]	6	6	6
lambdaH1AdjTr[ye.Trans(Bar(ye))]	2	2	2
lambdaH1AdjTr[yu.Trans(Bar(yu))]	6	6	6
$\frac{8}{3}$Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1S1bar2S2barBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaS1S1bar2S2bar)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
$-\frac{4}{3}$Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	1	1	1
$-\frac{33g2^2\text{lambdaH1Adj}}{2}$	1	1	1
lambdaH1AdjlambdaS1Adj	16	16	16
$-\frac{8}{3}$lambdaT3Bar(lambdaT3)	1	1	1
$-\frac{15gY^2\text{lambdaH1Adj}}{2}$	0	1	1
$-\frac{8}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	1
$-\frac{4}{3}$lambdaT3Bar(lambdaT3)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
$-\frac{51gY^2\text{lambdaH1Adj}}{2}$	0	0	0
$-\frac{3gY^2\text{lambdaH1Adj}}{2}$	1	0	0
$\frac{8}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	1	0	0
lambdaH1AdjTr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
$\frac{4}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).ye.Trans(Bar(ye))]	1	0	0
$-\frac{8}{9}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$g2^2gY^2$	0	4	-4
$g2^2$lambdaH1Adj	0	0	0
gY^2lambdaH1Adj	0	0	0
lambdaHlambdaH1Adj	2	2	2
lambdaH1lambdaH1Adj	8	8	8
lambda12AdjlambdaH2Adj	12	12	12
lambdaH1AdjlambdaS1	2	2	2
$\frac{16}{3}$lambdaT3Bar(lambdaT3)	0	0	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0	0	0
lambdaH1AdjTr[y1.Trans(Bar(y1))]	2	2	2
lambdaH1AdjTr[yd.Trans(Bar(yd))]	6	6	6
lambdaH1AdjTr[ye.Trans(Bar(ye))]	2	2	2
lambdaH1AdjTr[yu.Trans(Bar(yu))]	6	6	6
$\frac{8}{3}$Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1S1bar2S2barBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaS1S1bar2S2bar)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
$-\frac{4}{3}$Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	1	1	1
$-\frac{33g2^2\text{lambdaH1Adj}}{2}$	1	1	1
lambdaH1AdjlambdaS1Adj	16	16	16
$-\frac{8}{3}$lambdaT3Bar(lambdaT3)	0	0	0
$-\frac{15gY^2\text{lambdaH1Adj}}{2}$	0	1	1
$-\frac{8}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
$-\frac{4}{3}$lambdaT3Bar(lambdaT3)	1	1	1
$-\frac{4}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	1
$-\frac{51gY^2\text{lambdaH1Adj}}{2}$	0	0	0
$-\frac{3gY^2\text{lambdaH1Adj}}{2}$	1	0	0
$\frac{8}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaH1AdjTr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
$\frac{4}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).ye.Trans(Bar(ye))]	1	0	0
$-\frac{8}{9}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	1	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$g2^2gY^2$	-8
$g2^2$lambdaH1Adj	0
gY^2lambdaH1Adj	0
lambdaHlambdaH1Adj	2
lambdaH1lambdaH1Adj	8
lambda12AdjlambdaH2Adj	12
lambdaH1AdjlambdaS1	2
$\frac{16}{3}$lambdaT3Bar(lambdaT3)	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0
lambdaH1AdjTr[y1.Trans(Bar(y1))]	2
lambdaH1AdjTr[yd.Trans(Bar(yd))]	6
lambdaH1AdjTr[ye.Trans(Bar(ye))]	2
lambdaH1AdjTr[yu.Trans(Bar(yu))]	6
$\frac{8}{3}$Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0
lambdaS1S1bar2S2barBar(lambdaHHdagS1barS2barCross)	0
lambdaHHdagS1barS2barCrossBar(lambdaS1S1bar2S2bar)	0
lambdaT3GeneralQ110101I1Bar(lambdaS1S1bar2S2bar)	0

lambdaS1S1bar2S2barBar(lambdaT3GeneralQ110101I1)	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0
$-\frac{4}{3}$Tr[y1.Trans(Bar(y1)).ye.Trans(Bar(ye))]	1
$-\frac{33g2^2 lambdaH1Adj}{2}$	1
lambdaH1AdjlambdaS1Adj	16
$-\frac{8}{3}$lambdaT3Bar(lambdaT3)	0
$-\frac{15gY^2 lambdaH1Adj}{2}$	0
$-\frac{8}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0
$-\frac{4}{3}$lambdaT3Bar(lambdaT3)	1
$-\frac{4}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0
$-\frac{51gY^2 lambdaH1Adj}{2}$	1
$-\frac{3gY^2 lambdaH1Adj}{2}$	0
$\frac{8}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0
lambdaH1AdjTr[y1Conj.Trans(Bar(y1Conj))]	0
$\frac{4}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).ye.Trans(Bar(ye))]	0
$-\frac{8}{9}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0

$\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$g2^4$	6	6	6
gY^4	3	0	3
lambda12lambdaH1	2	2	2
$-\frac{33g2^2 lambdaH2}{2}$	1	1	1
$-\frac{15gY^2 lambdaH2}{2}$	1	0	1
lambdaHlambdaH2	6	6	6
$lambdaH2^2$	4	4	4
$lambdaH2Adj^2$	6	6	6
lambdaH2lambdaS2	8	8	8
lambdaH2lambdaS2Adj	4	4	4
lambdaT3Bar(lambdaT3)	4	4	4
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ000211I1)	4	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ110200I1)	4	0	0
lambdaH2Tr[y2.Trans(Bar(y2))]	2	2	2
lambdaH2Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaH2Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaH2Tr[yu.Trans(Bar(yu))]	6	6	6
Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	-2	-2	-2
$-\frac{3gY^2 lambdaH2}{2}$	0	1	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ000013I1)	0	20	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	4	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ110002I1)	0	20	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	16	0
Bar(lambdaT3GeneralQ110002I1)Tr[y2.Trans(Bar(y2Conj))]	0	4	0
lambdaT3GeneralQ110002I1Tr[y2Conj.Trans(Bar(y2))]	0	4	0
lambdaH2Tr[y2Conj.Trans(Bar(y2Conj))]	0	2	0
Tr[y2Conj.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	-2	0
lambda12CrosslambdaH1	0	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ110200I1)	0	0	0
$-\frac{51gY^2 lambdaH2}{2}$	0	0	0
$-\frac{9g2^2 lambdaH2}{2}$	0	0	0
$\frac{9g2^4}{4}$	0	0	0
$\frac{27gY^4}{4}$	0	0	0
$g2^2$lambdaH2	0	0	0

$gY^2 lambdaH2$	0	0	0
$\frac{3 lambdaH2Adj^2}{4}$	0	0	0
$\frac{3 gY^4}{4}$	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaS1barS2S2bar2)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$g2^4$	0	0	0
gY^4	0	0	0
lambda12lambdaH1	4	4	4
$-\frac{33 g2^2 lambdaH2}{2}$	0	0	0
$-\frac{15 gY^2 lambdaH2}{2}$	0	0	0
lambdaHlambdaH2	6	6	6
$lambdaH2^2$	4	4	4
$lambdaH2Adj^2$	0	0	0
lambdaH2lambdaS2	6	6	6
lambdaH2lambdaS2Adj	0	0	0
lambdaT3Bar(lambdaT3)	6	6	6
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaH2Tr[y2.Trans(Bar(y2))]	2	2	2
lambdaH2Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaH2Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaH2Tr[yu.Trans(Bar(yu))]	6	6	6
Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	-2	-2	-2
$-\frac{3 gY^2 lambdaH2}{2}$	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
Bar(lambdaT3GeneralQ110002I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ110002I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambdaH2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[y2Conj.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	0	0
lambda12CrosslambdaH1	0	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ110200I1)	0	0	0
$-\frac{51 gY^2 lambdaH2}{2}$	0	0	0
$-\frac{9 g2^2 lambdaH2}{2}$	0	0	0
$\frac{9 g2^4}{4}$	1	1	1
$\frac{27 gY^4}{4}$	1	0	0
$g2^2 lambdaH2$	-9	-9	-9
$gY^2 lambdaH2$	-15	-3	-3
$\frac{3 lambdaH2Adj^2}{4}$	1	1	1
$\frac{3 gY^4}{4}$	0	1	1
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	4	0
lambdaS1barS2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	3	0

lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	2	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	24	0
lambdaHHdagS1barS2barCrossBar(lambdaS1barS2S2bar2)	0	3	0
lambdaT3GeneralQ110101I1Bar(lambdaS1barS2S2bar2)	0	-3	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	2	0
lambdaS1barS2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	-3	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	4	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	24

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$g2^4$	0	0	0
gY^4	0	0	0
lambda12lambdaH1	4	4	4
$-\frac{33g2^2\text{lambdaH2}}{2}$	0	0	0
$-\frac{15gY^2\text{lambdaH2}}{2}$	0	0	0
lambdaHlambdaH2	6	6	6
lambdaH2^2	4	4	4
lambdaH2Adj^2	0	0	0
lambdaH2lambdaS2	6	6	6
lambdaH2lambdaS2Adj	0	0	0
lambdaT3Bar(lambdaT3)	6	6	6
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaH2Tr[y2.Trans(Bar(y2))]	3	3	3
lambdaH2Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaH2Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaH2Tr[yu.Trans(Bar(yu))]	6	6	6
Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	-3	-3	-3
$-\frac{3gY^2\text{lambdaH2}}{2}$	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
Bar(lambdaT3GeneralQ110002I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ110002I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambdaH2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[y2Conj.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	0	0
lambda12CrosslambdaH1	0	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ110200I1)	0	0	0
$-\frac{51gY^2\text{lambdaH2}}{2}$	0	0	0
$-\frac{9g2^2\text{lambdaH2}}{2}$	0	0	0
$\frac{9g2^4}{4}$	1	1	1
$\frac{27gY^4}{4}$	1	0	0
$g2^2\text{lambdaH2}$	-9	-9	-9
$gY^2\text{lambdaH2}$	-15	-3	-3
$\frac{3\text{lambdaH2Adj}^2}{4}$	1	1	1
$\frac{3gY^4}{4}$	0	1	1
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	4	0
lambdaS1barS2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	3	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	2	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	24	0
lambdaHHdagS1barS2barCrossBar(lambdaS1barS2S2bar2)	0	3	0
lambdaT3GeneralQ110101I1Bar(lambdaS1barS2S2bar2)	0	-3	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	2	0
lambdaS1barS2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	-3	0

lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	4	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	24

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$g2^4$	0	0	0
gY^4	3	12	0
lambda12lambdaH1	6	6	6
$-\frac{33g2^2 lambdaH2}{2}$	0	0	0
$-\frac{15gY^2 lambdaH2}{2}$	1	0	0
lambdaHlambdaH2	6	6	6
$lambdaH2^2$	4	4	4
$lambdaH2Adj^2$	0	0	0
lambdaH2lambdaS2	4	4	4
lambdaH2lambdaS2Adj	0	0	0
lambdaT3Bar(lambdaT3)	12	12	12
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ000211I1)	12	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	12	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ110200I1)	12	0	0
lambdaH2Tr[y2.Trans(Bar(y2))]	4	4	4
lambdaH2Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaH2Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaH2Tr[yu.Trans(Bar(yu))]	6	6	6
Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	-4	-4	-4
$-\frac{3gY^2 lambdaH2}{2}$	0	0	1
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ000013I1)	0	0	12
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	12
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ110002I1)	0	0	12
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	16
Bar(lambdaT3GeneralQ110002I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	8
lambdaT3GeneralQ110002I1Tr[y2Conj.Trans(Bar(y2))]	0	0	8
lambdaH2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	4
Tr[y2Conj.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	0	-4
lambda12CrosslambdaH1	0	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ110200I1)	0	0	0
$-\frac{51gY^2 lambdaH2}{2}$	0	1	0
$-\frac{9g2^2 lambdaH2}{2}$	1	1	1
$\frac{9g2^4}{4}$	0	0	0
$\frac{27gY^4}{4}$	0	0	0
$g2^2 lambdaH2$	0	0	0
$gY^2 lambdaH2$	0	0	0
$\frac{3 lambdaH2Adj^2}{4}$	0	0	0
$\frac{3gY^4}{4}$	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaS1barS2S2bar2)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)

$g2^4$	6	6	6
gY^4	3	12	0
lambda12lambdaH1	6	6	6
$-\frac{33g2^2 lambdaH2}{2}$	1	1	1
$-\frac{15gY^2 lambdaH2}{2}$	1	0	0
lambdaHlambdaH2	6	6	6
$lambdaH2^2$	4	4	4
$lambdaH2Adj^2$	6	6	6
lambdaH2lambdaS2	8	8	8
lambdaH2lambdaS2Adj	4	4	4
lambdaT3Bar(lambdaT3)	4	4	4
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ000211I1)	12	0	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ110200I1)	12	0	0
lambdaH2Tr[y2.Trans(Bar(y2))]	2	2	2
lambdaH2Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaH2Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaH2Tr[yu.Trans(Bar(yu))]	6	6	6
Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	-2	-2	-2
$-\frac{3gY^2 lambdaH2}{2}$	0	0	1
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ000013I1)	0	0	20
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	4
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ110002I1)	0	0	20
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	16
Bar(lambdaT3GeneralQ110002I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	4
lambdaT3GeneralQ110002I1Tr[y2Conj.Trans(Bar(y2))]	0	0	4
lambdaH2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	2
Tr[y2Conj.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	0	-2
lambda12CrosslambdaH1	4	4	4
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ000211I2)	4	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ110200I1)	4	0	0
$-\frac{51gY^2 lambdaH2}{2}$	0	1	0
$-\frac{9g2^2 lambdaH2}{2}$	0	0	0
$\frac{9g2^4}{4}$	0	0	0
$\frac{27gY^4}{4}$	0	0	0
$g2^2 lambdaH2$	0	0	0
$gY^2 lambdaH2$	0	0	0
$\frac{3 lambdaH2Adj^2}{4}$	0	0	0
$\frac{3gY^4}{4}$	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaS1barS2S2bar2)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$g2^4$	6
gY^4	3
lambda12lambdaH1	6
$-\frac{33g2^2 lambdaH2}{2}$	1
$-\frac{15gY^2 lambdaH2}{2}$	1

lambdaHlambdaH2	6
lambdaH2^2	4
lambdaH2Adj^2	6
lambdaH2lambdaS2	8
lambdaH2lambdaS2Adj	4
lambdaT3Bar(lambdaT3)	4
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ000211I1)	0
lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ110200I1)	0
lambdaH2Tr[y2.Trans(Bar(y2))]	2
lambdaH2Tr[yd.Trans(Bar(yd))]	6
lambdaH2Tr[ye.Trans(Bar(ye))]	2
lambdaH2Tr[yu.Trans(Bar(yu))]	6
Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	-2
$-\frac{3\text{gY}^2\text{lambdaH2}}{2}$	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ000013I1)	0
lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ110002I1)	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0
Bar(lambdaT3GeneralQ110002I1)Tr[y2.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ110002I1Tr[y2Conj.Trans(Bar(y2))]	0
lambdaH2Tr[y2Conj.Trans(Bar(y2Conj))]	0
Tr[y2Conj.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0
lambda12CrosslambdaH1	4
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ000211I2)	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ110200I1)	0
$-\frac{51\text{gY}^2\text{lambdaH2}}{2}$	0
$-\frac{9\text{g2}^2\text{lambdaH2}}{2}$	0
$\frac{9\text{g2}^4}{4}$	0
$\frac{27\text{gY}^4}{4}$	0
$\text{g2}^2\text{lambdaH2}$	0
$\text{gY}^2\text{lambdaH2}$	0
$\frac{3\text{lambdaH2Adj}^2}{4}$	0
$\frac{3\text{gY}^4}{4}$	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0
lambdaS1barS2S2bar2Bar(lambdaHHdagS1barS2barCross)	0
lambdaT3GeneralQ110101I1Bar(lambdaHHdagS1barS2barCross)	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0
lambdaHHdagS1barS2barCrossBar(lambdaS1barS2S2bar2)	0
lambdaT3GeneralQ110101I1Bar(lambdaS1barS2S2bar2)	0
lambdaHHdagS1barS2barCrossBar(lambdaT3GeneralQ110101I1)	0
lambdaS1barS2S2bar2Bar(lambdaT3GeneralQ110101I1)	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0

$\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\text{g2}^2\text{gY}^2$	4	0	-4
$-\frac{33\text{g2}^2\text{lambdaH2Adj}}{2}$	1	1	1
$-\frac{15\text{gY}^2\text{lambdaH2Adj}}{2}$	1	0	1
lambdaHlambdaH2Adj	2	2	2
lambdaH2lambdaH2Adj	8	8	8
lambdaH2AdjlambdaS2	2	2	2
lambdaH2AdjlambdaS2Adj	16	16	16
$\frac{8}{3}$lambdaT3Bar(lambdaT3)	1	1	1

$\frac{8}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	1	0	0
lambdaH2AdjTr[y2.Trans(Bar(y2))]	2	2	2
lambdaH2AdjTr[yd.Trans(Bar(yd))]	6	6	6
lambdaH2AdjTr[ye.Trans(Bar(ye))]	2	2	2
lambdaH2AdjTr[yu.Trans(Bar(yu))]	6	6	6
$\frac{4}{3}$Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	1	1	1
lambda12AdjlambdaH1Adj	0	0	0
$g2^2$lambdaH2Adj	0	0	0
gY^2lambdaH2Adj	0	0	0
lambdaT3Bar(lambdaT3)	0	0	0
Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaS1barS2S2bar2)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	0
$-\frac{8}{9}$lambdaT3Bar(lambdaT3)	0	0	0
$-\frac{8}{9}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
$-\frac{51gY^2lambdaH2Adj}{2}$	0	0	0
$-\frac{3gY^2lambdaH2Adj}{2}$	0	1	0
$-\frac{8}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	1	0
lambdaH2AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	2	0
$-\frac{4}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	1	0
$-\frac{4}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$g2^2gY^2$	18	6	-6
$-\frac{33g2^2lambdaH2Adj}{2}$	0	0	0
$-\frac{15gY^2lambdaH2Adj}{2}$	0	0	0
lambdaHlambdaH2Adj	2	2	2
lambdaH2lambdaH2Adj	8	8	8
lambdaH2AdjlambdaS2	2	2	2
lambdaH2AdjlambdaS2Adj	0	0	0
$\frac{8}{3}$lambdaT3Bar(lambdaT3)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaH2AdjTr[y2.Trans(Bar(y2))]	2	2	2
lambdaH2AdjTr[yd.Trans(Bar(yd))]	6	6	6
lambdaH2AdjTr[ye.Trans(Bar(ye))]	2	2	2
lambdaH2AdjTr[yu.Trans(Bar(yu))]	6	6	6
$\frac{4}{3}$Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	0	0	0
lambda12AdjlambdaH1Adj	1	1	1
$g2^2$lambdaH2Adj	-9	-9	-9
gY^2lambdaH2Adj	-15	-3	-3
lambdaT3Bar(lambdaT3)	16	16	16
Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	8	8	8
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	-8	0
lambdaS1barS2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	-4	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	64	0
lambdaHHdagS1barS2barCrossBar(lambdaS1barS2S2bar2)	0	-4	0
lambdaT3GeneralQ110101I1Bar(lambdaS1barS2S2bar2)	0	-4	0
lambdaS1barS2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	-4	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	8	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	-64
$-\frac{8}{9}$lambdaT3Bar(lambdaT3)	0	0	0
$-\frac{8}{9}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0

$-\frac{51gY^2lambdaH2Adj}{2}$	0	0	0
$-\frac{3gY^2lambdaH2Adj}{2}$	0	0	0
$-\frac{8}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambdaH2AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{4}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$g2^2gY^2$	18	6	-6
$-\frac{33g2^2lambdaH2Adj}{2}$	0	0	0
$-\frac{15gY^2lambdaH2Adj}{2}$	0	0	0
lambdaHlambdaH2Adj	2	2	2
lambdaH2lambdaH2Adj	8	8	8
lambdaH2AdjlambdaS2	2	2	2
lambdaH2AdjlambdaS2Adj	0	0	0
$\frac{8}{3}$lambdaT3Bar(lambdaT3)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaH2AdjTr[y2.Trans(Bar(y2))]	3	3	3
lambdaH2AdjTr[yd.Trans(Bar(yd))]	6	6	6
lambdaH2AdjTr[ye.Trans(Bar(ye))]	2	2	2
lambdaH2AdjTr[yu.Trans(Bar(yu))]	6	6	6
$\frac{4}{3}$Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	0	0	0
lambda12AdjlambdaH1Adj	1	1	1
$g2^2$lambdaH2Adj	-9	-9	-9
gY^2lambdaH2Adj	-15	-3	-3
lambdaT3Bar(lambdaT3)	16	16	16
Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	-4	-4	-4
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	-8	0
lambdaS1barS2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	-4	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	64	0
lambdaHHdagS1barS2barCrossBar(lambdaS1barS2S2bar2)	0	-4	0
lambdaT3GeneralQ110101I1Bar(lambdaS1barS2S2bar2)	0	-4	0
lambdaS1barS2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	-4	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	8	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	-64
$-\frac{8}{9}$lambdaT3Bar(lambdaT3)	0	0	0
$-\frac{8}{9}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
$-\frac{51gY^2lambdaH2Adj}{2}$	0	0	0
$-\frac{3gY^2lambdaH2Adj}{2}$	0	0	0
$-\frac{8}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambdaH2AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{4}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$g2^2gY^2$	0	0	0
$-\frac{33g2^2lambdaH2Adj}{2}$	0	0	0
$-\frac{15gY^2lambdaH2Adj}{2}$	0	0	0
lambdaHlambdaH2Adj	0	0	0
lambdaH2lambdaH2Adj	0	0	0
lambdaH2AdjlambdaS2	0	0	0
lambdaH2AdjlambdaS2Adj	0	0	0
$\frac{8}{3}$lambdaT3Bar(lambdaT3)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaH2AdjTr[y2.Trans(Bar(y2))]	0	0	0

lambdaH2AdjTr[yd.Trans(Bar(yd))]	0	0	0
lambdaH2AdjTr[ye.Trans(Bar(ye))]	0	0	0
lambdaH2AdjTr[yu.Trans(Bar(yu))]	0	0	0
$\frac{4}{3}$Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	0	0	0
lambda12Adjlambda$H1Adj	0	0	0
$g2^2$lambdaH2Adj	0	0	0
gY^2lambdaH2Adj	0	0	0
lambdaT3Bar(lambdaT3)	0	0	0
Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaS1barS2S2bar2)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	0
$-\frac{8}{9}$lambdaT3Bar(lambdaT3)	0	0	0
$-\frac{8}{9}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
$-\frac{51gY^2\text{lambdaH2Adj}}{2}$	0	0	0
$-\frac{3gY^2\text{lambdaH2Adj}}{2}$	0	0	0
$-\frac{8}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambdaH2AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{4}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$g2^2gY^2$	4	8	0
$-\frac{33g2^2\text{lambdaH2Adj}}{2}$	1	1	1
$-\frac{15gY^2\text{lambdaH2Adj}}{2}$	1	0	0
lambdaHlambdaH2Adj	2	2	2
lambdaH2lambdaH2Adj	8	8	8
lambdaH2Adjlambda$S2	2	2	2
lambdaH2Adjlambda$S2Adj	16	16	16
$\frac{8}{3}$lambdaT3Bar(lambdaT3)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaH2AdjTr[y2.Trans(Bar(y2))]	2	2	2
lambdaH2AdjTr[yd.Trans(Bar(yd))]	6	6	6
lambdaH2AdjTr[ye.Trans(Bar(ye))]	2	2	2
lambdaH2AdjTr[yu.Trans(Bar(yu))]	6	6	6
$\frac{4}{3}$Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	1	1	1
lambda12Adjlambda$H1Adj	12	12	12
$g2^2$lambdaH2Adj	0	0	0
gY^2lambdaH2Adj	0	0	0
lambdaT3Bar(lambdaT3)	0	0	0
Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	0	0
lambdaHHdagS1barS2barCrossBar(lambdaS1barS2S2bar2)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	0
$-\frac{8}{9}$lambdaT3Bar(lambdaT3)	1	1	1
$-\frac{8}{9}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	1	0	0
$-\frac{51gY^2\text{lambdaH2Adj}}{2}$	0	1	0

$-\frac{3gY^2lambdaH2Adj}{2}$	0	0	1
$-\frac{8}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambdaH2AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	2
$-\frac{4}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	0	1
$-\frac{4}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0	0	1

$\mathcal{S}$	E ($\alpha = -4$)
$g2^2gY^2$	-4
$-\frac{33g2^2lambdaH2Adj}{2}$	1
$-\frac{15gY^2lambdaH2Adj}{2}$	1
lambdaHlambdaH2Adj	2
lambdaH2lambdaH2Adj	8
lambdaH2AdjlambdaS2	2
lambdaH2AdjlambdaS2Adj	16
$\frac{8}{3}$lambdaT3Bar(lambdaT3)	0
$\frac{8}{3}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0
lambdaH2AdjTr[y2.Trans(Bar(y2))]	2
lambdaH2AdjTr[yd.Trans(Bar(yd))]	6
lambdaH2AdjTr[ye.Trans(Bar(ye))]	2
lambdaH2AdjTr[yu.Trans(Bar(yu))]	6
$\frac{4}{3}$Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	1
lambda12AdjlambdaH1Adj	12
$g2^2$lambdaH2Adj	0
gY^2lambdaH2Adj	0
lambdaT3Bar(lambdaT3)	0
Tr[y2.Trans(Bar(y2)).ye.Trans(Bar(ye))]	0
lambdaHHdagS1barS2barCrossBar(lambdaHHdagS1barS2barCross)	0
lambdaS1barS2S2bar2Bar(lambdaHHdagS1barS2barCross)	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0
lambdaHHdagS1barS2barCrossBar(lambdaS1barS2S2bar2)	0
lambdaT3GeneralQ110101I1Bar(lambdaS1barS2S2bar2)	0
lambdaS1barS2S2bar2Bar(lambdaT3GeneralQ110101I1)	0
lambdaT3GeneralQ110101I1Bar(lambdaT3GeneralQ110101I1)	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0
$-\frac{8}{9}$lambdaT3Bar(lambdaT3)	1
$-\frac{8}{9}$lambdaT3GeneralQ021010I1Bar(lambdaT3GeneralQ021010I1)	0
$-\frac{51gY^2lambdaH2Adj}{2}$	0
$-\frac{3gY^2lambdaH2Adj}{2}$	0
$-\frac{8}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0
lambdaH2AdjTr[y2Conj.Trans(Bar(y2Conj))]	0
$-\frac{4}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0
$-\frac{4}{3}$lambdaT3GeneralQ020101I1Bar(lambdaT3GeneralQ020101I1)	0

$\beta^{(1)}_{\lambda_7}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$g2^2$lambdaHHdagS1barS1bar	0	0	0
gY^2lambdaHHdagS1barS1bar	0	0	0
lambdaHlambdaHHdagS1barS1bar	0	0	0
lambdaH1lambdaHHdagS1barS1bar	0	0	0
lambdaH1AdjlambdaHHdagS1barS1bar	0	0	0
lambdaHHdagS1barS1barlambdaS1	0	0	0
lambdaHHdagS2S2lambdaS1bar2S2bar2	0	0	0
lambdaHHdagS1barS1barTr[y1.Trans(Bar(y1))]	0	0	0
$-\frac{1}{3}$Bar(lambdaT3)Tr[y2.Trans(Bar(y1))]	0	0	0
lambdaHHdagS1barS1barTr[yd.Trans(Bar(yd))]	0	0	0

lambdaHHdagS1barS1barTr[ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS1barTr[yu.Trans(Bar(yu))]	0	0	0
$-\frac{1}{2}$Bar(lambdaT3)Tr[y2.Trans(Bar(y1))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
g2^{2}lambdaHHdagS1barS1bar	0	-9	0
gY^{2}lambdaHHdagS1barS1bar	0	-3	0
lambdaHlambdaHHdagS1barS1bar	0	2	0
lambdaH1lambdaHHdagS1barS1bar	0	8	0
lambdaH1AdjlambdaHHdagS1barS1bar	0	-4	0
lambdaHHdagS1barS1barlambdaS1	0	2	0
lambdaHHdagS2S2lambdaS1bar2S2bar2	0	4	0
lambdaHHdagS1barS1barTr[y1.Trans(Bar(y1))]	0	2	0
$-\frac{1}{3}$Bar(lambdaT3)Tr[y2.Trans(Bar(y1))]	0	1	0
lambdaHHdagS1barS1barTr[yd.Trans(Bar(yd))]	0	6	0
lambdaHHdagS1barS1barTr[ye.Trans(Bar(ye))]	0	2	0
lambdaHHdagS1barS1barTr[yu.Trans(Bar(yu))]	0	6	0
$-\frac{1}{2}$Bar(lambdaT3)Tr[y2.Trans(Bar(y1))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
g2^{2}lambdaHHdagS1barS1bar	0	-9	0
gY^{2}lambdaHHdagS1barS1bar	0	-3	0
lambdaHlambdaHHdagS1barS1bar	0	2	0
lambdaH1lambdaHHdagS1barS1bar	0	8	0
lambdaH1AdjlambdaHHdagS1barS1bar	0	-4	0
lambdaHHdagS1barS1barlambdaS1	0	2	0
lambdaHHdagS2S2lambdaS1bar2S2bar2	0	4	0
lambdaHHdagS1barS1barTr[y1.Trans(Bar(y1))]	0	3	0
$-\frac{1}{3}$Bar(lambdaT3)Tr[y2.Trans(Bar(y1))]	0	0	0
lambdaHHdagS1barS1barTr[yd.Trans(Bar(yd))]	0	6	0
lambdaHHdagS1barS1barTr[ye.Trans(Bar(ye))]	0	2	0
lambdaHHdagS1barS1barTr[yu.Trans(Bar(yu))]	0	6	0
$-\frac{1}{2}$Bar(lambdaT3)Tr[y2.Trans(Bar(y1))]	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
g2^{2}lambdaHHdagS1barS1bar	0	0	0
gY^{2}lambdaHHdagS1barS1bar	0	0	0
lambdaHlambdaHHdagS1barS1bar	0	0	0
lambdaH1lambdaHHdagS1barS1bar	0	0	0
lambdaH1AdjlambdaHHdagS1barS1bar	0	0	0
lambdaHHdagS1barS1barlambdaS1	0	0	0
lambdaHHdagS2S2lambdaS1bar2S2bar2	0	0	0
lambdaHHdagS1barS1barTr[y1.Trans(Bar(y1))]	0	0	0
$-\frac{1}{3}$Bar(lambdaT3)Tr[y2.Trans(Bar(y1))]	0	0	0
lambdaHHdagS1barS1barTr[yd.Trans(Bar(yd))]	0	0	0
lambdaHHdagS1barS1barTr[ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS1barTr[yu.Trans(Bar(yu))]	0	0	0
$-\frac{1}{2}$Bar(lambdaT3)Tr[y2.Trans(Bar(y1))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
g2^{2}lambdaHHdagS1barS1bar	0	0	0
gY^{2}lambdaHHdagS1barS1bar	0	0	0
lambdaHlambdaHHdagS1barS1bar	0	0	0

lambdaH1lambdaHHdagS1barS1bar	0	0	0
lambdaH1AdjlambdaHHdagS1barS1bar	0	0	0
lambdaHHdagS1barS1barlambdaS1	0	0	0
lambdaHHdagS2S2lambdaS1bar2S2bar2	0	0	0
lambdaHHdagS1barS1barTr[y1.Trans(Bar(y1))]	0	0	0
$-\frac{1}{3}$Bar(lambdaT3)Tr[y2.Trans(Bar(y1))]	0	0	0
lambdaHHdagS1barS1barTr[yd.Trans(Bar(yd))]	0	0	0
lambdaHHdagS1barS1barTr[ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS1barTr[yu.Trans(Bar(yu))]	0	0	0
$-\frac{1}{2}$Bar(lambdaT3)Tr[y2.Trans(Bar(y1))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
g2^{2}lambdaHHdagS1barS1bar	0
gY^{2}lambdaHHdagS1barS1bar	0
lambdaHlambdaHHdagS1barS1bar	0
lambdaH1lambdaHHdagS1barS1bar	0
lambdaH1AdjlambdaHHdagS1barS1bar	0
lambdaHHdagS1barS1barlambdaS1	0
lambdaHHdagS2S2lambdaS1bar2S2bar2	0
lambdaHHdagS1barS1barTr[y1.Trans(Bar(y1))]	0
$-\frac{1}{3}$Bar(lambdaT3)Tr[y2.Trans(Bar(y1))]	0
lambdaHHdagS1barS1barTr[yd.Trans(Bar(yd))]	0
lambdaHHdagS1barS1barTr[ye.Trans(Bar(ye))]	0
lambdaHHdagS1barS1barTr[yu.Trans(Bar(yu))]	0
$-\frac{1}{2}$Bar(lambdaT3)Tr[y2.Trans(Bar(y1))]	0

$\beta^{(1)}_{\lambda_{12}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
g2^{2}lambdaHHdagS1barS2barCross	0	0	0
gY^{2}lambdaHHdagS1barS2barCross	0	0	0
lambda12lambdaHHdagS1barS2barCross	0	0	0
$-\frac{\text{lambda12AdjlambdaHHdagS1barS2barCross}}{2}$	0	0	0
lambdaHlambdaHHdagS1barS2barCross	0	0	0
lambdaH1lambdaHHdagS1barS2barCross	0	0	0
lambdaH1AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaH2lambdaHHdagS1barS2barCross	0	0	0
lambdaH2AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaH2lambdaS1barS2S2bar2	0	0	0
$-\frac{\text{lambdaH2AdjlambdaS1barS2S2bar2}}{2}$	0	0	0
lambdaH1lambdaS1S1bar2S2bar	0	0	0
$\frac{\text{lambdaH1AdjlambdaS1S1bar2S2bar}}{2}$	0	0	0
lambda12AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaHlambdaT3GeneralQ110101I1	0	0	0
lambdaH1AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaS1bar2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaHHdagS1barS2barCrossTr[y1.Trans(Bar(y1))]	0	0	0
lambdaHHdagS1barS2barCrossTr[y2.Trans(Bar(y2))]	0	0	0
lambdaHHdagS1barS2barCrossTr[yd.Trans(Bar(yd))]	0	0	0
lambdaHHdagS1barS2barCrossTr[ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS2barCrossTr[yu.Trans(Bar(yu))]	0	0	0
$\frac{3}{2}$lambdaHHdagS1barS2barCrossTr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaHHdagS1barS2barCrossTr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
g2^{2}lambdaHHdagS1barS2barCross	0	-9	0
gY^{2}lambdaHHdagS1barS2barCross	0	-3	0
lambda12lambdaHHdagS1barS2barCross	0	2	0
$_\frac{\text{lambda12AdjlambdaHHdagS1barS2barCross}}{2}$	0	1	0
lambdaHlambdaHHdagS1barS2barCross	0	4	0
lambdaH1lambdaHHdagS1barS2barCross	0	4	0
lambdaH1AdjlambdaHHdagS1barS2barCross	0	2	0
lambdaH2lambdaHHdagS1barS2barCross	0	4	0
lambdaH2AdjlambdaHHdagS1barS2barCross	0	-2	0
lambdaH2lambdaS1barS2S2bar2	0	6	0
$_\frac{\text{lambdaH2AdjlambdaS1barS2S2bar2}}{2}$	0	1	0
lambdaH1lambdaS1S1bar2S2bar	0	6	0
$\frac{\text{lambdaH1AdjlambdaS1S1bar2S2bar}}{2}$	0	1	0
lambda12AdjlambdaT3GeneralQ110101I1	0	1	0
lambdaHlambdaT3GeneralQ110101I1	0	-2	0
lambdaH1AdjlambdaT3GeneralQ110101I1	0	1	0
lambdaH2AdjlambdaT3GeneralQ110101I1	0	-1	0
lambdaS1bar2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	8	0
lambdaS1bar2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	-4	0
lambdaHHdagS1barS2barCrossTr[y1.Trans(Bar(y1))]	0	1	0
lambdaHHdagS1barS2barCrossTr[y2.Trans(Bar(y2))]	0	1	0
lambdaHHdagS1barS2barCrossTr[yd.Trans(Bar(yd))]	0	6	0
lambdaHHdagS1barS2barCrossTr[ye.Trans(Bar(ye))]	0	2	0
lambdaHHdagS1barS2barCrossTr[yu.Trans(Bar(yu))]	0	6	0
$\frac{3}{2}$lambdaHHdagS1barS2barCrossTr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaHHdagS1barS2barCrossTr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
g2^{2}lambdaHHdagS1barS2barCross	0	-9	0
gY^{2}lambdaHHdagS1barS2barCross	0	-3	0
lambda12lambdaHHdagS1barS2barCross	0	2	0
$_\frac{\text{lambda12AdjlambdaHHdagS1barS2barCross}}{2}$	0	1	0
lambdaHlambdaHHdagS1barS2barCross	0	4	0
lambdaH1lambdaHHdagS1barS2barCross	0	4	0
lambdaH1AdjlambdaHHdagS1barS2barCross	0	2	0
lambdaH2lambdaHHdagS1barS2barCross	0	4	0
lambdaH2AdjlambdaHHdagS1barS2barCross	0	-2	0
lambdaH2lambdaS1barS2S2bar2	0	6	0
$_\frac{\text{lambdaH2AdjlambdaS1barS2S2bar2}}{2}$	0	1	0
lambdaH1lambdaS1S1bar2S2bar	0	6	0
$\frac{\text{lambdaH1AdjlambdaS1S1bar2S2bar}}{2}$	0	1	0
lambda12AdjlambdaT3GeneralQ110101I1	0	1	0
lambdaHlambdaT3GeneralQ110101I1	0	-2	0
lambdaH1AdjlambdaT3GeneralQ110101I1	0	1	0
lambdaH2AdjlambdaT3GeneralQ110101I1	0	-1	0
lambdaS1bar2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	8	0
lambdaS1bar2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	-4	0
lambdaHHdagS1barS2barCrossTr[y1.Trans(Bar(y1))]	0	0	0
lambdaHHdagS1barS2barCrossTr[y2.Trans(Bar(y2))]	0	0	0
lambdaHHdagS1barS2barCrossTr[yd.Trans(Bar(yd))]	0	6	0
lambdaHHdagS1barS2barCrossTr[ye.Trans(Bar(ye))]	0	2	0
lambdaHHdagS1barS2barCrossTr[yu.Trans(Bar(yu))]	0	6	0
$\frac{3}{2}$lambdaHHdagS1barS2barCrossTr[y1.Trans(Bar(y1))]	0	1	0
$\frac{3}{2}$lambdaHHdagS1barS2barCrossTr[y2.Trans(Bar(y2))]	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$g2^2$lambdaHHdagS1barS2barCross	0	0	0
gY^2lambdaHHdagS1barS2barCross	0	0	0
lambda12lambdaHHdagS1barS2barCross	0	0	0
$_\frac{\text{lambda12AdjlambdaHHdagS1barS2barCross}}{2}$	0	0	0
lambdaHlambdaHHdagS1barS2barCross	0	0	0
lambdaH1lambdaHHdagS1barS2barCross	0	0	0
lambdaH1AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaH2lambdaHHdagS1barS2barCross	0	0	0
lambdaH2AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaH2lambdaS1barS2S2bar2	0	0	0
$_\frac{\text{lambdaH2AdjlambdaS1barS2S2bar2}}{2}$	0	0	0
lambdaH1lambdaS1S1bar2S2bar	0	0	0
$\frac{\text{lambdaH1AdjlambdaS1S1bar2S2bar}}{2}$	0	0	0
lambda12AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaHlambdaT3GeneralQ110101I1	0	0	0
lambdaH1AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaS1bar2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaHHdagS1barS2barCrossTr[y1.Trans(Bar(y1))]	0	0	0
lambdaHHdagS1barS2barCrossTr[y2.Trans(Bar(y2))]	0	0	0
lambdaHHdagS1barS2barCrossTr[yd.Trans(Bar(yd))]	0	0	0
lambdaHHdagS1barS2barCrossTr[ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS2barCrossTr[yu.Trans(Bar(yu))]	0	0	0
$\frac{3}{2}$lambdaHHdagS1barS2barCrossTr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaHHdagS1barS2barCrossTr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$g2^2$lambdaHHdagS1barS2barCross	0	0	0
gY^2lambdaHHdagS1barS2barCross	0	0	0
lambda12lambdaHHdagS1barS2barCross	0	0	0
$_\frac{\text{lambda12AdjlambdaHHdagS1barS2barCross}}{2}$	0	0	0
lambdaHlambdaHHdagS1barS2barCross	0	0	0
lambdaH1lambdaHHdagS1barS2barCross	0	0	0
lambdaH1AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaH2lambdaHHdagS1barS2barCross	0	0	0
lambdaH2AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaH2lambdaS1barS2S2bar2	0	0	0
$_\frac{\text{lambdaH2AdjlambdaS1barS2S2bar2}}{2}$	0	0	0
lambdaH1lambdaS1S1bar2S2bar	0	0	0
$\frac{\text{lambdaH1AdjlambdaS1S1bar2S2bar}}{2}$	0	0	0
lambda12AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaHlambdaT3GeneralQ110101I1	0	0	0
lambdaH1AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaS1bar2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaHHdagS1barS2barCrossTr[y1.Trans(Bar(y1))]	0	0	0
lambdaHHdagS1barS2barCrossTr[y2.Trans(Bar(y2))]	0	0	0
lambdaHHdagS1barS2barCrossTr[yd.Trans(Bar(yd))]	0	0	0
lambdaHHdagS1barS2barCrossTr[ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS1barS2barCrossTr[yu.Trans(Bar(yu))]	0	0	0
$\frac{3}{2}$lambdaHHdagS1barS2barCrossTr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaHHdagS1barS2barCrossTr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
g2^{2}lambdaHHdagS1barS2barCross	0
gY^{2}lambdaHHdagS1barS2barCross	0
lambda12lambdaHHdagS1barS2barCross	0
$_\frac{\text{lambda12AdjlambdaHHdagS1barS2barCross}}{2}$	0
lambdaHlambdaHHdagS1barS2barCross	0
lambdaH1lambdaHHdagS1barS2barCross	0
lambdaH1AdjlambdaHHdagS1barS2barCross	0
lambdaH2lambdaHHdagS1barS2barCross	0
lambdaH2AdjlambdaHHdagS1barS2barCross	0
lambdaH2lambdaS1barS2S2bar2	0
$_\frac{\text{lambdaH2AdjlambdaS1barS2S2bar2}}{2}$	0
lambdaH1lambdaS1S1bar2S2bar	0
$\frac{\text{lambdaH1AdjlambdaS1S1bar2S2bar}}{2}$	0
lambda12AdjlambdaT3GeneralQ110101I1	0
lambdaHlambdaT3GeneralQ110101I1	0
lambdaH1AdjlambdaT3GeneralQ110101I1	0
lambdaH2AdjlambdaT3GeneralQ110101I1	0
lambdaS1bar2S2bar2Bar(lambdaHHdagS1barS2barCross)	0
lambdaS1bar2S2bar2Bar(lambdaT3GeneralQ110101I1)	0
lambdaHHdagS1barS2barCrossTr[y1.Trans(Bar(y1))]	0
lambdaHHdagS1barS2barCrossTr[y2.Trans(Bar(y2))]	0
lambdaHHdagS1barS2barCrossTr[yd.Trans(Bar(yd))]	0
lambdaHHdagS1barS2barCrossTr[ye.Trans(Bar(ye))]	0
lambdaHHdagS1barS2barCrossTr[yu.Trans(Bar(yu))]	0
$\frac{3}{2}$lambdaHHdagS1barS2barCrossTr[y1.Trans(Bar(y1))]	0
$\frac{3}{2}$lambdaHHdagS1barS2barCrossTr[y2.Trans(Bar(y2))]	0

$\beta^{(1)}_{\lambda_8}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
g2^{2}lambdaHHdagS2S2	0	0	0
gY^{2}lambdaHHdagS2S2	0	0	0
lambdaHlambdaHHdagS2S2	0	0	0
lambdaH2lambdaHHdagS2S2	0	0	0
lambdaH2AdjlambdaHHdagS2S2	0	0	0
lambdaHHdagS2S2lambdaS2	0	0	0
lambdaHHdagS1barS1barBar(lambdaS1bar2S2bar2)	0	0	0
Bar(lambdaT3)Tr[y1.Trans(Bar(y2))]	0	0	0
lambdaHHdagS2S2Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaHHdagS2S2Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaHHdagS2S2Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS2S2Tr[yu.Trans(Bar(yu))]	0	0	0
$\frac{3}{2}$Bar(lambdaT3)Tr[y1.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
g2^{2}lambdaHHdagS2S2	0	-9	0
gY^{2}lambdaHHdagS2S2	0	-3	0
lambdaHlambdaHHdagS2S2	0	2	0
lambdaH2lambdaHHdagS2S2	0	8	0
lambdaH2AdjlambdaHHdagS2S2	0	4	0
lambdaHHdagS2S2lambdaS2	0	2	0
lambdaHHdagS1barS1barBar(lambdaS1bar2S2bar2)	0	4	0
Bar(lambdaT3)Tr[y1.Trans(Bar(y2))]	0	1	0
lambdaHHdagS2S2Tr[y2.Trans(Bar(y2))]	0	2	0

lambdaHHdagS2S2Tr[yd.Trans(Bar(yd))]	0	6	0
lambdaHHdagS2S2Tr[ye.Trans(Bar(ye))]	0	2	0
lambdaHHdagS2S2Tr[yu.Trans(Bar(yu))]	0	6	0
$\frac{3}{2}$Bar(lambdaT3)Tr[y1.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
g2^{2}lambdaHHdagS2S2	0	-9	0
gY^{2}lambdaHHdagS2S2	0	-3	0
lambdaHlambdaHHdagS2S2	0	2	0
lambdaH2lambdaHHdagS2S2	0	8	0
lambdaH2AdjlambdaHHdagS2S2	0	4	0
lambdaHHdagS2S2lambdaS2	0	2	0
lambdaHHdagS1barS1barBar(lambdaS1bar2S2bar2)	0	4	0
Bar(lambdaT3)Tr[y1.Trans(Bar(y2))]	0	0	0
lambdaHHdagS2S2Tr[y2.Trans(Bar(y2))]	0	3	0
lambdaHHdagS2S2Tr[yd.Trans(Bar(yd))]	0	6	0
lambdaHHdagS2S2Tr[ye.Trans(Bar(ye))]	0	2	0
lambdaHHdagS2S2Tr[yu.Trans(Bar(yu))]	0	6	0
$\frac{3}{2}$Bar(lambdaT3)Tr[y1.Trans(Bar(y2))]	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
g2^{2}lambdaHHdagS2S2	0	0	0
gY^{2}lambdaHHdagS2S2	0	0	0
lambdaHlambdaHHdagS2S2	0	0	0
lambdaH2lambdaHHdagS2S2	0	0	0
lambdaH2AdjlambdaHHdagS2S2	0	0	0
lambdaHHdagS2S2lambdaS2	0	0	0
lambdaHHdagS1barS1barBar(lambdaS1bar2S2bar2)	0	0	0
Bar(lambdaT3)Tr[y1.Trans(Bar(y2))]	0	0	0
lambdaHHdagS2S2Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaHHdagS2S2Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaHHdagS2S2Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS2S2Tr[yu.Trans(Bar(yu))]	0	0	0
$\frac{3}{2}$Bar(lambdaT3)Tr[y1.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
g2^{2}lambdaHHdagS2S2	0	0	0
gY^{2}lambdaHHdagS2S2	0	0	0
lambdaHlambdaHHdagS2S2	0	0	0
lambdaH2lambdaHHdagS2S2	0	0	0
lambdaH2AdjlambdaHHdagS2S2	0	0	0
lambdaHHdagS2S2lambdaS2	0	0	0
lambdaHHdagS1barS1barBar(lambdaS1bar2S2bar2)	0	0	0
Bar(lambdaT3)Tr[y1.Trans(Bar(y2))]	0	0	0
lambdaHHdagS2S2Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaHHdagS2S2Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaHHdagS2S2Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaHHdagS2S2Tr[yu.Trans(Bar(yu))]	0	0	0
$\frac{3}{2}$Bar(lambdaT3)Tr[y1.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
g2^{2}lambdaHHdagS2S2	0
gY^{2}lambdaHHdagS2S2	0

lambdaHlambdaHHdagS2S2	0
lambdaH2lambdaHHdagS2S2	0
lambdaH2AdjlambdaHHdagS2S2	0
lambdaHHdagS2S2lambdaS2	0
lambdaHHdagS1barS1barBar(lambdaS1bar2S2bar2)	0
Bar(lambdaT3)Tr[y1.Trans(Bar(y2))]	0
lambdaHHdagS2S2Tr[y2.Trans(Bar(y2))]	0
lambdaHHdagS2S2Tr[yd.Trans(Bar(yd))]	0
lambdaHHdagS2S2Tr[ye.Trans(Bar(ye))]	0
lambdaHHdagS2S2Tr[yu.Trans(Bar(yu))]	0
$\frac{3}{2}$Bar(lambdaT3)Tr[y1.Trans(Bar(y2))]	0

$\beta^{(1)}_{\lambda^{(1)}_{S_1}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambda12^2	6	6	6
lambdaH1^2	4	4	4
lambdaS1^2	10	10	10
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	12	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	288	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	108	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	8	0	0
lambdaS1Tr[y1.Trans(Bar(y1))]	8	8	8
lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	24	0	0
Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	24	0	0
lambdaS1Tr[y1Conj.Trans(Bar(y1Conj))]	8	0	0
Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	-8	-8	-8
Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	-16	0	0
Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	-16	0	0
Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	-8	0	0
gY^4	0	12	192
gY^{2}lambdaS1	0	-12	-48
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	24	0
$\frac{9g2^4}{4}$	0	0	0
$\frac{3g2^2gY^2}{2}$	0	0	0
$\frac{3gY^4}{4}$	0	0	0
$\frac{\text{lambda12Adj}^2}{4}$	0	0	0
$\frac{\text{lambdaH1Adj}^2}{4}$	0	0	0
g2^{2}lambdaS1	0	0	0
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	0	0
$\frac{27g2^2gY^2}{2}$	0	0	0
$\frac{243gY^4}{4}$	0	0	0
g2^4	0	0	0
lambdaS1lambdaS1Adj	0	0	0
lambdaS1Adj^2	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
$\frac{832}{3}$lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	0	0	0
$\frac{352}{3}$lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
$\frac{16}{3}$lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
$\frac{32}{3}$lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
$\frac{32}{3}$Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	0	0	0
$-\frac{248}{45}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0

$-\frac{248}{45}$Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	0	0	0
$-\frac{8}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
$g2^2gY^2$	0	0	0
lambda12lambda12Cross	0	0	0
$\frac{20\text{lambda12Cross}^2}{3}$	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
$\frac{20}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$lambda12^2$	4	4	4
$lambdaH1^2$	4	4	4
$lambdaS1^2$	12	12	12
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaS1Tr[y1.Trans(Bar(y1))]	4	4	4
lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaS1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	-4	-4	-4
Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	0	0	0
Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
gY^4	0	0	0
$gY^2lambdaS1$	-3	-3	-27
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{9g2^4}{4}$	1	1	1
$\frac{3g2^2gY^2}{2}$	1	1	0
$\frac{3gY^4}{4}$	1	1	0
$\frac{\text{lambda12Adj}^2}{4}$	1	1	1
$\frac{\text{lambdaH1Adj}^2}{4}$	1	1	1
$g2^2lambdaS1$	-9	-9	-9
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	8	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	8	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	8	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	24	0
$\frac{27g2^2gY^2}{2}$	0	0	1
$\frac{243gY^4}{4}$	0	0	1
$g2^4$	0	0	0
lambdaS1lambdaS1Adj	0	0	0
$lambdaS1Adj^2$	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
$\frac{832}{3}$lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	0	0	0
$\frac{352}{3}$lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
$\frac{16}{3}$lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
$\frac{32}{3}$lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
$\frac{32}{3}$Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	0	0	0
$-\frac{248}{45}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
$-\frac{248}{45}$Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	0	0	0
$-\frac{8}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
$g2^2gY^2$	0	0	0

lambda12lambda12Cross	0	0	0
$\frac{20\text{lambda12Cross}^2}{3}$	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
$\frac{20}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambda12^2	4	4	4
lambdaH1^2	4	4	4
lambdaS1^2	12	12	12
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaS1Tr[y1.Trans(Bar(y1))]	6	6	6
lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaS1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	-5	-5	-5
Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	0	0	0
Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
gY^4	0	0	0
$\text{gY}^2\text{lambdaS1}$	-3	-3	-27
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{9\text{g2}^4}{4}$	1	1	1
$\frac{3\text{g2}^2\text{gY}^2}{2}$	1	1	0
$\frac{3\text{gY}^4}{4}$	1	1	0
$\frac{\text{lambda12Adj}^2}{4}$	1	1	1
$\frac{\text{lambdaH1Adj}^2}{4}$	1	1	1
$\text{g2}^2\text{lambdaS1}$	-9	-9	-9
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	8	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	8	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	8	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	24	0
$\frac{27\text{g2}^2\text{gY}^2}{2}$	0	0	1
$\frac{243\text{gY}^4}{4}$	0	0	1
g2^4	0	0	0
lambdaS1lambdaS1Adj	0	0	0
lambdaS1Adj^2	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
$\frac{832}{3}$lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	0	0	0
$\frac{352}{3}$lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
$\frac{16}{3}$lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
$\frac{32}{3}$lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
$\frac{32}{3}$Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	0	0	0
$-\frac{248}{45}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
$-\frac{248}{45}$Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	0	0	0
$-\frac{8}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
$\text{g2}^2\text{gY}^2$	0	0	0
lambda12lambda12Cross	0	0	0
$\frac{20\text{lambda12Cross}^2}{3}$	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0

$\frac{8}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
$\frac{20}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambda12^2	2	2	2
lambdaH1^2	4	4	4
lambdaS1^2	14	14	14
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaS1Tr[y1.Trans(Bar(y1))]	4	4	4
lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaS1Tr[y1Conj.Trans(Bar(y1Conj))]	4	0	0
Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	0	0	0
Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	0	0	0
Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
gY^4	0	12	12
gY^{2}lambdaS1	0	-12	-12
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	8
$\frac{9g2^4}{4}$	0	0	0
$\frac{3g2^2gY^2}{2}$	0	0	0
$\frac{3gY^4}{4}$	0	0	0
$\frac{lambda12Adj^2}{4}$	0	0	0
$\frac{lambdaH1Adj^2}{4}$	0	0	0
g2^{2}lambdaS1	-24	-24	-24
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	0	0
$\frac{27g2^2gY^2}{2}$	0	0	0
$\frac{243gY^4}{4}$	0	0	0
g2^4	22	22	22
lambdaS1lambdaS1Adj	8	8	8
lambdaS1Adj^2	16	16	16
$\frac{8}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	1	0	0
$\frac{832}{3}$lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	1	0	0
$\frac{352}{3}$lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	1	0	0
$\frac{16}{3}$lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	1	0	0
$\frac{32}{3}$lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	1	0	0
$\frac{32}{3}$Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	1	0	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	1	1	1
$-\frac{248}{45}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	1	0	0
$-\frac{248}{45}$Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	1	0	0
$-\frac{8}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	1	0	0
g2^{2}gY^2	0	8	8
lambda12lambda12Cross	0	0	0
$\frac{20lambda12Cross^2}{3}$	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0

lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
$\frac{20}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambda12^2	6	6	6
lambdaH1^2	4	4	4
lambdaS1^2	14	14	14
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	8	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	0	0	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaS1Tr[y1.Trans(Bar(y1))]	4	4	4
lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaS1Tr[y1Conj.Trans(Bar(y1Conj))]	4	0	0
Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	0	0	0
Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	0	0	0
Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
gY^4	0	12	12
gY^{2}lambdaS1	0	-12	-12
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	24
$\frac{9g2^4}{4}$	0	0	0
$\frac{3g2^2gY^2}{2}$	0	0	0
$\frac{3gY^4}{4}$	0	0	0
$\frac{lambda12Adj^2}{4}$	0	0	0
$\frac{lambdaH1Adj^2}{4}$	0	0	0
g2^{2}lambdaS1	-24	-24	-24
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	0	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0	0	0
$\frac{27g2^2gY^2}{2}$	0	0	0
$\frac{243gY^4}{4}$	0	0	0
g2^4	22	22	22
lambdaS1lambdaS1Adj	8	8	8
lambdaS1Adj^2	16	16	16
$\frac{8}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
$\frac{832}{3}$lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	1	0	0
$\frac{352}{3}$lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	1	0	0
$\frac{16}{3}$lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	1	0	0
$\frac{32}{3}$lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	1	0	0
$\frac{32}{3}$Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	1	0	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	1	1	1
$-\frac{248}{45}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	1	0	0
$-\frac{248}{45}$Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	1	0	0
$-\frac{8}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	1	0	0
g2^{2}gY^2	0	8	8
lambda12lambda12Cross	8	8	8
$\frac{20lambda12Cross^2}{3}$	1	1	1
$\frac{8}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	1	0	0
$\frac{8}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	1	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	4	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	8
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	8
$\frac{20}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	1

$\mathcal{S}$	E ($\alpha = -4$)
lambda12^2	6
lambdaH1^2	4
lambdaS1^2	14
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	0
lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	0
lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0
lambdaS1Tr[y1.Trans(Bar(y1))]	4
lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	0
Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	0
lambdaS1Tr[y1Conj.Trans(Bar(y1Conj))]	0
Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	0
Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0
Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	0
Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	0
gY^4	192
$\text{gY}^2\text{lambdaS1}$	-48
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0
$\frac{9\text{g2}^4}{4}$	0
$\frac{3\text{g2}^2\text{gY}^2}{2}$	0
$\frac{3\text{gY}^4}{4}$	0
$\frac{\text{lambda12Adj}^2}{4}$	0
$\frac{\text{lambdaH1Adj}^2}{4}$	0
$\text{g2}^2\text{lambdaS1}$	-24
lambdaT3GeneralQ022000I1Bar(lambdaT3GeneralQ022000I1)	0
lambdaHHdagS1barS1barBar(lambdaHHdagS1barS1bar)	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0
lambdaS1S1bar2S2barBar(lambdaS1S1bar2S2bar)	0
$\frac{27\text{g2}^2\text{gY}^2}{2}$	0
$\frac{243\text{gY}^4}{4}$	0
g2^4	22
lambdaS1lambdaS1Adj	8
lambdaS1Adj^2	16
$\frac{8}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0
$\frac{832}{3}$lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	0
$\frac{352}{3}$lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	0
$\frac{16}{3}$lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0
$\frac{32}{3}$lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	0
$\frac{32}{3}$Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	0
$-\frac{8}{3}$Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	1
$-\frac{248}{45}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0
$-\frac{248}{45}$Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	0
$-\frac{8}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	0
$\text{g2}^2\text{gY}^2$	32
lambda12lambda12Cross	8
$\frac{20\text{lambda12Cross}^2}{3}$	1
$\frac{8}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0
$\frac{8}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0
$\frac{20}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0

$\beta^{(1)}_{\lambda^{(A)}_{S_1}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$g2^4$	0	0	0
$lambdaH1Adj^2$	0	0	0
$g2^2 lambdaS1Adj$	0	0	0
$lambdaS1lambdaS1Adj$	0	0	0
$lambdaS1Adj^2$	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
$-\frac{224}{3}$lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	0	0	0
$-\frac{44}{3}$lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
$-\frac{8}{3}$lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaS1AdjTr[y1.Trans(Bar(y1))]	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
$-\frac{4}{3}$Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaS1AdjTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	0	0	0
$\frac{76}{45}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
$\frac{76}{45}$Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	0	0	0
$-\frac{2}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
$g2^2 gY^2$	0	0	0
$gY^2 lambdaS1Adj$	0	0	0
$lambda12Adj^2$	0	0	0
$-\frac{4 lambda12Cross^2}{3}$	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$g2^4$	0	0	0
$lambdaH1Adj^2$	0	0	0
$g2^2 lambdaS1Adj$	0	0	0
$lambdaS1lambdaS1Adj$	0	0	0
$lambdaS1Adj^2$	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
$-\frac{224}{3}$lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)	0	0	0
$-\frac{44}{3}$lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
$-\frac{8}{3}$lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaS1AdjTr[y1.Trans(Bar(y1))]	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
$-\frac{4}{3}$Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaS1AdjTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
$-\frac{2}{3}$Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]	0	0	0
$\frac{76}{45}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
$\frac{76}{45}$Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]	0	0	0
$-\frac{2}{3}$Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
$g2^2 gY^2$	0	0	0
$gY^2 lambdaS1Adj$	0	0	0
$lambda12Adj^2$	0	0	0
$-\frac{4 lambda12Cross^2}{3}$	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)

$g2^4$	0	0	0
$lambdaH1Adj^2$	0	0	0
$g2^2 lambdaS1Adj$	0	0	0
$lambdaS1 lambdaS1Adj$	0	0	0
$lambdaS1Adj^2$	0	0	0
$-\frac{4}{3} lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)$	0	0	0
$-\frac{224}{3} lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)$	0	0	0
$-\frac{44}{3} lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)$	0	0	0
$-\frac{8}{3} lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)$	0	0	0
$lambdaS1AdjTr[y1.Trans(Bar(y1))]$	0	0	0
$-\frac{4}{3} lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]$	0	0	0
$-\frac{4}{3} Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]$	0	0	0
$lambdaS1AdjTr[y1Conj.Trans(Bar(y1Conj))]$	0	0	0
$-\frac{2}{3} Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]$	0	0	0
$\frac{76}{45} Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]$	0	0	0
$\frac{76}{45} Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]$	0	0	0
$-\frac{2}{3} Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]$	0	0	0
$g2^2 gY^2$	0	0	0
$gY^2 lambdaS1Adj$	0	0	0
$lambda12Adj^2$	0	0	0
$-\frac{4 lambda12Cross^2}{3}$	0	0	0
$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)$	0	0	0
$-\frac{4}{3} lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)$	0	0	0
$-\frac{4}{3} lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)$	0	0	0
$-\frac{4}{3} lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$g2^4$	-2	-2	-2
$lambdaH1Adj^2$	3	3	3
$g2^2 lambdaS1Adj$	-24	-24	-24
$lambdaS1 lambdaS1Adj$	12	12	12
$lambdaS1Adj^2$	20	20	20
$-\frac{4}{3} lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)$	1	0	0
$-\frac{224}{3} lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)$	1	0	0
$-\frac{44}{3} lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)$	1	0	0
$-\frac{8}{3} lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)$	1	0	0
$lambdaS1AdjTr[y1.Trans(Bar(y1))]$	4	4	4
$-\frac{4}{3} lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]$	1	0	0
$-\frac{4}{3} Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]$	1	0	0
$lambdaS1AdjTr[y1Conj.Trans(Bar(y1Conj))]$	4	0	0
$-\frac{2}{3} Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]$	1	1	1
$\frac{76}{45} Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]$	1	0	0
$\frac{76}{45} Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]$	1	0	0
$-\frac{2}{3} Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]$	1	0	0
$g2^2 gY^2$	0	8	8
$gY^2 lambdaS1Adj$	0	-12	-12
$lambda12Adj^2$	0	0	0
$-\frac{4 lambda12Cross^2}{3}$	0	0	0
$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)$	0	0	0
$-\frac{4}{3} lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)$	0	0	0
$-\frac{4}{3} lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)$	0	0	0
$-\frac{4}{3} lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$g2^4$	-2	-2	-2
$lambdaH1Adj^2$	3	3	3

$g2^2 lambdaS1Adj$	-24	-24	-24
$lambdaS1lambdaS1Adj$	12	12	12
$lambdaS1Adj^2$	20	20	20
$-\frac{4}{3} lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)$	0	0	0
$-\frac{224}{3} lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)$	1	0	0
$-\frac{44}{3} lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)$	1	0	0
$-\frac{8}{3} lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)$	1	0	0
$lambdaS1AdjTr[y1.Trans(Bar(y1))]$	4	4	4
$-\frac{4}{3} lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]$	1	0	0
$-\frac{4}{3} Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]$	1	0	0
$lambdaS1AdjTr[y1Conj.Trans(Bar(y1Conj))]$	4	0	0
$-\frac{2}{3} Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]$	1	1	1
$\frac{76}{45} Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]$	1	0	0
$\frac{76}{45} Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]$	1	0	0
$-\frac{2}{3} Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]$	1	0	0
$g2^2 gY^2$	0	8	8
$gY^2 lambdaS1Adj$	0	-12	-12
$lambda12Adj^2$	12	12	12
$-\frac{4 lambda12Cross^2}{3}$	1	1	1
$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)$	-4	0	0
$-\frac{4}{3} lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)$	1	0	0
$-\frac{4}{3} lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)$	1	0	0
$-\frac{4}{3} lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)$	0	0	1

$\mathcal{S}$	E ($\alpha = -4$)
$g2^4$	-2
$lambdaH1Adj^2$	3
$g2^2 lambdaS1Adj$	-24
$lambdaS1lambdaS1Adj$	12
$lambdaS1Adj^2$	20
$-\frac{4}{3} lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)$	0
$-\frac{224}{3} lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000400I1)$	0
$-\frac{44}{3} lambdaT3GeneralQ001300I1Bar(lambdaT3GeneralQ001300I1)$	0
$-\frac{8}{3} lambdaT3GeneralQ110200I1Bar(lambdaT3GeneralQ110200I1)$	0
$lambdaS1AdjTr[y1.Trans(Bar(y1))]$	4
$-\frac{4}{3} lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1Conj))]$	0
$-\frac{4}{3} Bar(lambdaT3GeneralQ001300I1)Tr[y1Conj.Trans(Bar(y1))]$	0
$lambdaS1AdjTr[y1Conj.Trans(Bar(y1Conj))]$	0
$-\frac{2}{3} Tr[y1.Trans(Bar(y1)).y1.Trans(Bar(y1))]$	1
$\frac{76}{45} Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]$	0
$\frac{76}{45} Tr[y1.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1))]$	0
$-\frac{2}{3} Tr[y1Conj.Trans(Bar(y1Conj)).y1Conj.Trans(Bar(y1Conj))]$	0
$g2^2 gY^2$	32
$gY^2 lambdaS1Adj$	-48
$lambda12Adj^2$	12
$-\frac{4 lambda12Cross^2}{3}$	1
$lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)$	0
$-\frac{4}{3} lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)$	0
$-\frac{4}{3} lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)$	0
$-\frac{4}{3} lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)$	0

$\beta^{(1)}_{\lambda_{11}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$lambdaH1lambdaHHdagS1barS2barCross$	0	0	0
$\frac{lambdaH1AdjlambdaHHdagS1barS2barCross}{2}$	0	0	0

lambda12lambdaS1barS2S2bar2	0	0	0
$_\frac{\text{lambda12AdjlambdaS1barS2S2bar2}}{2}$	0	0	0
g2^{2}lambdaS1S1bar2S2bar	0	0	0
gY^{2}lambdaS1S1bar2S2bar	0	0	0
lambda12lambdaS1S1bar2S2bar	0	0	0
$_\frac{\text{5lambda12AdjlambdaS1S1bar2S2bar}}{2}$	0	0	0
lambdaS1lambdaS1S1bar2S2bar	0	0	0
lambdaH1lambdaT3GeneralQ110101I1	0	0	0
$\frac{\text{lambdaH1AdjlambdaT3GeneralQ110101I1}}{2}$	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barTr[y1.Trans(Bar(y1))]	0	0	0
lambdaS1S1bar2S2barTr[y2.Trans(Bar(y2))]	0	0	0
$\frac{9}{2}$lambdaS1S1bar2S2barTr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaS1S1bar2S2barTr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambdaH1lambdaHHdagS1barS2barCross	0	2	0
$\frac{\text{lambdaH1AdjlambdaHHdagS1barS2barCross}}{2}$	0	1	0
lambda12lambdaS1barS2S2bar2	0	6	0
$_\frac{\text{lambda12AdjlambdaS1barS2S2bar2}}{2}$	0	1	0
g2^{2}lambdaS1S1bar2S2bar	0	-9	0
gY^{2}lambdaS1S1bar2S2bar	0	-3	0
lambda12lambdaS1S1bar2S2bar	0	6	0
$_\frac{\text{5lambda12AdjlambdaS1S1bar2S2bar}}{2}$	0	1	0
lambdaS1lambdaS1S1bar2S2bar	0	12	0
lambdaH1lambdaT3GeneralQ110101I1	0	-2	0
$\frac{\text{lambdaH1AdjlambdaT3GeneralQ110101I1}}{2}$	0	1	0
lambdaS1bar2S2bar2Bar(lambdaS1barS2S2bar2)	0	4	0
lambdaS1bar2S2bar2Bar(lambdaS1S1bar2S2bar)	0	20	0
lambdaS1S1bar2S2barTr[y1.Trans(Bar(y1))]	0	3	0
lambdaS1S1bar2S2barTr[y2.Trans(Bar(y2))]	0	1	0
$\frac{9}{2}$lambdaS1S1bar2S2barTr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaS1S1bar2S2barTr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambdaH1lambdaHHdagS1barS2barCross	0	2	0
$\frac{\text{lambdaH1AdjlambdaHHdagS1barS2barCross}}{2}$	0	1	0
lambda12lambdaS1barS2S2bar2	0	6	0
$_\frac{\text{lambda12AdjlambdaS1barS2S2bar2}}{2}$	0	1	0
g2^{2}lambdaS1S1bar2S2bar	0	-9	0
gY^{2}lambdaS1S1bar2S2bar	0	-3	0
lambda12lambdaS1S1bar2S2bar	0	6	0
$_\frac{\text{5lambda12AdjlambdaS1S1bar2S2bar}}{2}$	0	1	0
lambdaS1lambdaS1S1bar2S2bar	0	12	0
lambdaH1lambdaT3GeneralQ110101I1	0	-2	0
$\frac{\text{lambdaH1AdjlambdaT3GeneralQ110101I1}}{2}$	0	1	0
lambdaS1bar2S2bar2Bar(lambdaS1barS2S2bar2)	0	4	0
lambdaS1bar2S2bar2Bar(lambdaS1S1bar2S2bar)	0	20	0
lambdaS1S1bar2S2barTr[y1.Trans(Bar(y1))]	0	0	0
lambdaS1S1bar2S2barTr[y2.Trans(Bar(y2))]	0	0	0
$\frac{9}{2}$lambdaS1S1bar2S2barTr[y1.Trans(Bar(y1))]	0	1	0
$\frac{3}{2}$lambdaS1S1bar2S2barTr[y2.Trans(Bar(y2))]	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambdaH1lambdaHHdagS1barS2barCross	0	0	0
$\frac{\text{lambdaH1AdjlambdaHHdagS1barS2barCross}}{2}$	0	0	0
lambda12lambdaS1barS2S2bar2	0	0	0
$-\frac{\text{lambda12AdjlambdaS1barS2S2bar2}}{2}$	0	0	0
$g2^2$lambdaS1S1bar2S2bar	0	0	0
gY^2lambdaS1S1bar2S2bar	0	0	0
lambda12lambdaS1S1bar2S2bar	0	0	0
$-\frac{\text{5lambda12AdjlambdaS1S1bar2S2bar}}{2}$	0	0	0
lambdaS1lambdaS1S1bar2S2bar	0	0	0
lambdaH1lambdaT3GeneralQ110101I1	0	0	0
$\frac{\text{lambdaH1AdjlambdaT3GeneralQ110101I1}}{2}$	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barTr[y1.Trans(Bar(y1))]	0	0	0
lambdaS1S1bar2S2barTr[y2.Trans(Bar(y2))]	0	0	0
$\frac{9}{2}$lambdaS1S1bar2S2barTr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaS1S1bar2S2barTr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambdaH1lambdaHHdagS1barS2barCross	0	0	0
$\frac{\text{lambdaH1AdjlambdaHHdagS1barS2barCross}}{2}$	0	0	0
lambda12lambdaS1barS2S2bar2	0	0	0
$-\frac{\text{lambda12AdjlambdaS1barS2S2bar2}}{2}$	0	0	0
$g2^2$lambdaS1S1bar2S2bar	0	0	0
gY^2lambdaS1S1bar2S2bar	0	0	0
lambda12lambdaS1S1bar2S2bar	0	0	0
$-\frac{\text{5lambda12AdjlambdaS1S1bar2S2bar}}{2}$	0	0	0
lambdaS1lambdaS1S1bar2S2bar	0	0	0
lambdaH1lambdaT3GeneralQ110101I1	0	0	0
$\frac{\text{lambdaH1AdjlambdaT3GeneralQ110101I1}}{2}$	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barTr[y1.Trans(Bar(y1))]	0	0	0
lambdaS1S1bar2S2barTr[y2.Trans(Bar(y2))]	0	0	0
$\frac{9}{2}$lambdaS1S1bar2S2barTr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaS1S1bar2S2barTr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
lambdaH1lambdaHHdagS1barS2barCross	0
$\frac{\text{lambdaH1AdjlambdaHHdagS1barS2barCross}}{2}$	0
lambda12lambdaS1barS2S2bar2	0
$-\frac{\text{lambda12AdjlambdaS1barS2S2bar2}}{2}$	0
$g2^2$lambdaS1S1bar2S2bar	0
gY^2lambdaS1S1bar2S2bar	0
lambda12lambdaS1S1bar2S2bar	0
$-\frac{\text{5lambda12AdjlambdaS1S1bar2S2bar}}{2}$	0
lambdaS1lambdaS1S1bar2S2bar	0
lambdaH1lambdaT3GeneralQ110101I1	0
$\frac{\text{lambdaH1AdjlambdaT3GeneralQ110101I1}}{2}$	0
lambdaS1bar2S2bar2Bar(lambdaS1barS2S2bar2)	0
lambdaS1bar2S2bar2Bar(lambdaS1S1bar2S2bar)	0
lambdaS1S1bar2S2barTr[y1.Trans(Bar(y1))]	0
lambdaS1S1bar2S2barTr[y2.Trans(Bar(y2))]	0

$\frac{9}{2}$lambdaS1S1bar2S2barTr[y1.Trans(Bar(y1))]	0
$\frac{3}{2}$lambdaS1S1bar2S2barTr[y2.Trans(Bar(y2))]	0

$\beta^{(1)}_{\lambda_9}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambdaHHdagS1barS2barCross^2	0	0	0
g2^{2}lambdaS1bar2S2bar2	0	0	0
gY^{2}lambdaS1bar2S2bar2	0	0	0
lambda12lambdaS1bar2S2bar2	0	0	0
lambda12AdjlambdaS1bar2S2bar2	0	0	0
lambdaS1lambdaS1bar2S2bar2	0	0	0
lambdaS1barS2S2bar2^2	0	0	0
lambdaS1barS2S2bar2lambdaS1S1bar2S2bar	0	0	0
lambdaS1S1bar2S2bar^2	0	0	0
lambdaS1bar2S2bar2lambdaS2	0	0	0
lambdaT3GeneralQ110101I1^2	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS2S2)	0	0	0
lambdaS1bar2S2bar2Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaS1bar2S2bar2Tr[y2.Trans(Bar(y2))]	0	0	0
$\frac{2}{3}$Tr[y2.Trans(Bar(y1)).y2.Trans(Bar(y1))]	0	0	0
$\frac{5}{6}$Tr[y2.Trans(Bar(y1)).y2.Trans(Bar(y1))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambdaHHdagS1barS2barCross^2	0	1	0
g2^{2}lambdaS1bar2S2bar2	0	-9	0
gY^{2}lambdaS1bar2S2bar2	0	-3	0
lambda12lambdaS1bar2S2bar2	0	8	0
lambda12AdjlambdaS1bar2S2bar2	0	-4	0
lambdaS1lambdaS1bar2S2bar2	0	2	0
lambdaS1barS2S2bar2^2	0	5	0
lambdaS1barS2S2bar2lambdaS1S1bar2S2bar	0	2	0
lambdaS1S1bar2S2bar^2	0	5	0
lambdaS1bar2S2bar2lambdaS2	0	2	0
lambdaT3GeneralQ110101I1^2	0	1	0
lambdaHHdagS1barS1barBar(lambdaHHdagS2S2)	0	4	0
lambdaS1bar2S2bar2Tr[y1.Trans(Bar(y1))]	0	2	0
lambdaS1bar2S2bar2Tr[y2.Trans(Bar(y2))]	0	2	0
$\frac{2}{3}$Tr[y2.Trans(Bar(y1)).y2.Trans(Bar(y1))]	0	1	0
$\frac{5}{6}$Tr[y2.Trans(Bar(y1)).y2.Trans(Bar(y1))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambdaHHdagS1barS2barCross^2	0	1	0
g2^{2}lambdaS1bar2S2bar2	0	-9	0
gY^{2}lambdaS1bar2S2bar2	0	-3	0
lambda12lambdaS1bar2S2bar2	0	8	0
lambda12AdjlambdaS1bar2S2bar2	0	-4	0
lambdaS1lambdaS1bar2S2bar2	0	2	0
lambdaS1barS2S2bar2^2	0	5	0
lambdaS1barS2S2bar2lambdaS1S1bar2S2bar	0	2	0
lambdaS1S1bar2S2bar^2	0	5	0
lambdaS1bar2S2bar2lambdaS2	0	2	0
lambdaT3GeneralQ110101I1^2	0	1	0
lambdaHHdagS1barS1barBar(lambdaHHdagS2S2)	0	4	0

lambdaS1bar2S2bar2Tr[y1.Trans(Bar(y1))]	0	3	0
lambdaS1bar2S2bar2Tr[y2.Trans(Bar(y2))]	0	3	0
$\frac{2}{3}$Tr[y2.Trans(Bar(y1)).y2.Trans(Bar(y1))]	0	0	0
$\frac{5}{6}$Tr[y2.Trans(Bar(y1)).y2.Trans(Bar(y1))]	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambdaHHdagS1barS2barCross^2	0	0	0
g2^{2}lambdaS1bar2S2bar2	0	0	0
gY^{2}lambdaS1bar2S2bar2	0	0	0
lambda12lambdaS1bar2S2bar2	0	0	0
lambda12AdjlambdaS1bar2S2bar2	0	0	0
lambdaS1lambdaS1bar2S2bar2	0	0	0
lambdaS1barS2S2bar2^2	0	0	0
lambdaS1barS2S2bar2lambdaS1S1bar2S2bar	0	0	0
lambdaS1S1bar2S2bar^2	0	0	0
lambdaS1bar2S2bar2lambdaS2	0	0	0
lambdaT3GeneralQ110101I1^2	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS2S2)	0	0	0
lambdaS1bar2S2bar2Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaS1bar2S2bar2Tr[y2.Trans(Bar(y2))]	0	0	0
$\frac{2}{3}$Tr[y2.Trans(Bar(y1)).y2.Trans(Bar(y1))]	0	0	0
$\frac{5}{6}$Tr[y2.Trans(Bar(y1)).y2.Trans(Bar(y1))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambdaHHdagS1barS2barCross^2	0	0	0
g2^{2}lambdaS1bar2S2bar2	0	0	0
gY^{2}lambdaS1bar2S2bar2	0	0	0
lambda12lambdaS1bar2S2bar2	0	0	0
lambda12AdjlambdaS1bar2S2bar2	0	0	0
lambdaS1lambdaS1bar2S2bar2	0	0	0
lambdaS1barS2S2bar2^2	0	0	0
lambdaS1barS2S2bar2lambdaS1S1bar2S2bar	0	0	0
lambdaS1S1bar2S2bar^2	0	0	0
lambdaS1bar2S2bar2lambdaS2	0	0	0
lambdaT3GeneralQ110101I1^2	0	0	0
lambdaHHdagS1barS1barBar(lambdaHHdagS2S2)	0	0	0
lambdaS1bar2S2bar2Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaS1bar2S2bar2Tr[y2.Trans(Bar(y2))]	0	0	0
$\frac{2}{3}$Tr[y2.Trans(Bar(y1)).y2.Trans(Bar(y1))]	0	0	0
$\frac{5}{6}$Tr[y2.Trans(Bar(y1)).y2.Trans(Bar(y1))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
lambdaHHdagS1barS2barCross^2	0
g2^{2}lambdaS1bar2S2bar2	0
gY^{2}lambdaS1bar2S2bar2	0
lambda12lambdaS1bar2S2bar2	0
lambda12AdjlambdaS1bar2S2bar2	0
lambdaS1lambdaS1bar2S2bar2	0
lambdaS1barS2S2bar2^2	0
lambdaS1barS2S2bar2lambdaS1S1bar2S2bar	0
lambdaS1S1bar2S2bar^2	0
lambdaS1bar2S2bar2lambdaS2	0
lambdaT3GeneralQ110101I1^2	0
lambdaHHdagS1barS1barBar(lambdaHHdagS2S2)	0

lambdaS1bar2S2bar2Tr[y1.Trans(Bar(y1))]	0
lambdaS1bar2S2bar2Tr[y2.Trans(Bar(y2))]	0
$\frac{2}{3}$Tr[y2.Trans(Bar(y1)).y2.Trans(Bar(y1))]	0
$\frac{5}{6}$Tr[y2.Trans(Bar(y1)).y2.Trans(Bar(y1))]	0

$\beta^{(1)}_{\lambda_{10}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambdaH2lambdaHHdagS1barS2barCross	0	0	0
$-\frac{\text{lambdaH2AdjlambdaHHdagS1barS2barCross}}{2}$	0	0	0
g2^{2}lambdaS1barS2S2bar2	0	0	0
gY^{2}lambdaS1barS2S2bar2	0	0	0
lambda12lambdaS1barS2S2bar2	0	0	0
$-\frac{\text{5lambda12AdjlambdaS1barS2S2bar2}}{2}$	0	0	0
lambda12lambdaS1S1bar2S2bar	0	0	0
$-\frac{\text{lambda12AdjlambdaS1S1bar2S2bar}}{2}$	0	0	0
lambdaS1barS2S2bar2lambdaS2	0	0	0
lambdaH2lambdaT3GeneralQ110101I1	0	0	0
$-\frac{\text{lambdaH2AdjlambdaT3GeneralQ110101I1}}{2}$	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1barS2S2bar2Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaS1barS2S2bar2Tr[y2.Trans(Bar(y2))]	0	0	0
$\frac{3}{2}$lambdaS1barS2S2bar2Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{9}{2}$lambdaS1barS2S2bar2Tr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambdaH2lambdaHHdagS1barS2barCross	0	2	0
$-\frac{\text{lambdaH2AdjlambdaHHdagS1barS2barCross}}{2}$	0	1	0
g2^{2}lambdaS1barS2S2bar2	0	-9	0
gY^{2}lambdaS1barS2S2bar2	0	-3	0
lambda12lambdaS1barS2S2bar2	0	6	0
$-\frac{\text{5lambda12AdjlambdaS1barS2S2bar2}}{2}$	0	1	0
lambda12lambdaS1S1bar2S2bar	0	6	0
$-\frac{\text{lambda12AdjlambdaS1S1bar2S2bar}}{2}$	0	1	0
lambdaS1barS2S2bar2lambdaS2	0	12	0
lambdaH2lambdaT3GeneralQ110101I1	0	-2	0
$-\frac{\text{lambdaH2AdjlambdaT3GeneralQ110101I1}}{2}$	0	1	0
lambdaS1bar2S2bar2Bar(lambdaS1barS2S2bar2)	0	20	0
lambdaS1bar2S2bar2Bar(lambdaS1S1bar2S2bar)	0	4	0
lambdaS1barS2S2bar2Tr[y1.Trans(Bar(y1))]	0	1	0
lambdaS1barS2S2bar2Tr[y2.Trans(Bar(y2))]	0	3	0
$\frac{3}{2}$lambdaS1barS2S2bar2Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{9}{2}$lambdaS1barS2S2bar2Tr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambdaH2lambdaHHdagS1barS2barCross	0	2	0
$-\frac{\text{lambdaH2AdjlambdaHHdagS1barS2barCross}}{2}$	0	1	0
g2^{2}lambdaS1barS2S2bar2	0	-9	0
gY^{2}lambdaS1barS2S2bar2	0	-3	0
lambda12lambdaS1barS2S2bar2	0	6	0
$-\frac{\text{5lambda12AdjlambdaS1barS2S2bar2}}{2}$	0	1	0
lambda12lambdaS1S1bar2S2bar	0	6	0
$-\frac{\text{lambda12AdjlambdaS1S1bar2S2bar}}{2}$	0	1	0

lambdaS1barS2S2bar2lambdaS2	0	12	0
lambdaH2lambdaT3GeneralQ110101I1	0	−2	0
$_\frac{\text{lambdaH2AdjlambdaT3GeneralQ110101I1}}{2}$	0	1	0
lambdaS1bar2S2bar2Bar(lambdaS1barS2S2bar2)	0	20	0
lambdaS1bar2S2bar2Bar(lambdaS1S1bar2S2bar)	0	4	0
lambdaS1barS2S2bar2Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaS1barS2S2bar2Tr[y2.Trans(Bar(y2))]	0	0	0
$\frac{3}{2}$lambdaS1barS2S2bar2Tr[y1.Trans(Bar(y1))]	0	1	0
$\frac{9}{2}$lambdaS1barS2S2bar2Tr[y2.Trans(Bar(y2))]	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambdaH2lambdaHHdagS1barS2barCross	0	0	0
$_\frac{\text{lambdaH2AdjlambdaHHdagS1barS2barCross}}{2}$	0	0	0
g2^{2}lambdaS1barS2S2bar2	0	0	0
gY^{2}lambdaS1barS2S2bar2	0	0	0
lambda12lambdaS1barS2S2bar2	0	0	0
$_\frac{\text{5lambda12AdjlambdaS1barS2S2bar2}}{2}$	0	0	0
lambda12lambdaS1S1bar2S2bar	0	0	0
$_\frac{\text{lambda12AdjlambdaS1S1bar2S2bar}}{2}$	0	0	0
lambdaS1barS2S2bar2lambdaS2	0	0	0
lambdaH2lambdaT3GeneralQ110101I1	0	0	0
$_\frac{\text{lambdaH2AdjlambdaT3GeneralQ110101I1}}{2}$	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1barS2S2bar2Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaS1barS2S2bar2Tr[y2.Trans(Bar(y2))]	0	0	0
$\frac{3}{2}$lambdaS1barS2S2bar2Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{9}{2}$lambdaS1barS2S2bar2Tr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambdaH2lambdaHHdagS1barS2barCross	0	0	0
$_\frac{\text{lambdaH2AdjlambdaHHdagS1barS2barCross}}{2}$	0	0	0
g2^{2}lambdaS1barS2S2bar2	0	0	0
gY^{2}lambdaS1barS2S2bar2	0	0	0
lambda12lambdaS1barS2S2bar2	0	0	0
$_\frac{\text{5lambda12AdjlambdaS1barS2S2bar2}}{2}$	0	0	0
lambda12lambdaS1S1bar2S2bar	0	0	0
$_\frac{\text{lambda12AdjlambdaS1S1bar2S2bar}}{2}$	0	0	0
lambdaS1barS2S2bar2lambdaS2	0	0	0
lambdaH2lambdaT3GeneralQ110101I1	0	0	0
$_\frac{\text{lambdaH2AdjlambdaT3GeneralQ110101I1}}{2}$	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1barS2S2bar2Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaS1barS2S2bar2Tr[y2.Trans(Bar(y2))]	0	0	0
$\frac{3}{2}$lambdaS1barS2S2bar2Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{9}{2}$lambdaS1barS2S2bar2Tr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
lambdaH2lambdaHHdagS1barS2barCross	0
$_\frac{\text{lambdaH2AdjlambdaHHdagS1barS2barCross}}{2}$	0
g2^{2}lambdaS1barS2S2bar2	0
gY^{2}lambdaS1barS2S2bar2	0
lambda12lambdaS1barS2S2bar2	0

$_\frac{5\text{lambda12Adjlambda}S1\text{bar}S2S2\text{bar2}}{2}$	0
lambda12lambda$S1S1$bar2S2bar	0
$_\frac{\text{lambda12Adjlambda}S1S1\text{bar2}S2\text{bar}}{2}$	0
lambda$S1$bar$S2S2$bar2lambda$S2$	0
lambdaH2lambdaT3GeneralQ110101I1	0
$_\frac{\text{lambdaH2Adjlambda}T3\text{GeneralQ}110101I1}{2}$	0
lambda$S1$bar$2S2$bar2Bar(lambda$S1$bar$S2S2$bar2)	0
lambda$S1$bar$2S2$bar2Bar(lambda$S1S1$bar$2S2$bar)	0
lambda$S1$bar$S2S2$bar2Tr[y1.Trans(Bar(y1))]	0
lambda$S1$bar$S2S2$bar2Tr[y2.Trans(Bar(y2))]	0
$\frac{3}{2}$lambda$S1$bar$S2S2$bar2Tr[y1.Trans(Bar(y1))]	0
$\frac{9}{2}$lambda$S1$bar$S2S2$bar2Tr[y2.Trans(Bar(y2))]	0

$\beta^{(1)}_{\lambda^{(1)}_{S_2}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$g2^4$	22	22	22
$g2^2gY^2$	8	0	8
gY^4	12	0	12
lambda12^2	2	2	2
lambdaH2^2	4	4	4
$g2^2$lambda$S2$	-24	-24	-24
gY^2lambda$S2$	-12	0	-12
$\text{lambda}S2^2$	14	14	14
lambda$S2$lambda$S2$Adj	8	8	8
$\text{lambda}S2\text{Adj}^2$	16	16	16
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	8	0	0
lambda$S2$Tr[y2.Trans(Bar(y2))]	4	4	4
$-\frac{8}{3}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	1	1	1
$\frac{832}{3}$lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	1	0
$\frac{352}{3}$lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	1	0
$\frac{8}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	1	0
$\frac{16}{3}$lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	1	0
$\frac{32}{3}$Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	1	0
$\frac{32}{3}$lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	1	0
lambda$S2$Tr[y2Conj.Trans(Bar(y2Conj))]	0	4	0
$-\frac{248}{45}$Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	1	0
$-\frac{248}{45}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	1	0
$-\frac{8}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	1	0
lambda12lambda12Cross	0	0	0
$\frac{20\text{lambda12Cross}^2}{3}$	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
$\frac{20}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0
Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	0	0	0
$\frac{9g2^4}{4}$	0	0	0
$\frac{27g2^2gY^2}{2}$	0	0	0
$\frac{243gY^4}{4}$	0	0	0
$\frac{\text{lambda12Adj}^2}{4}$	0	0	0
$\frac{\text{lambdaH2Adj}^2}{4}$	0	0	0
$\frac{3g2^2gY^2}{2}$	0	0	0
$\frac{3gY^4}{4}$	0	0	0

lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	0
Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$g2^4$	0	0	0
$g2^2gY^2$	0	0	0
gY^4	0	0	0
$lambda12^2$	4	4	4
$lambdaH2^2$	4	4	4
$g2^2lambdaS2$	-9	-9	-9
$gY^2lambdaS2$	-27	-3	-3
$lambdaS2^2$	12	12	12
lambdaS2lambdaS2Adj	0	0	0
$lambdaS2Adj^2$	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaS2Tr[y2.Trans(Bar(y2))]	4	4	4
$-\frac{8}{3}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	0	0	0
$\frac{832}{3}$lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	0
$\frac{352}{3}$lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{16}{3}$lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
$\frac{32}{3}$Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
$\frac{32}{3}$lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambdaS2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{248}{45}$Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{248}{45}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	0
$-\frac{8}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
lambda12lambda12Cross	0	0	0
$\frac{20lambda12Cross^2}{3}$	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
$\frac{20}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0
Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	-4	-4	-4
$\frac{9g2^4}{4}$	1	1	1
$\frac{27g2^2gY^2}{2}$	1	0	0
$\frac{243gY^4}{4}$	1	0	0
$\frac{lambda12Adj^2}{4}$	1	1	1
$\frac{lambdaH2Adj^2}{4}$	1	1	1
$\frac{3g2^2gY^2}{2}$	0	1	1
$\frac{3gY^4}{4}$	0	1	1
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	8	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	8	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	24	0

lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	8
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	0
Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$g2^4$	0	0	0
$g2^2gY^2$	0	0	0
gY^4	0	0	0
$lambda12^2$	4	4	4
$lambdaH2^2$	4	4	4
$g2^2lambdaS2$	-9	-9	-9
$gY^2lambdaS2$	-27	-3	-3
$lambdaS2^2$	12	12	12
lambdaS2lambdaS2Adj	0	0	0
$lambdaS2Adj^2$	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaS2Tr[y2.Trans(Bar(y2))]	6	6	6
$-\frac{8}{3}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	0	0	0
$\frac{832}{3}$lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	0
$\frac{352}{3}$lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{16}{3}$lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
$\frac{32}{3}$Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
$\frac{32}{3}$lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambdaS2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{248}{45}$Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{248}{45}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	0
$-\frac{8}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
lambda12lambda12Cross	0	0	0
$\frac{20lambda12Cross^2}{3}$	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
$\frac{20}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0
Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	-5	-5	-5
$\frac{9g2^4}{4}$	1	1	1
$\frac{27g2^2gY^2}{2}$	1	0	0
$\frac{243gY^4}{4}$	1	0	0
$\frac{lambda12Adj^2}{4}$	1	1	1
$\frac{lambdaH2Adj^2}{4}$	1	1	1
$\frac{3g2^2gY^2}{2}$	0	1	1
$\frac{3gY^4}{4}$	0	1	1
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	8	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	8	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	24	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	8
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	0

lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	0
Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$g2^4$	0	0	0
$g2^2gY^2$	0	0	0
gY^4	12	192	0
$lambda12^2$	6	6	6
$lambdaH2^2$	4	4	4
$g2^2lambdaS2$	0	0	0
$gY^2lambdaS2$	-12	-48	0
$lambdaS2^2$	10	10	10
lambdaS2lambdaS2Adj	0	0	0
$lambdaS2Adj^2$	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	24	0	0
lambdaS2Tr[y2.Trans(Bar(y2))]	8	8	8
$-\frac{8}{3}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	0	0	0
$\frac{832}{3}$lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	0
$\frac{352}{3}$lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{16}{3}$lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
$\frac{32}{3}$Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
$\frac{32}{3}$lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambdaS2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	8
$-\frac{248}{45}$Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{248}{45}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	0
$-\frac{8}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
lambda12lambda12Cross	0	0	0
$\frac{20lambda12Cross^2}{3}$	0	0	0
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
$\frac{20}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	12
$\frac{8}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{8}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	0
Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	-8	-8	-8
$\frac{9g2^4}{4}$	0	0	0
$\frac{27g2^2gY^2}{2}$	0	0	0
$\frac{243gY^4}{4}$	0	0	0
$\frac{lambda12Adj^2}{4}$	0	0	0
$\frac{lambdaH2Adj^2}{4}$	0	0	0
$\frac{3g2^2gY^2}{2}$	0	0	0
$\frac{3gY^4}{4}$	0	0	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	288
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	108
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	8
Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	24
lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	24

Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	−16
Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	−16
Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	−8

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$g2^4$	22	22	22
$g2^2gY^2$	8	32	0
gY^4	12	192	0
$lambda12^2$	6	6	6
$lambdaH2^2$	4	4	4
$g2^2lambdaS2$	−24	−24	−24
$gY^2lambdaS2$	−12	−48	0
$lambdaS2^2$	14	14	14
lambdaS2lambdaS2Adj	8	8	8
$lambdaS2Adj^2$	16	16	16
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	24	0	0
lambdaS2Tr[y2.Trans(Bar(y2))]	4	4	4
$-\frac{8}{3}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	1	1	1
$\frac{832}{3}$lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	1
$\frac{352}{3}$lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	1
$\frac{8}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$\frac{16}{3}$lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	1
$\frac{32}{3}$Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	1
$\frac{32}{3}$lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	1
lambdaS2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	4
$-\frac{248}{45}$Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	1
$-\frac{248}{45}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	1
$-\frac{8}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	1
lambda12lambda12Cross	8	8	8
$\frac{20lambda12Cross^2}{3}$	1	1	1
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	8	0	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	8	0	0
$\frac{20}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	1	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	8
$\frac{8}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	1
$\frac{8}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	1
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0	0	4
Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	0	0	0
$\frac{9g2^4}{4}$	0	0	0
$\frac{27g2^2gY^2}{2}$	0	0	0
$\frac{243gY^4}{4}$	0	0	0
$\frac{lambda12Adj^2}{4}$	0	0	0
$\frac{lambdaH2Adj^2}{4}$	0	0	0
$\frac{3g2^2gY^2}{2}$	0	0	0
$\frac{3gY^4}{4}$	0	0	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	0

Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$g2^4$	22
$g2^2gY^2$	8
gY^4	12
$lambda12^2$	6
$lambdaH2^2$	4
$g2^2lambdaS2$	-24
$gY^2lambdaS2$	-12
$lambdaS2^2$	14
lambdaS2lambdaS2Adj	8
$lambdaS2Adj^2$	16
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I1)	0
lambdaS2Tr[y2.Trans(Bar(y2))]	4
$-\frac{8}{3}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	1
$\frac{832}{3}$lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0
$\frac{352}{3}$lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0
$\frac{8}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0
$\frac{16}{3}$lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0
$\frac{32}{3}$Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0
$\frac{32}{3}$lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0
lambdaS2Tr[y2Conj.Trans(Bar(y2Conj))]	0
$-\frac{248}{45}$Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0
$-\frac{248}{45}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0
$-\frac{8}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0
lambda12lambda12Cross	8
$\frac{20lambda12Cross^2}{3}$	1
lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I1)	0
lambdaT3GeneralQ000211I1Bar(lambdaT3GeneralQ000211I2)	0
$\frac{20}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0
$\frac{8}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0
$\frac{8}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I2)	0
Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	0
$\frac{9g2^4}{4}$	0
$\frac{27g2^2gY^2}{2}$	0
$\frac{243gY^4}{4}$	0
$\frac{lambda12Adj^2}{4}$	0
$\frac{lambdaH2Adj^2}{4}$	0
$\frac{3g2^2gY^2}{2}$	0
$\frac{3gY^4}{4}$	0
lambdaHHdagS2S2Bar(lambdaHHdagS2S2)	0
lambdaS1bar2S2bar2Bar(lambdaS1bar2S2bar2)	0
lambdaS1barS2S2bar2Bar(lambdaS1barS2S2bar2)	0
lambdaT3GeneralQ020002I1Bar(lambdaT3GeneralQ020002I1)	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0
lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0
Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0
Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0
Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0
Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0

$\beta^{(1)}_{\lambda^{(A)}_{S_2}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$g2^4$	-2	-2	-2
$g2^2gY^2$	8	0	8
$lambdaH2Adj^2$	3	3	3
$g2^2lambdaS2Adj$	-24	-24	-24
$gY^2lambdaS2Adj$	-12	0	-12
lambdaS2lambdaS2Adj	12	12	12
$lambdaS2Adj^2$	20	20	20
lambdaS2AdjTr[y2.Trans(Bar(y2))]	4	4	4
$-\frac{2}{3}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	1	1	1
$-\frac{224}{3}$lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	1	0
$-\frac{44}{3}$lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	1	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	1	0
$-\frac{8}{3}$lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	1	0
$-\frac{4}{3}$Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	1	0
$-\frac{4}{3}$lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	1	0
lambdaS2AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	4	0
$\frac{76}{45}$Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	1	0
$\frac{76}{45}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	1	0
$-\frac{2}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	1	0
$lambda12Adj^2$	0	0	0
$-\frac{4lambda12Cross^2}{3}$	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$g2^4$	0	0	0
$g2^2gY^2$	0	0	0
$lambdaH2Adj^2$	0	0	0
$g2^2lambdaS2Adj$	0	0	0
$gY^2lambdaS2Adj$	0	0	0
lambdaS2lambdaS2Adj	0	0	0
$lambdaS2Adj^2$	0	0	0
lambdaS2AdjTr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{2}{3}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{224}{3}$lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	0
$-\frac{44}{3}$lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$-\frac{8}{3}$lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
$-\frac{4}{3}$Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambdaS2AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$\frac{76}{45}$Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
$\frac{76}{45}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	0
$-\frac{2}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
$lambda12Adj^2$	0	0	0
$-\frac{4lambda12Cross^2}{3}$	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$g2^4$	0	0	0
$g2^2gY^2$	0	0	0
$lambdaH2Adj^2$	0	0	0
$g2^2lambdaS2Adj$	0	0	0
$gY^2lambdaS2Adj$	0	0	0
lambdaS2lambdaS2Adj	0	0	0
$lambdaS2Adj^2$	0	0	0
lambdaS2AdjTr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{2}{3}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{224}{3}$lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	0
$-\frac{44}{3}$lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$-\frac{8}{3}$lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
$-\frac{4}{3}$Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambdaS2AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$\frac{76}{45}$Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
$\frac{76}{45}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	0
$-\frac{2}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
$lambda12Adj^2$	0	0	0
$-\frac{4lambda12Cross^2}{3}$	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$g2^4$	0	0	0
$g2^2gY^2$	0	0	0
$lambdaH2Adj^2$	0	0	0
$g2^2lambdaS2Adj$	0	0	0
$gY^2lambdaS2Adj$	0	0	0
lambdaS2lambdaS2Adj	0	0	0
$lambdaS2Adj^2$	0	0	0
lambdaS2AdjTr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{2}{3}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	0	0	0
$-\frac{224}{3}$lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	0
$-\frac{44}{3}$lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$-\frac{8}{3}$lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
$-\frac{4}{3}$Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambdaS2AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$\frac{76}{45}$Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
$\frac{76}{45}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	0
$-\frac{2}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
$lambda12Adj^2$	0	0	0
$-\frac{4lambda12Cross^2}{3}$	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)

$g2^4$	-2	-2	-2
$g2^2gY^2$	8	32	0
$lambdaH2Adj^2$	3	3	3
$g2^2lambdaS2Adj$	-24	-24	-24
$gY^2lambdaS2Adj$	-12	-48	0
lambdaS2lambdaS2Adj	12	12	12
$lambdaS2Adj^2$	20	20	20
lambdaS2AdjTr[y2.Trans(Bar(y2))]	4	4	4
$-\frac{2}{3}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	1	1	1
$-\frac{224}{3}$lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0	0	1
$-\frac{44}{3}$lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0	0	1
$-\frac{4}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
$-\frac{8}{3}$lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0	0	1
$-\frac{4}{3}$Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	1
$-\frac{4}{3}$lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0	0	1
lambdaS2AdjTr[y2Conj.Trans(Bar(y2Conj))]	0	0	4
$\frac{76}{45}$Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0	0	1
$\frac{76}{45}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0	0	1
$-\frac{2}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	1
$lambda12Adj^2$	12	12	12
$-\frac{4lambda12Cross^2}{3}$	1	1	1
$-\frac{4}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	1	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0	0	-4
$-\frac{4}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0	0	1
$-\frac{4}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0	0	1

$\mathcal{S}$	E ($\alpha = -4$)
$g2^4$	-2
$g2^2gY^2$	8
$lambdaH2Adj^2$	3
$g2^2lambdaS2Adj$	-24
$gY^2lambdaS2Adj$	-12
lambdaS2lambdaS2Adj	12
$lambdaS2Adj^2$	20
lambdaS2AdjTr[y2.Trans(Bar(y2))]	4
$-\frac{2}{3}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2))]	1
$-\frac{224}{3}$lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000004I1)	0
$-\frac{44}{3}$lambdaT3GeneralQ000013I1Bar(lambdaT3GeneralQ000013I1)	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0
$-\frac{8}{3}$lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ110002I1)	0
$-\frac{4}{3}$Bar(lambdaT3GeneralQ000013I1)Tr[y2.Trans(Bar(y2Conj))]	0
$-\frac{4}{3}$lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2))]	0
lambdaS2AdjTr[y2Conj.Trans(Bar(y2Conj))]	0
$\frac{76}{45}$Tr[y2.Trans(Bar(y2)).y2Conj.Trans(Bar(y2Conj))]	0
$\frac{76}{45}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2))]	0
$-\frac{2}{3}$Tr[y2Conj.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0
$lambda12Adj^2$	12
$-\frac{4lambda12Cross^2}{3}$	1
$-\frac{4}{3}$lambdaT3GeneralQ000211I2Bar(lambdaT3GeneralQ000211I2)	0
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I1)	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ001102I1)	0
$-\frac{4}{3}$lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ001102I2)	0

$\beta^{(1)}_{\lambda_5}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)

$_\frac{21g2^2\text{lambdaT3}}{2}$	1	1	1
$_\frac{9gY^2\text{lambdaT3}}{2}$	1	1	0
lambda12lambdaT3	2	2	2
lambdaHlambdaT3	2	2	2
lambdaH1lambdaT3	4	4	4
lambdaH2lambdaT3	4	4	4
lambdaH2AdjlambdaT3	4	4	4
Bar(lambdaT3GeneralQ000211I1)Bar(lambdaT3GeneralQ021010I1)	4	0	0
Bar(lambdaT3GeneralQ021010I1)Bar(lambdaT3GeneralQ110200I1)	8	0	0
lambdaT3Tr[y1.Trans(Bar(y1))]	2	2	2
Bar(lambdaT3GeneralQ021010I1)Tr[y1.Trans(Bar(y1Conj))]	4	0	0
lambdaT3Tr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
lambdaT3Tr[y2.Trans(Bar(y2))]	1	1	1
lambdaT3Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaT3Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaT3Tr[yu.Trans(Bar(yu))]	6	6	6
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ020101I1)	0	4	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ020101I1)	0	8	0
Bar(lambdaT3GeneralQ020101I1)Tr[y2.Trans(Bar(y2Conj))]	0	2	0
lambdaT3Tr[y2Conj.Trans(Bar(y2Conj))]	0	1	0
$_\frac{33gY^2\text{lambdaT3}}{2}$	0	0	1
lambdaH1AdjlambdaT3	0	0	0
$_\frac{33g2^2\text{lambdaT3}}{2}$	0	0	0
$_\frac{4\text{lambda12AdjlambdaT3}}{3}$	0	0	0
lambda12CrosslambdaT3	0	0	0
$_\frac{4\text{lambdaH2AdjlambdaT3}}{3}$	0	0	0
Bar(lambdaT3GeneralQ000211I2)Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ020101I1)	0	0	0
$g2^2$lambdaT3	0	0	0
gY^2lambdaT3	0	0	0
$\frac{\text{lambda12AdjlambdaT3}}{6}$	0	0	0
$\frac{2\text{lambdaH1AdjlambdaT3}}{3}$	0	0	0
Bar(lambdaHHdagS2S2)Tr[y1.Trans(Bar(y2))]	0	0	0
$-\frac{2}{3}$Bar(lambdaHHdagS1barS1bar)Tr[y2.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaT3Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaT3Tr[y2.Trans(Bar(y2))]	0	0	0
Bar(lambdaHHdagS1barS1bar)Tr[y2.Trans(Bar(y1))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$_\frac{21g2^2\text{lambdaT3}}{2}$	0	0	0
$_\frac{9gY^2\text{lambdaT3}}{2}$	0	0	0
lambda12lambdaT3	2	2	2
lambdaHlambdaT3	2	2	2
lambdaH1lambdaT3	4	4	4
lambdaH2lambdaT3	4	4	4
lambdaH2AdjlambdaT3	2	2	2
Bar(lambdaT3GeneralQ000211I1)Bar(lambdaT3GeneralQ021010I1)	0	0	0
Bar(lambdaT3GeneralQ021010I1)Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaT3Tr[y1.Trans(Bar(y1))]	1	1	1
Bar(lambdaT3GeneralQ021010I1)Tr[y1.Trans(Bar(y1Conj))]	0	0	0
lambdaT3Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaT3Tr[y2.Trans(Bar(y2))]	1	1	1
lambdaT3Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaT3Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaT3Tr[yu.Trans(Bar(yu))]	6	6	6
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ020101I1)	0	0	0

Bar(lambdaT3GeneralQ020101I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{33gY^2lambdaT3}{2}$	0	0	0
lambdaH1AdjlambdaT3	0	0	0
$-\frac{33g2^2lambdaT3}{2}$	0	0	0
$-\frac{4lambda12AdjlambdaT3}{3}$	0	0	0
lambda12CrosslambdaT3	0	0	0
$-\frac{4lambdaH2AdjlambdaT3}{3}$	0	0	0
Bar(lambdaT3GeneralQ000211I2)Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ020101I1)	0	0	0
$g2^2$lambdaT3	−9	−9	−9
gY^2lambdaT3	−9	−3	−9
$\frac{lambda12AdjlambdaT3}{6}$	1	1	1
$\frac{2lambdaH1AdjlambdaT3}{3}$	1	1	1
Bar(lambdaHHdagS2S2)Tr[y1.Trans(Bar(y2))]	0	2	0
$-\frac{2}{3}$Bar(lambdaHHdagS1barS1bar)Tr[y2.Trans(Bar(y1))]	0	1	0
$\frac{3}{2}$lambdaT3Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaT3Tr[y2.Trans(Bar(y2))]	0	0	0
Bar(lambdaHHdagS1barS1bar)Tr[y2.Trans(Bar(y1))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$-\frac{21g2^2lambdaT3}{2}$	0	0	0
$-\frac{9gY^2lambdaT3}{2}$	0	0	0
lambda12lambdaT3	2	2	2
lambdaHlambdaT3	2	2	2
lambdaH1lambdaT3	4	4	4
lambdaH2lambdaT3	4	4	4
lambdaH2AdjlambdaT3	2	2	2
Bar(lambdaT3GeneralQ000211I1)Bar(lambdaT3GeneralQ021010I1)	0	0	0
Bar(lambdaT3GeneralQ021010I1)Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaT3Tr[y1.Trans(Bar(y1))]	0	0	0
Bar(lambdaT3GeneralQ021010I1)Tr[y1.Trans(Bar(y1Conj))]	0	0	0
lambdaT3Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaT3Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaT3Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaT3Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaT3Tr[yu.Trans(Bar(yu))]	6	6	6
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ020101I1)	0	0	0
Bar(lambdaT3GeneralQ020101I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{33gY^2lambdaT3}{2}$	0	0	0
lambdaH1AdjlambdaT3	0	0	0
$-\frac{33g2^2lambdaT3}{2}$	0	0	0
$-\frac{4lambda12AdjlambdaT3}{3}$	0	0	0
lambda12CrosslambdaT3	0	0	0
$-\frac{4lambdaH2AdjlambdaT3}{3}$	0	0	0
Bar(lambdaT3GeneralQ000211I2)Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ020101I1)	0	0	0
$g2^2$lambdaT3	−9	−9	−9
gY^2lambdaT3	−9	−3	−9
$\frac{lambda12AdjlambdaT3}{6}$	1	1	1
$\frac{2lambdaH1AdjlambdaT3}{3}$	1	1	1
Bar(lambdaHHdagS2S2)Tr[y1.Trans(Bar(y2))]	0	3	0
$-\frac{2}{3}$Bar(lambdaHHdagS1barS1bar)Tr[y2.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaT3Tr[y1.Trans(Bar(y1))]	1	1	1
$\frac{3}{2}$lambdaT3Tr[y2.Trans(Bar(y2))]	1	1	1

Bar(lambdaHHdagS1barS1bar)Tr[y2.Trans(Bar(y1))]	0	−1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$-\frac{21g2^2\text{lambdaT3}}{2}$	1	1	1
$-\frac{9gY^2\text{lambdaT3}}{2}$	1	0	1
lambda12lambdaT3	2	2	2
lambdaHlambdaT3	2	2	2
lambdaH1lambdaT3	4	4	4
lambdaH2lambdaT3	4	4	4
lambdaH2AdjlambdaT3	0	0	0
Bar(lambdaT3GeneralQ000211I1)Bar(lambdaT3GeneralQ021010I1)	4	0	0
Bar(lambdaT3GeneralQ021010I1)Bar(lambdaT3GeneralQ110200I1)	8	0	0
lambdaT3Tr[y1.Trans(Bar(y1))]	1	1	1
Bar(lambdaT3GeneralQ021010I1)Tr[y1.Trans(Bar(y1Conj))]	2	0	0
lambdaT3Tr[y1Conj.Trans(Bar(y1Conj))]	1	0	0
lambdaT3Tr[y2.Trans(Bar(y2))]	2	2	2
lambdaT3Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaT3Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaT3Tr[yu.Trans(Bar(yu))]	6	6	6
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ020101I1)	0	0	4
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ020101I1)	0	0	8
Bar(lambdaT3GeneralQ020101I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	4
lambdaT3Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	2
$-\frac{33gY^2\text{lambdaT3}}{2}$	0	1	0
lambdaH1AdjlambdaT3	−4	−4	−4
$-\frac{33g2^2\text{lambdaT3}}{2}$	0	0	0
$-\frac{4\text{lambda12AdjlambdaT3}}{3}$	0	0	0
lambda12CrosslambdaT3	0	0	0
$-\frac{4\text{lambdaH2AdjlambdaT3}}{3}$	0	0	0
Bar(lambdaT3GeneralQ000211I2)Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ020101I1)	0	0	0
$g2^2$lambdaT3	0	0	0
gY^2lambdaT3	0	0	0
$\frac{\text{lambda12AdjlambdaT3}}{6}$	0	0	0
$\frac{2\text{lambdaH1AdjlambdaT3}}{3}$	0	0	0
Bar(lambdaHHdagS2S2)Tr[y1.Trans(Bar(y2))]	0	0	0
$-\frac{2}{3}$Bar(lambdaHHdagS1barS1bar)Tr[y2.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaT3Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaT3Tr[y2.Trans(Bar(y2))]	0	0	0
Bar(lambdaHHdagS1barS1bar)Tr[y2.Trans(Bar(y1))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$-\frac{21g2^2\text{lambdaT3}}{2}$	0	0	0
$-\frac{9gY^2\text{lambdaT3}}{2}$	1	0	1
lambda12lambdaT3	2	2	2
lambdaHlambdaT3	2	2	2
lambdaH1lambdaT3	4	4	4
lambdaH2lambdaT3	4	4	4
lambdaH2AdjlambdaT3	0	0	0
Bar(lambdaT3GeneralQ000211I1)Bar(lambdaT3GeneralQ021010I1)	4	0	0
Bar(lambdaT3GeneralQ021010I1)Bar(lambdaT3GeneralQ110200I1)	8	0	0
lambdaT3Tr[y1.Trans(Bar(y1))]	1	1	1
Bar(lambdaT3GeneralQ021010I1)Tr[y1.Trans(Bar(y1Conj))]	2	0	0
lambdaT3Tr[y1Conj.Trans(Bar(y1Conj))]	1	0	0
lambdaT3Tr[y2.Trans(Bar(y2))]	1	1	1

lambdaT3Tr[yd.Trans(Bar(yd))]	6	6	6
lambdaT3Tr[ye.Trans(Bar(ye))]	2	2	2
lambdaT3Tr[yu.Trans(Bar(yu))]	6	6	6
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ020101I1)	0	0	4
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ020101I1)	0	0	8
Bar(lambdaT3GeneralQ020101I1)Tr[y2.Trans(Bar(y2Conj))]	0	0	2
lambdaT3Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	1
$-\frac{33gY^2lambdaT3}{2}$	0	1	0
lambdaH1AdjlambdaT3	-2	-2	-2
$-\frac{33g2^2lambdaT3}{2}$	1	1	1
$-\frac{4lambda12AdjlambdaT3}{3}$	1	1	1
lambda12CrosslambdaT3	-2	-2	-2
$-\frac{4lambdaH2AdjlambdaT3}{3}$	1	1	1
Bar(lambdaT3GeneralQ000211I2)Bar(lambdaT3GeneralQ021010I1)	-2	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ020101I1)	0	0	-2
$g2^2lambdaT3$	0	0	0
$gY^2lambdaT3$	0	0	0
$\frac{lambda12AdjlambdaT3}{6}$	0	0	0
$\frac{2lambdaH1AdjlambdaT3}{3}$	0	0	0
Bar(lambdaHHdagS2S2)Tr[y1.Trans(Bar(y2))]	0	0	0
$-\frac{2}{3}$Bar(lambdaHHdagS1barS1bar)Tr[y2.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaT3Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaT3Tr[y2.Trans(Bar(y2))]	0	0	0
Bar(lambdaHHdagS1barS1bar)Tr[y2.Trans(Bar(y1))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$-\frac{21g2^2lambdaT3}{2}$	0
$-\frac{9gY^2lambdaT3}{2}$	0
lambda12lambdaT3	2
lambdaHlambdaT3	2
lambdaH1lambdaT3	4
lambdaH2lambdaT3	4
lambdaH2AdjlambdaT3	0
Bar(lambdaT3GeneralQ000211I1)Bar(lambdaT3GeneralQ021010I1)	0
Bar(lambdaT3GeneralQ021010I1)Bar(lambdaT3GeneralQ110200I1)	0
lambdaT3Tr[y1.Trans(Bar(y1))]	1
Bar(lambdaT3GeneralQ021010I1)Tr[y1.Trans(Bar(y1Conj))]	0
lambdaT3Tr[y1Conj.Trans(Bar(y1Conj))]	0
lambdaT3Tr[y2.Trans(Bar(y2))]	1
lambdaT3Tr[yd.Trans(Bar(yd))]	6
lambdaT3Tr[ye.Trans(Bar(ye))]	2
lambdaT3Tr[yu.Trans(Bar(yu))]	6
lambdaT3GeneralQ001102I1Bar(lambdaT3GeneralQ020101I1)	0
lambdaT3GeneralQ110002I1Bar(lambdaT3GeneralQ020101I1)	0
Bar(lambdaT3GeneralQ020101I1)Tr[y2.Trans(Bar(y2Conj))]	0
lambdaT3Tr[y2Conj.Trans(Bar(y2Conj))]	0
$-\frac{33gY^2lambdaT3}{2}$	1
lambdaH1AdjlambdaT3	-2
$-\frac{33g2^2lambdaT3}{2}$	1
$-\frac{4lambda12AdjlambdaT3}{3}$	1
lambda12CrosslambdaT3	-2
$-\frac{4lambdaH2AdjlambdaT3}{3}$	1
Bar(lambdaT3GeneralQ000211I2)Bar(lambdaT3GeneralQ021010I1)	0
lambdaT3GeneralQ001102I2Bar(lambdaT3GeneralQ020101I1)	0
$g2^2lambdaT3$	0
$gY^2lambdaT3$	0
$\frac{lambda12AdjlambdaT3}{6}$	0

$\frac{\text{2lambdaH1AdjlambdaT3}}{3}$	0
Bar(lambdaHHdagS2S2)Tr[y1.Trans(Bar(y2))]	0
$-\frac{2}{3}$Bar(lambdaHHdagS1barS1bar)Tr[y2.Trans(Bar(y1))]	0
$\frac{3}{2}$lambdaT3Tr[y1.Trans(Bar(y1))]	0
$\frac{3}{2}$lambdaT3Tr[y2.Trans(Bar(y2))]	0
Bar(lambdaHHdagS1barS1bar)Tr[y2.Trans(Bar(y1))]	0

$\beta^{(1)}_{lambdaT3GeneralQ000004I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
g2^{2}lambdaT3GeneralQ000004I1	0	-24	0
lambdaS2lambdaT3GeneralQ000004I1	0	12	0
lambdaS2AdjlambdaT3GeneralQ000004I1	0	-8	0
lambdaT3GeneralQ000013I1^2	0	11	0
lambdaT3GeneralQ001102I1^2	0	1	0
lambdaT3GeneralQ110002I1^2	0	2	0
lambdaT3GeneralQ000004I1Tr[y2.Trans(Bar(y2))]	0	4	0
lambdaT3GeneralQ000013I1Tr[y2.Trans(Bar(y2Conj))]	0	2	0
lambdaT3GeneralQ000004I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	4	0
$-\frac{23}{15}$Tr[y2.Trans(Bar(y2Conj)).y2.Trans(Bar(y2Conj))]	0	1	0
lambdaT3GeneralQ001102I1lambdaT3GeneralQ001102I2	0	0	0
lambdaT3GeneralQ001102I2^2	0	0	0
Tr[y2.Trans(Bar(y2Conj)).y2.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
g2^{2}lambdaT3GeneralQ000004I1	0	0	0
lambdaS2lambdaT3GeneralQ000004I1	0	0	0
lambdaS2AdjlambdaT3GeneralQ000004I1	0	0	0
lambdaT3GeneralQ000013I1^2	0	0	0
lambdaT3GeneralQ001102I1^2	0	0	0
lambdaT3GeneralQ110002I1^2	0	0	0
lambdaT3GeneralQ000004I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ000013I1Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ000004I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{23}{15}$Tr[y2.Trans(Bar(y2Conj)).y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I1lambdaT3GeneralQ001102I2	0	0	0
lambdaT3GeneralQ001102I2^2	0	0	0
Tr[y2.Trans(Bar(y2Conj)).y2.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
g2^{2}lambdaT3GeneralQ000004I1	0	0	0
lambdaS2lambdaT3GeneralQ000004I1	0	0	0
lambdaS2AdjlambdaT3GeneralQ000004I1	0	0	0
lambdaT3GeneralQ000013I1^2	0	0	0
lambdaT3GeneralQ001102I1^2	0	0	0
lambdaT3GeneralQ110002I1^2	0	0	0
lambdaT3GeneralQ000004I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ000013I1Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ000004I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{23}{15}$Tr[y2.Trans(Bar(y2Conj)).y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I1lambdaT3GeneralQ001102I2	0	0	0
lambdaT3GeneralQ001102I2^2	0	0	0
Tr[y2.Trans(Bar(y2Conj)).y2.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
g2^{2}lambdaT3GeneralQ000004I1	0	0	0
lambdaS2lambdaT3GeneralQ000004I1	0	0	12
lambdaS2AdjlambdaT3GeneralQ000004I1	0	0	0
lambdaT3GeneralQ000013I1^2	0	0	9
lambdaT3GeneralQ001102I1^2	0	0	3
lambdaT3GeneralQ110002I1^2	0	0	2
lambdaT3GeneralQ000004I1Tr[y2.Trans(Bar(y2))]	0	0	8
lambdaT3GeneralQ000013I1Tr[y2.Trans(Bar(y2Conj))]	0	0	4
lambdaT3GeneralQ000004I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	8
$-\frac{23}{15}$Tr[y2.Trans(Bar(y2Conj)).y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I1lambdaT3GeneralQ001102I2	0	0	0
lambdaT3GeneralQ001102I2^2	0	0	0
Tr[y2.Trans(Bar(y2Conj)).y2.Trans(Bar(y2Conj))]	0	0	-4

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
g2^{2}lambdaT3GeneralQ000004I1	0	0	-24
lambdaS2lambdaT3GeneralQ000004I1	0	0	12
lambdaS2AdjlambdaT3GeneralQ000004I1	0	0	-8
lambdaT3GeneralQ000013I1^2	0	0	11
lambdaT3GeneralQ001102I1^2	0	0	3
lambdaT3GeneralQ110002I1^2	0	0	2
lambdaT3GeneralQ000004I1Tr[y2.Trans(Bar(y2))]	0	0	4
lambdaT3GeneralQ000013I1Tr[y2.Trans(Bar(y2Conj))]	0	0	2
lambdaT3GeneralQ000004I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	4
$-\frac{23}{15}$Tr[y2.Trans(Bar(y2Conj)).y2.Trans(Bar(y2Conj))]	0	0	1
lambdaT3GeneralQ001102I1lambdaT3GeneralQ001102I2	0	0	2
lambdaT3GeneralQ001102I2^2	0	0	1
Tr[y2.Trans(Bar(y2Conj)).y2.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
g2^{2}lambdaT3GeneralQ000004I1	0
lambdaS2lambdaT3GeneralQ000004I1	0
lambdaS2AdjlambdaT3GeneralQ000004I1	0
lambdaT3GeneralQ000013I1^2	0
lambdaT3GeneralQ001102I1^2	0
lambdaT3GeneralQ110002I1^2	0
lambdaT3GeneralQ000004I1Tr[y2.Trans(Bar(y2))]	0
lambdaT3GeneralQ000013I1Tr[y2.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ000004I1Tr[y2Conj.Trans(Bar(y2Conj))]	0
$-\frac{23}{15}$Tr[y2.Trans(Bar(y2Conj)).y2.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ001102I1lambdaT3GeneralQ001102I2	0
lambdaT3GeneralQ001102I2^2	0
Tr[y2.Trans(Bar(y2Conj)).y2.Trans(Bar(y2Conj))]	0

$\beta^{(1)}_{lambdaT3GeneralQ000013I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
g2^{2}lambdaT3GeneralQ000013I1	0	-24	0
lambdaS2lambdaT3GeneralQ000013I1	0	22	0
lambda12lambdaT3GeneralQ001102I1	0	2	0
lambdaH2lambdaT3GeneralQ110002I1	0	4	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000013I1)	0	88	0

lambdaT3GeneralQ000013I1Tr[y2.Trans(Bar(y2))]	0	4	0
lambdaS2Tr[y2.Trans(Bar(y2Conj))]	0	2	0
lambdaT3GeneralQ000004I1Tr[y2Conj.Trans(Bar(y2))]	0	8	0
lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	4	0
$-\frac{46}{15}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2Conj))]	0	1	0
$-\frac{46}{15}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	1	0
lambda12CrosslambdaT3GeneralQ001102I1	0	0	0
lambda12lambdaT3GeneralQ001102I2	0	0	0
lambda12CrosslambdaT3GeneralQ001102I2	0	0	0
Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2Conj))]	0	0	0
Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$g2^2$lambdaT3GeneralQ000013I1	0	0	0
lambdaS2lambdaT3GeneralQ000013I1	0	0	0
lambda12lambdaT3GeneralQ001102I1	0	0	0
lambdaH2lambdaT3GeneralQ110002I1	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ000013I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaS2Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ000004I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{46}{15}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2Conj))]	0	0	0
$-\frac{46}{15}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
lambda12CrosslambdaT3GeneralQ001102I1	0	0	0
lambda12lambdaT3GeneralQ001102I2	0	0	0
lambda12CrosslambdaT3GeneralQ001102I2	0	0	0
Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2Conj))]	0	0	0
Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$g2^2$lambdaT3GeneralQ000013I1	0	0	0
lambdaS2lambdaT3GeneralQ000013I1	0	0	0
lambda12lambdaT3GeneralQ001102I1	0	0	0
lambdaH2lambdaT3GeneralQ110002I1	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ000013I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaS2Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ000004I1Tr[y2Conj.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{46}{15}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2Conj))]	0	0	0
$-\frac{46}{15}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
lambda12CrosslambdaT3GeneralQ001102I1	0	0	0
lambda12lambdaT3GeneralQ001102I2	0	0	0
lambda12CrosslambdaT3GeneralQ001102I2	0	0	0
Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2Conj))]	0	0	0
Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$g2^2$lambdaT3GeneralQ000013I1	0	0	0
lambdaS2lambdaT3GeneralQ000013I1	0	0	18
lambda12lambdaT3GeneralQ001102I1	0	0	6
lambdaH2lambdaT3GeneralQ110002I1	0	0	4
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000013I1)	0	0	72
lambdaT3GeneralQ000013I1Tr[y2.Trans(Bar(y2))]	0	0	8

lambdaS2Tr[y2.Trans(Bar(y2Conj))]	0	0	4
lambdaT3GeneralQ000004I1Tr[y2Conj.Trans(Bar(y2))]	0	0	16
lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	8
$-\frac{46}{15}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2Conj))]	0	0	0
$-\frac{46}{15}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0
lambda12CrosslambdaT3GeneralQ001102I1	0	0	0
lambda12lambdaT3GeneralQ001102I2	0	0	0
lambda12CrosslambdaT3GeneralQ001102I2	0	0	0
Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2Conj))]	0	0	−8
Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	−8

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$g2^2$lambdaT3GeneralQ000013I1	0	0	−24
lambdaS2lambdaT3GeneralQ000013I1	0	0	22
lambda12lambdaT3GeneralQ001102I1	0	0	6
lambdaH2lambdaT3GeneralQ110002I1	0	0	4
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000013I1)	0	0	88
lambdaT3GeneralQ000013I1Tr[y2.Trans(Bar(y2))]	0	0	4
lambdaS2Tr[y2.Trans(Bar(y2Conj))]	0	0	2
lambdaT3GeneralQ000004I1Tr[y2Conj.Trans(Bar(y2))]	0	0	8
lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	4
$-\frac{46}{15}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2Conj))]	0	0	1
$-\frac{46}{15}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	1
lambda12CrosslambdaT3GeneralQ001102I1	0	0	4
lambda12lambdaT3GeneralQ001102I2	0	0	2
lambda12CrosslambdaT3GeneralQ001102I2	0	0	4
Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2Conj))]	0	0	0
Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$g2^2$lambdaT3GeneralQ000013I1	0
lambdaS2lambdaT3GeneralQ000013I1	0
lambda12lambdaT3GeneralQ001102I1	0
lambdaH2lambdaT3GeneralQ110002I1	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ000013I1)	0
lambdaT3GeneralQ000013I1Tr[y2.Trans(Bar(y2))]	0
lambdaS2Tr[y2.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ000004I1Tr[y2Conj.Trans(Bar(y2))]	0
lambdaT3GeneralQ000013I1Tr[y2Conj.Trans(Bar(y2Conj))]	0
$-\frac{46}{15}$Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2Conj))]	0
$-\frac{46}{15}$Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0
lambda12CrosslambdaT3GeneralQ001102I1	0
lambda12lambdaT3GeneralQ001102I2	0
lambda12CrosslambdaT3GeneralQ001102I2	0
Tr[y2.Trans(Bar(y2)).y2.Trans(Bar(y2Conj))]	0
Tr[y2.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(y2Conj))]	0

$\beta^{(1)}_{lambdaT3GeneralQ000211I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$g2^2$lambdaT3GeneralQ000211I1	−12	0	0
gY^2lambdaT3GeneralQ000211I1	−6	0	0
lambda12lambdaT3GeneralQ000211I1	8	0	0
lambdaS1lambdaT3GeneralQ000211I1	2	0	0
lambdaS2lambdaT3GeneralQ000211I1	8	0	0

lambdaS2AdjlambdaT3GeneralQ000211I1	4	0	0
lambda12lambdaT3GeneralQ001300I1	6	0	0
lambdaH2lambdaT3GeneralQ110200I1	4	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I1)	24	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambdaT3GeneralQ000211I1Tr[y1.Trans(Bar(y1))]	4	0	0
lambda12Tr[y1Conj.Trans(Bar(y1))]	4	0	0
lambdaT3GeneralQ000211I1Tr[y1Conj.Trans(Bar(y1Conj))]	4	0	0
lambdaT3GeneralQ000211I1Tr[y2.Trans(Bar(y2))]	2	0	0
Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	-4	0	0
lambdaS1AdjlambdaT3GeneralQ000211I1	0	0	0
lambda12CrosslambdaT3GeneralQ000211I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ000211I2	0	0	0
lambdaS2lambdaT3GeneralQ000211I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ001300I1	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
$-\frac{22}{15}$Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$g2^2$lambdaT3GeneralQ000211I1	0	0	0
gY^2lambdaT3GeneralQ000211I1	0	0	0
lambda12lambdaT3GeneralQ000211I1	0	0	0
lambdaS1lambdaT3GeneralQ000211I1	0	0	0
lambdaS2lambdaT3GeneralQ000211I1	0	0	0
lambdaS2AdjlambdaT3GeneralQ000211I1	0	0	0
lambda12lambdaT3GeneralQ001300I1	0	0	0
lambdaH2lambdaT3GeneralQ110200I1	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ000211I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambda12Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000211I1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaT3GeneralQ000211I1Tr[y2.Trans(Bar(y2))]	0	0	0
Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
lambdaS1AdjlambdaT3GeneralQ000211I1	0	0	0
lambda12CrosslambdaT3GeneralQ000211I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ000211I2	0	0	0
lambdaS2lambdaT3GeneralQ000211I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ001300I1	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
$-\frac{22}{15}$Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$g2^2$lambdaT3GeneralQ000211I1	0	0	0
gY^2lambdaT3GeneralQ000211I1	0	0	0
lambda12lambdaT3GeneralQ000211I1	0	0	0
lambdaS1lambdaT3GeneralQ000211I1	0	0	0
lambdaS2lambdaT3GeneralQ000211I1	0	0	0
lambdaS2AdjlambdaT3GeneralQ000211I1	0	0	0
lambda12lambdaT3GeneralQ001300I1	0	0	0
lambdaH2lambdaT3GeneralQ110200I1	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I1)	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ000211I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambda12Tr[y1Conj.Trans(Bar(y1))]	0	0	0

lambdaT3GeneralQ000211I1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaT3GeneralQ000211I1Tr[y2.Trans(Bar(y2))]	0	0	0
Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
lambdaS1AdjlambdaT3GeneralQ000211I1	0	0	0
lambda12CrosslambdaT3GeneralQ000211I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ000211I2	0	0	0
lambdaS2lambdaT3GeneralQ000211I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ001300I1	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
$-\frac{22}{15}$Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
g2^{2}lambdaT3GeneralQ000211I1	-12	0	0
gY^{2}lambdaT3GeneralQ000211I1	-6	0	0
lambda12lambdaT3GeneralQ000211I1	8	0	0
lambdaS1lambdaT3GeneralQ000211I1	2	0	0
lambdaS2lambdaT3GeneralQ000211I1	4	0	0
lambdaS2AdjlambdaT3GeneralQ000211I1	0	0	0
lambda12lambdaT3GeneralQ001300I1	10	0	0
lambdaH2lambdaT3GeneralQ110200I1	4	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I1)	40	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambdaT3GeneralQ000211I1Tr[y1.Trans(Bar(y1))]	2	0	0
lambda12Tr[y1Conj.Trans(Bar(y1))]	2	0	0
lambdaT3GeneralQ000211I1Tr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
lambdaT3GeneralQ000211I1Tr[y2.Trans(Bar(y2))]	4	0	0
Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	-4	0	0
lambdaS1AdjlambdaT3GeneralQ000211I1	-4	0	0
lambda12CrosslambdaT3GeneralQ000211I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ000211I2	0	0	0
lambdaS2lambdaT3GeneralQ000211I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ001300I1	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
$-\frac{22}{15}$Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
g2^{2}lambdaT3GeneralQ000211I1	-24	0	0
gY^{2}lambdaT3GeneralQ000211I1	-6	0	0
lambda12lambdaT3GeneralQ000211I1	8	0	0
lambdaS1lambdaT3GeneralQ000211I1	2	0	0
lambdaS2lambdaT3GeneralQ000211I1	8	0	0
lambdaS2AdjlambdaT3GeneralQ000211I1	4	0	0
lambda12lambdaT3GeneralQ001300I1	10	0	0
lambdaH2lambdaT3GeneralQ110200I1	4	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I1)	40	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	2	0	0
lambdaT3GeneralQ000211I1Tr[y1.Trans(Bar(y1))]	2	0	0
lambda12Tr[y1Conj.Trans(Bar(y1))]	2	0	0
lambdaT3GeneralQ000211I1Tr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
lambdaT3GeneralQ000211I1Tr[y2.Trans(Bar(y2))]	2	0	0
Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0
lambdaS1AdjlambdaT3GeneralQ000211I1	-4	0	0
lambda12CrosslambdaT3GeneralQ000211I2	4	0	0
lambdaS1AdjlambdaT3GeneralQ000211I2	-2	0	0
lambdaS2lambdaT3GeneralQ000211I2	2	0	0

lambdaS2AdjlambdaT3GeneralQ000211I2	2	0	0
lambda12CrosslambdaT3GeneralQ001300I1	4	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I2)	8	0	0
$-\frac{22}{15}$Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	1	0	0

$\mathcal{S}$	E ($\alpha=-4$)
g2^{2}lambdaT3GeneralQ000211I1	0
gY^{2}lambdaT3GeneralQ000211I1	0
lambda12lambdaT3GeneralQ000211I1	0
lambdaS1lambdaT3GeneralQ000211I1	0
lambdaS2lambdaT3GeneralQ000211I1	0
lambdaS2AdjlambdaT3GeneralQ000211I1	0
lambda12lambdaT3GeneralQ001300I1	0
lambdaH2lambdaT3GeneralQ110200I1	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I1)	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	0
lambdaT3GeneralQ000211I1Tr[y1.Trans(Bar(y1))]	0
lambda12Tr[y1Conj.Trans(Bar(y1))]	0
lambdaT3GeneralQ000211I1Tr[y1Conj.Trans(Bar(y1Conj))]	0
lambdaT3GeneralQ000211I1Tr[y2.Trans(Bar(y2))]	0
Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0
lambdaS1AdjlambdaT3GeneralQ000211I1	0
lambda12CrosslambdaT3GeneralQ000211I2	0
lambdaS1AdjlambdaT3GeneralQ000211I2	0
lambdaS2lambdaT3GeneralQ000211I2	0
lambdaS2AdjlambdaT3GeneralQ000211I2	0
lambda12CrosslambdaT3GeneralQ001300I1	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I2)	0
$-\frac{22}{15}$Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0

$\beta^{(1)}_{lambdaT3GeneralQ000211I2}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
lambda12CrosslambdaT3GeneralQ000211I1	0	0	0
g2^{2}lambdaT3GeneralQ000211I2	0	0	0
gY^{2}lambdaT3GeneralQ000211I2	0	0	0
lambda12lambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ000211I2	0	0	0
lambdaS1lambdaT3GeneralQ000211I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ000211I2	0	0	0
lambdaS2lambdaT3GeneralQ000211I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ001300I1	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ000211I2Tr[y1.Trans(Bar(y1))]	0	0	0
lambda12CrossTr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000211I2Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaT3GeneralQ000211I2Tr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{8}{5}$Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
lambda12CrosslambdaT3GeneralQ000211I1	0	0	0
g2^{2}lambdaT3GeneralQ000211I2	0	0	0

gY^2lambdaT3GeneralQ000211I2	0	0	0
lambda12lambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ000211I2	0	0	0
lambdaS1lambdaT3GeneralQ000211I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ000211I2	0	0	0
lambdaS2lambdaT3GeneralQ000211I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ001300I1	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ000211I2Tr[y1.Trans(Bar(y1))]	0	0	0
lambda12CrossTr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000211I2Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaT3GeneralQ000211I2Tr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{8}{5}$Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambda12CrosslambdaT3GeneralQ000211I1	0	0	0
$g2^2$lambdaT3GeneralQ000211I2	0	0	0
gY^2lambdaT3GeneralQ000211I2	0	0	0
lambda12lambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ000211I2	0	0	0
lambdaS1lambdaT3GeneralQ000211I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ000211I2	0	0	0
lambdaS2lambdaT3GeneralQ000211I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ001300I1	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ000211I2Tr[y1.Trans(Bar(y1))]	0	0	0
lambda12CrossTr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000211I2Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaT3GeneralQ000211I2Tr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{8}{5}$Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambda12CrosslambdaT3GeneralQ000211I1	0	0	0
$g2^2$lambdaT3GeneralQ000211I2	0	0	0
gY^2lambdaT3GeneralQ000211I2	0	0	0
lambda12lambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ000211I2	0	0	0
lambdaS1lambdaT3GeneralQ000211I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ000211I2	0	0	0
lambdaS2lambdaT3GeneralQ000211I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ001300I1	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I2)	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ000211I2Tr[y1.Trans(Bar(y1))]	0	0	0
lambda12CrossTr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000211I2Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaT3GeneralQ000211I2Tr[y2.Trans(Bar(y2))]	0	0	0
$-\frac{8}{5}$Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)

lambda12CrosslambdaT3GeneralQ000211I1	16	0	0
g2^{2}lambdaT3GeneralQ000211I2	−24	0	0
gY^{2}lambdaT3GeneralQ000211I2	−6	0	0
lambda12lambdaT3GeneralQ000211I2	8	0	0
lambda12CrosslambdaT3GeneralQ000211I2	20	0	0
lambdaS1lambdaT3GeneralQ000211I2	2	0	0
lambdaS1AdjlambdaT3GeneralQ000211I2	2	0	0
lambdaS2lambdaT3GeneralQ000211I2	2	0	0
lambdaS2AdjlambdaT3GeneralQ000211I2	−2	0	0
lambda12CrosslambdaT3GeneralQ001300I1	8	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I2)	16	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	−2	0	0
lambdaT3GeneralQ000211I2Tr[y1.Trans(Bar(y1))]	2	0	0
lambda12CrossTr[y1Conj.Trans(Bar(y1))]	4	0	0
lambdaT3GeneralQ000211I2Tr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
lambdaT3GeneralQ000211I2Tr[y2.Trans(Bar(y2))]	2	0	0
$-\frac{8}{5}$Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	1	0	0

$\mathcal{S}$	E ($\alpha = -4$)
lambda12CrosslambdaT3GeneralQ000211I1	0
g2^{2}lambdaT3GeneralQ000211I2	0
gY^{2}lambdaT3GeneralQ000211I2	0
lambda12lambdaT3GeneralQ000211I2	0
lambda12CrosslambdaT3GeneralQ000211I2	0
lambdaS1lambdaT3GeneralQ000211I2	0
lambdaS1AdjlambdaT3GeneralQ000211I2	0
lambdaS2lambdaT3GeneralQ000211I2	0
lambdaS2AdjlambdaT3GeneralQ000211I2	0
lambda12CrosslambdaT3GeneralQ001300I1	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ000211I2)	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	0
lambdaT3GeneralQ000211I2Tr[y1.Trans(Bar(y1))]	0
lambda12CrossTr[y1Conj.Trans(Bar(y1))]	0
lambdaT3GeneralQ000211I2Tr[y1Conj.Trans(Bar(y1Conj))]	0
lambdaT3GeneralQ000211I2Tr[y2.Trans(Bar(y2))]	0
$-\frac{8}{5}$Tr[y1Conj.Trans(Bar(y1)).y2.Trans(Bar(y2))]	0

$\beta^{(1)}_{lambdaT3GeneralQ000400I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambdaT3GeneralQ000211I1^2	3	0	0
lambdaS1lambdaT3GeneralQ000400I1	12	0	0
lambdaT3GeneralQ001300I1^2	9	0	0
lambdaT3GeneralQ110200I1^2	2	0	0
lambdaT3GeneralQ000400I1Tr[y1.Trans(Bar(y1))]	8	0	0
lambdaT3GeneralQ001300I1Tr[y1Conj.Trans(Bar(y1))]	4	0	0
lambdaT3GeneralQ000400I1Tr[y1Conj.Trans(Bar(y1Conj))]	8	0	0
Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	−4	0	0
g2^{2}lambdaT3GeneralQ000400I1	0	0	0
lambdaS1AdjlambdaT3GeneralQ000400I1	0	0	0
$-\frac{23}{15}$Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000211I1lambdaT3GeneralQ000211I2	0	0	0
lambdaT3GeneralQ000211I2^2	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambdaT3GeneralQ000211I1^2	0	0	0
lambdaS1lambdaT3GeneralQ000400I1	0	0	0
lambdaT3GeneralQ001300I1^2	0	0	0
lambdaT3GeneralQ110200I1^2	0	0	0
lambdaT3GeneralQ000400I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ001300I1Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000400I1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
g2^{2}lambdaT3GeneralQ000400I1	0	0	0
lambdaS1AdjlambdaT3GeneralQ000400I1	0	0	0
$-\frac{23}{15}$Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000211I1lambdaT3GeneralQ000211I2	0	0	0
lambdaT3GeneralQ000211I2^2	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambdaT3GeneralQ000211I1^2	0	0	0
lambdaS1lambdaT3GeneralQ000400I1	0	0	0
lambdaT3GeneralQ001300I1^2	0	0	0
lambdaT3GeneralQ110200I1^2	0	0	0
lambdaT3GeneralQ000400I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ001300I1Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000400I1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
g2^{2}lambdaT3GeneralQ000400I1	0	0	0
lambdaS1AdjlambdaT3GeneralQ000400I1	0	0	0
$-\frac{23}{15}$Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000211I1lambdaT3GeneralQ000211I2	0	0	0
lambdaT3GeneralQ000211I2^2	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambdaT3GeneralQ000211I1^2	1	0	0
lambdaS1lambdaT3GeneralQ000400I1	12	0	0
lambdaT3GeneralQ001300I1^2	11	0	0
lambdaT3GeneralQ110200I1^2	2	0	0
lambdaT3GeneralQ000400I1Tr[y1.Trans(Bar(y1))]	4	0	0
lambdaT3GeneralQ001300I1Tr[y1Conj.Trans(Bar(y1))]	2	0	0
lambdaT3GeneralQ000400I1Tr[y1Conj.Trans(Bar(y1Conj))]	4	0	0
Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
g2^{2}lambdaT3GeneralQ000400I1	-24	0	0
lambdaS1AdjlambdaT3GeneralQ000400I1	-8	0	0
$-\frac{23}{15}$Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	1	0	0
lambdaT3GeneralQ000211I1lambdaT3GeneralQ000211I2	0	0	0
lambdaT3GeneralQ000211I2^2	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambdaT3GeneralQ000211I1^2	3	0	0
lambdaS1lambdaT3GeneralQ000400I1	12	0	0
lambdaT3GeneralQ001300I1^2	11	0	0
lambdaT3GeneralQ110200I1^2	2	0	0
lambdaT3GeneralQ000400I1Tr[y1.Trans(Bar(y1))]	4	0	0
lambdaT3GeneralQ001300I1Tr[y1Conj.Trans(Bar(y1))]	2	0	0
lambdaT3GeneralQ000400I1Tr[y1Conj.Trans(Bar(y1Conj))]	4	0	0
Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0

g2^{2}lambdaT3GeneralQ000400I1	-24	0	0
lambdaS1AdjlambdaT3GeneralQ000400I1	-8	0	0
$-\frac{23}{15}$Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	1	0	0
lambdaT3GeneralQ000211I1lambdaT3GeneralQ000211I2	2	0	0
lambdaT3GeneralQ000211I2^2	1	0	0

$\mathcal{S}$	E ($\alpha = -4$)
lambdaT3GeneralQ000211I1^2	0
lambdaS1lambdaT3GeneralQ000400I1	0
lambdaT3GeneralQ001300I1^2	0
lambdaT3GeneralQ110200I1^2	0
lambdaT3GeneralQ000400I1Tr[y1.Trans(Bar(y1))]	0
lambdaT3GeneralQ001300I1Tr[y1Conj.Trans(Bar(y1))]	0
lambdaT3GeneralQ000400I1Tr[y1Conj.Trans(Bar(y1Conj))]	0
Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0
g2^{2}lambdaT3GeneralQ000400I1	0
lambdaS1AdjlambdaT3GeneralQ000400I1	0
$-\frac{23}{15}$Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0
lambdaT3GeneralQ000211I1lambdaT3GeneralQ000211I2	0
lambdaT3GeneralQ000211I2^2	0

$\beta^{(1)}_{lambdaT3GeneralQ001102I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambda12lambdaT3GeneralQ000013I1	0	10	0
g2^{2}lambdaT3GeneralQ001102I1	0	-12	0
gY^{2}lambdaT3GeneralQ001102I1	0	-6	0
lambda12lambdaT3GeneralQ001102I1	0	8	0
lambdaS1lambdaT3GeneralQ001102I1	0	4	0
lambdaS2lambdaT3GeneralQ001102I1	0	2	0
lambdaS2AdjlambdaT3GeneralQ001102I1	0	-4	0
lambdaT3lambdaT3GeneralQ020101I1	0	4	0
lambdaH1lambdaT3GeneralQ110002I1	0	4	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I1)	0	40	0
lambdaT3GeneralQ001102I1Tr[y1.Trans(Bar(y1))]	0	4	0
lambdaT3GeneralQ001102I1Tr[y2.Trans(Bar(y2))]	0	2	0
lambda12Tr[y2.Trans(Bar(y2Conj))]	0	2	0
lambdaT3GeneralQ001102I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	2	0
Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	-4	0
lambdaS1AdjlambdaT3GeneralQ001102I1	0	0	0
lambda12CrosslambdaT3GeneralQ000013I1	0	0	0
lambda12CrosslambdaT3GeneralQ001102I2	0	0	0
lambdaS1lambdaT3GeneralQ001102I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambda12lambdaT3GeneralQ000013I1	0	0	0
g2^{2}lambdaT3GeneralQ001102I1	0	0	0
gY^{2}lambdaT3GeneralQ001102I1	0	0	0
lambda12lambdaT3GeneralQ001102I1	0	0	0
lambdaS1lambdaT3GeneralQ001102I1	0	0	0

lambdaS2lambdaT3GeneralQ001102I1	0	0	0
lambdaS2AdjlambdaT3GeneralQ001102I1	0	0	0
lambdaT3lambdaT3GeneralQ020101I1	0	0	0
lambdaH1lambdaT3GeneralQ110002I1	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ001102I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ001102I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambda12Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	0
lambdaS1AdjlambdaT3GeneralQ001102I1	0	0	0
lambda12CrosslambdaT3GeneralQ000013I1	0	0	0
lambda12CrosslambdaT3GeneralQ001102I2	0	0	0
lambdaS1lambdaT3GeneralQ001102I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambda12lambdaT3GeneralQ000013I1	0	0	0
$g2^2$lambdaT3GeneralQ001102I1	0	0	0
gY^2lambdaT3GeneralQ001102I1	0	0	0
lambda12lambdaT3GeneralQ001102I1	0	0	0
lambdaS1lambdaT3GeneralQ001102I1	0	0	0
lambdaS2lambdaT3GeneralQ001102I1	0	0	0
lambdaS2AdjlambdaT3GeneralQ001102I1	0	0	0
lambdaT3lambdaT3GeneralQ020101I1	0	0	0
lambdaH1lambdaT3GeneralQ110002I1	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ001102I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ001102I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambda12Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	0
lambdaS1AdjlambdaT3GeneralQ001102I1	0	0	0
lambda12CrosslambdaT3GeneralQ000013I1	0	0	0
lambda12CrosslambdaT3GeneralQ001102I2	0	0	0
lambdaS1lambdaT3GeneralQ001102I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambda12lambdaT3GeneralQ000013I1	0	0	6
$g2^2$lambdaT3GeneralQ001102I1	0	0	-12
gY^2lambdaT3GeneralQ001102I1	0	0	-6
lambda12lambdaT3GeneralQ001102I1	0	0	8
lambdaS1lambdaT3GeneralQ001102I1	0	0	8
lambdaS2lambdaT3GeneralQ001102I1	0	0	2
lambdaS2AdjlambdaT3GeneralQ001102I1	0	0	0
lambdaT3lambdaT3GeneralQ020101I1	0	0	4
lambdaH1lambdaT3GeneralQ110002I1	0	0	4
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I1)	0	0	24
lambdaT3GeneralQ001102I1Tr[y1.Trans(Bar(y1))]	0	0	2
lambdaT3GeneralQ001102I1Tr[y2.Trans(Bar(y2))]	0	0	4

lambda12Tr[y2.Trans(Bar(y2Conj))]	0	0	4
lambdaT3GeneralQ001102I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	4
Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	-4
lambdaS1AdjlambdaT3GeneralQ001102I1	0	0	4
lambda12CrosslambdaT3GeneralQ000013I1	0	0	0
lambda12CrosslambdaT3GeneralQ001102I2	0	0	0
lambdaS1lambdaT3GeneralQ001102I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambda12lambdaT3GeneralQ000013I1	0	0	10
$g2^2$lambdaT3GeneralQ001102I1	0	0	-24
gY^2lambdaT3GeneralQ001102I1	0	0	-6
lambda12lambdaT3GeneralQ001102I1	0	0	8
lambdaS1lambdaT3GeneralQ001102I1	0	0	8
lambdaS2lambdaT3GeneralQ001102I1	0	0	2
lambdaS2AdjlambdaT3GeneralQ001102I1	0	0	-4
lambdaT3lambdaT3GeneralQ020101I1	0	0	2
lambdaH1lambdaT3GeneralQ110002I1	0	0	4
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I1)	0	0	40
lambdaT3GeneralQ001102I1Tr[y1.Trans(Bar(y1))]	0	0	2
lambdaT3GeneralQ001102I1Tr[y2.Trans(Bar(y2))]	0	0	2
lambda12Tr[y2.Trans(Bar(y2Conj))]	0	0	2
lambdaT3GeneralQ001102I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	2
Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	0
lambdaS1AdjlambdaT3GeneralQ001102I1	0	0	4
lambda12CrosslambdaT3GeneralQ000013I1	0	0	4
lambda12CrosslambdaT3GeneralQ001102I2	0	0	4
lambdaS1lambdaT3GeneralQ001102I2	0	0	2
lambdaS1AdjlambdaT3GeneralQ001102I2	0	0	2
lambdaS2AdjlambdaT3GeneralQ001102I2	0	0	-2
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I2)	0	0	8
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	1

$\mathcal{S}$	E ($\alpha = -4$)
lambda12lambdaT3GeneralQ000013I1	0
$g2^2$lambdaT3GeneralQ001102I1	0
gY^2lambdaT3GeneralQ001102I1	0
lambda12lambdaT3GeneralQ001102I1	0
lambdaS1lambdaT3GeneralQ001102I1	0
lambdaS2lambdaT3GeneralQ001102I1	0
lambdaS2AdjlambdaT3GeneralQ001102I1	0
lambdaT3lambdaT3GeneralQ020101I1	0
lambdaH1lambdaT3GeneralQ110002I1	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I1)	0
lambdaT3GeneralQ001102I1Tr[y1.Trans(Bar(y1))]	0
lambdaT3GeneralQ001102I1Tr[y2.Trans(Bar(y2))]	0
lambda12Tr[y2.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ001102I1Tr[y2Conj.Trans(Bar(y2Conj))]	0
Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0
lambdaS1AdjlambdaT3GeneralQ001102I1	0
lambda12CrosslambdaT3GeneralQ000013I1	0
lambda12CrosslambdaT3GeneralQ001102I2	0
lambdaS1lambdaT3GeneralQ001102I2	0

lambdaS1AdjlambdaT3GeneralQ001102I2	0
lambdaS2AdjlambdaT3GeneralQ001102I2	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I2)	0
$-\frac{22}{15}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0

$\beta^{(1)}_{lambdaT3GeneralQ001102I2}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambda12CrosslambdaT3GeneralQ000013I1	0	0	0
lambda12CrosslambdaT3GeneralQ001102I1	0	0	0
g2^{2}lambdaT3GeneralQ001102I2	0	0	0
gY^{2}lambdaT3GeneralQ001102I2	0	0	0
lambda12lambdaT3GeneralQ001102I2	0	0	0
lambda12CrosslambdaT3GeneralQ001102I2	0	0	0
lambdaS1lambdaT3GeneralQ001102I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaS2lambdaT3GeneralQ001102I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaT3lambdaT3GeneralQ020101I1	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ001102I2Tr[y2.Trans(Bar(y2))]	0	0	0
lambda12CrossTr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{8}{5}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambda12CrosslambdaT3GeneralQ000013I1	0	0	0
lambda12CrosslambdaT3GeneralQ001102I1	0	0	0
g2^{2}lambdaT3GeneralQ001102I2	0	0	0
gY^{2}lambdaT3GeneralQ001102I2	0	0	0
lambda12lambdaT3GeneralQ001102I2	0	0	0
lambda12CrosslambdaT3GeneralQ001102I2	0	0	0
lambdaS1lambdaT3GeneralQ001102I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaS2lambdaT3GeneralQ001102I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaT3lambdaT3GeneralQ020101I1	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ001102I2Tr[y2.Trans(Bar(y2))]	0	0	0
lambda12CrossTr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{8}{5}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambda12CrosslambdaT3GeneralQ000013I1	0	0	0
lambda12CrosslambdaT3GeneralQ001102I1	0	0	0
g2^{2}lambdaT3GeneralQ001102I2	0	0	0
gY^{2}lambdaT3GeneralQ001102I2	0	0	0
lambda12lambdaT3GeneralQ001102I2	0	0	0
lambda12CrosslambdaT3GeneralQ001102I2	0	0	0
lambdaS1lambdaT3GeneralQ001102I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ001102I2	0	0	0

lambdaS2lambdaT3GeneralQ001102I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaT3lambdaT3GeneralQ020101I1	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ001102I2Tr[y2.Trans(Bar(y2))]	0	0	0
lambda12CrossTr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{8}{5}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambda12CrosslambdaT3GeneralQ000013I1	0	0	0
lambda12CrosslambdaT3GeneralQ001102I1	0	0	0
g2^{2}lambdaT3GeneralQ001102I2	0	0	0
gY^{2}lambdaT3GeneralQ001102I2	0	0	0
lambda12lambdaT3GeneralQ001102I2	0	0	0
lambda12CrosslambdaT3GeneralQ001102I2	0	0	0
lambdaS1lambdaT3GeneralQ001102I2	0	0	0
lambdaS1AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaS2lambdaT3GeneralQ001102I2	0	0	0
lambdaS2AdjlambdaT3GeneralQ001102I2	0	0	0
lambdaT3lambdaT3GeneralQ020101I1	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ001102I2Tr[y2.Trans(Bar(y2))]	0	0	0
lambda12CrossTr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ001102I2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
$-\frac{8}{5}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambda12CrosslambdaT3GeneralQ000013I1	0	0	8
lambda12CrosslambdaT3GeneralQ001102I1	0	0	16
g2^{2}lambdaT3GeneralQ001102I2	0	0	-24
gY^{2}lambdaT3GeneralQ001102I2	0	0	-6
lambda12lambdaT3GeneralQ001102I2	0	0	8
lambda12CrosslambdaT3GeneralQ001102I2	0	0	20
lambdaS1lambdaT3GeneralQ001102I2	0	0	2
lambdaS1AdjlambdaT3GeneralQ001102I2	0	0	-2
lambdaS2lambdaT3GeneralQ001102I2	0	0	2
lambdaS2AdjlambdaT3GeneralQ001102I2	0	0	2
lambdaT3lambdaT3GeneralQ020101I1	0	0	-2
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I2)	0	0	16
lambdaT3GeneralQ001102I2Tr[y1.Trans(Bar(y1))]	0	0	2
lambdaT3GeneralQ001102I2Tr[y2.Trans(Bar(y2))]	0	0	2
lambda12CrossTr[y2.Trans(Bar(y2Conj))]	0	0	4
lambdaT3GeneralQ001102I2Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	2
$-\frac{8}{5}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0	0	1

$\mathcal{S}$	E ($\alpha = -4$)
lambda12CrosslambdaT3GeneralQ000013I1	0
lambda12CrosslambdaT3GeneralQ001102I1	0
g2^{2}lambdaT3GeneralQ001102I2	0
gY^{2}lambdaT3GeneralQ001102I2	0
lambda12lambdaT3GeneralQ001102I2	0
lambda12CrosslambdaT3GeneralQ001102I2	0

lambdaS1lambdaT3GeneralQ001102I2	0
lambdaS1AdjlambdaT3GeneralQ001102I2	0
lambdaS2lambdaT3GeneralQ001102I2	0
lambdaS2AdjlambdaT3GeneralQ001102I2	0
lambdaT3lambdaT3GeneralQ020101I1	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ001102I2)	0
lambdaT3GeneralQ001102I2Tr[y1.Trans(Bar(y1))]	0
lambdaT3GeneralQ001102I2Tr[y2.Trans(Bar(y2))]	0
lambda12CrossTr[y2.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ001102I2Tr[y2Conj.Trans(Bar(y2Conj))]	0
$-\frac{8}{5}$Tr[y1.Trans(Bar(y1)).y2.Trans(Bar(y2Conj))]	0

$\beta^{(1)}_{lambdaT3GeneralQ001300I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambda12lambdaT3GeneralQ000211I1	6	0	0
lambdaS1lambdaT3GeneralQ001300I1	18	0	0
lambdaH1lambdaT3GeneralQ110200I1	4	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ001300I1)	72	0	0
lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1))]	8	0	0
lambdaT3GeneralQ000400I1Tr[y1.Trans(Bar(y1Conj))]	16	0	0
lambdaS1Tr[y1Conj.Trans(Bar(y1))]	4	0	0
lambdaT3GeneralQ001300I1Tr[y1Conj.Trans(Bar(y1Conj))]	8	0	0
Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	-8	0	0
Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	-8	0	0
g2^{2}lambdaT3GeneralQ001300I1	0	0	0
$-\frac{46}{15}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
$-\frac{46}{15}$Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambda12CrosslambdaT3GeneralQ000211I1	0	0	0
lambda12lambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ000211I2	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambda12lambdaT3GeneralQ000211I1	0	0	0
lambdaS1lambdaT3GeneralQ001300I1	0	0	0
lambdaH1lambdaT3GeneralQ110200I1	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000400I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
lambdaS1Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ001300I1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
g2^{2}lambdaT3GeneralQ001300I1	0	0	0
$-\frac{46}{15}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
$-\frac{46}{15}$Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambda12CrosslambdaT3GeneralQ000211I1	0	0	0
lambda12lambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ000211I2	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambda12lambdaT3GeneralQ000211I1	0	0	0
lambdaS1lambdaT3GeneralQ001300I1	0	0	0
lambdaH1lambdaT3GeneralQ110200I1	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ001300I1)	0	0	0

lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ000400I1Tr[y1.Trans(Bar(y1Conj))]	0	0	0
lambdaS1Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ001300I1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
$g2^2$lambdaT3GeneralQ001300I1	0	0	0
$-\frac{46}{15}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
$-\frac{46}{15}$Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambda12CrosslambdaT3GeneralQ000211I1	0	0	0
lambda12lambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ000211I2	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambda12lambdaT3GeneralQ000211I1	2	0	0
lambdaS1lambdaT3GeneralQ001300I1	22	0	0
lambdaH1lambdaT3GeneralQ110200I1	4	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ001300I1)	88	0	0
lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1))]	4	0	0
lambdaT3GeneralQ000400I1Tr[y1.Trans(Bar(y1Conj))]	8	0	0
lambdaS1Tr[y1Conj.Trans(Bar(y1))]	2	0	0
lambdaT3GeneralQ001300I1Tr[y1Conj.Trans(Bar(y1Conj))]	4	0	0
Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
$g2^2$lambdaT3GeneralQ001300I1	-24	0	0
$-\frac{46}{15}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	1	0	0
$-\frac{46}{15}$Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	1	0	0
lambda12CrosslambdaT3GeneralQ000211I1	0	0	0
lambda12lambdaT3GeneralQ000211I2	0	0	0
lambda12CrosslambdaT3GeneralQ000211I2	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambda12lambdaT3GeneralQ000211I1	6	0	0
lambdaS1lambdaT3GeneralQ001300I1	22	0	0
lambdaH1lambdaT3GeneralQ110200I1	4	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ001300I1)	88	0	0
lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1))]	4	0	0
lambdaT3GeneralQ000400I1Tr[y1.Trans(Bar(y1Conj))]	8	0	0
lambdaS1Tr[y1Conj.Trans(Bar(y1))]	2	0	0
lambdaT3GeneralQ001300I1Tr[y1Conj.Trans(Bar(y1Conj))]	4	0	0
Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0	0	0
Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0	0	0
$g2^2$lambdaT3GeneralQ001300I1	-24	0	0
$-\frac{46}{15}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	1	0	0
$-\frac{46}{15}$Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	1	0	0
lambda12CrosslambdaT3GeneralQ000211I1	4	0	0
lambda12lambdaT3GeneralQ000211I2	2	0	0
lambda12CrosslambdaT3GeneralQ000211I2	4	0	0

$\mathcal{S}$	E ($\alpha = -4$)
lambda12lambdaT3GeneralQ000211I1	0
lambdaS1lambdaT3GeneralQ001300I1	0
lambdaH1lambdaT3GeneralQ110200I1	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ001300I1)	0
lambdaT3GeneralQ001300I1Tr[y1.Trans(Bar(y1))]	0

lambdaT3GeneralQ000400I1Tr[y1.Trans(Bar(y1Conj))]	0
lambdaS1Tr[y1Conj.Trans(Bar(y1))]	0
lambdaT3GeneralQ001300I1Tr[y1Conj.Trans(Bar(y1Conj))]	0
Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0
Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0
$g2^2$lambdaT3GeneralQ001300I1	0
$-\frac{46}{15}$Tr[y1.Trans(Bar(y1)).y1Conj.Trans(Bar(y1))]	0
$-\frac{46}{15}$Tr[y1Conj.Trans(Bar(y1)).y1Conj.Trans(Bar(y1Conj))]	0
lambda12CrosslambdaT3GeneralQ000211I1	0
lambda12lambdaT3GeneralQ000211I2	0
lambda12CrosslambdaT3GeneralQ000211I2	0

$\beta^{(1)}_{lambdaT3GeneralQ020002I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$g2^2$lambdaT3GeneralQ020002I1	0	0	0
gY^2lambdaT3GeneralQ020002I1	0	0	0
lambdaHlambdaT3GeneralQ020002I1	0	0	0
lambdaH2lambdaT3GeneralQ020002I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ020002I1	0	0	0
lambdaS2lambdaT3GeneralQ020002I1	0	0	0
lambdaT3GeneralQ020002I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ020002I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ020002I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ020002I1Tr[yu.Trans(Bar(yu))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$g2^2$lambdaT3GeneralQ020002I1	0	0	-9
gY^2lambdaT3GeneralQ020002I1	0	0	-3
lambdaHlambdaT3GeneralQ020002I1	0	0	2
lambdaH2lambdaT3GeneralQ020002I1	0	0	8
lambdaH2AdjlambdaT3GeneralQ020002I1	0	0	-4
lambdaS2lambdaT3GeneralQ020002I1	0	0	2
lambdaT3GeneralQ020002I1Tr[y2.Trans(Bar(y2))]	0	0	2
lambdaT3GeneralQ020002I1Tr[yd.Trans(Bar(yd))]	0	0	6
lambdaT3GeneralQ020002I1Tr[ye.Trans(Bar(ye))]	0	0	2
lambdaT3GeneralQ020002I1Tr[yu.Trans(Bar(yu))]	0	0	6

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$g2^2$lambdaT3GeneralQ020002I1	0	0	-9
gY^2lambdaT3GeneralQ020002I1	0	0	-3
lambdaHlambdaT3GeneralQ020002I1	0	0	2
lambdaH2lambdaT3GeneralQ020002I1	0	0	8
lambdaH2AdjlambdaT3GeneralQ020002I1	0	0	-4
lambdaS2lambdaT3GeneralQ020002I1	0	0	2
lambdaT3GeneralQ020002I1Tr[y2.Trans(Bar(y2))]	0	0	3
lambdaT3GeneralQ020002I1Tr[yd.Trans(Bar(yd))]	0	0	6
lambdaT3GeneralQ020002I1Tr[ye.Trans(Bar(ye))]	0	0	2
lambdaT3GeneralQ020002I1Tr[yu.Trans(Bar(yu))]	0	0	6

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$g2^2$lambdaT3GeneralQ020002I1	0	0	0

gY^2lambdaT3GeneralQ020002I1	0	0	0
lambdaHlambdaT3GeneralQ020002I1	0	0	0
lambdaH2lambdaT3GeneralQ020002I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ020002I1	0	0	0
lambdaS2lambdaT3GeneralQ020002I1	0	0	0
lambdaT3GeneralQ020002I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ020002I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ020002I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ020002I1Tr[yu.Trans(Bar(yu))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$g2^2$lambdaT3GeneralQ020002I1	0	0	0
gY^2lambdaT3GeneralQ020002I1	0	0	0
lambdaHlambdaT3GeneralQ020002I1	0	0	0
lambdaH2lambdaT3GeneralQ020002I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ020002I1	0	0	0
lambdaS2lambdaT3GeneralQ020002I1	0	0	0
lambdaT3GeneralQ020002I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ020002I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ020002I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ020002I1Tr[yu.Trans(Bar(yu))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$g2^2$lambdaT3GeneralQ020002I1	0
gY^2lambdaT3GeneralQ020002I1	0
lambdaHlambdaT3GeneralQ020002I1	0
lambdaH2lambdaT3GeneralQ020002I1	0
lambdaH2AdjlambdaT3GeneralQ020002I1	0
lambdaS2lambdaT3GeneralQ020002I1	0
lambdaT3GeneralQ020002I1Tr[y2.Trans(Bar(y2))]	0
lambdaT3GeneralQ020002I1Tr[yd.Trans(Bar(yd))]	0
lambdaT3GeneralQ020002I1Tr[ye.Trans(Bar(ye))]	0
lambdaT3GeneralQ020002I1Tr[yu.Trans(Bar(yu))]	0

$\beta^{(1)}_{lambdaT3GeneralQ020101I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$-\frac{21g2^2\text{lambdaT3GeneralQ020101I1}}{2}$	0	1	0
$-\frac{9gY^2\text{lambdaT3GeneralQ020101I1}}{2}$	0	1	0
lambda12lambdaT3GeneralQ020101I1	0	2	0
lambdaHlambdaT3GeneralQ020101I1	0	2	0
lambdaH1lambdaT3GeneralQ020101I1	0	4	0
lambdaH2lambdaT3GeneralQ020101I1	0	4	0
lambdaH2AdjlambdaT3GeneralQ020101I1	0	-4	0
lambdaT3GeneralQ001102I1Bar(lambdaT3)	0	4	0
lambdaT3GeneralQ110002I1Bar(lambdaT3)	0	8	0
lambdaT3GeneralQ020101I1Tr[y1.Trans(Bar(y1))]	0	2	0
lambdaT3GeneralQ020101I1Tr[y2.Trans(Bar(y2))]	0	1	0
Bar(lambdaT3)Tr[y2.Trans(Bar(y2Conj))]	0	2	0
lambdaT3GeneralQ020101I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	1	0
lambdaT3GeneralQ020101I1Tr[yd.Trans(Bar(yd))]	0	6	0
lambdaT3GeneralQ020101I1Tr[ye.Trans(Bar(ye))]	0	2	0
lambdaT3GeneralQ020101I1Tr[yu.Trans(Bar(yu))]	0	6	0
lambdaH1AdjlambdaT3GeneralQ020101I1	0	0	0

$_\frac{33g2^2\text{lambdaT3GeneralQ020101I1}}{2}$	0	0	0
lambda12AdjlambdaT3GeneralQ020101I1	0	0	0
lambda12CrosslambdaT3GeneralQ020101I1	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3)	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$_\frac{21g2^2\text{lambdaT3GeneralQ020101I1}}{2}$	0	0	0
$_\frac{9gY^2\text{lambdaT3GeneralQ020101I1}}{2}$	0	0	0
lambda12lambdaT3GeneralQ020101I1	0	0	0
lambdaHlambdaT3GeneralQ020101I1	0	0	0
lambdaH1lambdaT3GeneralQ020101I1	0	0	0
lambdaH2lambdaT3GeneralQ020101I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ020101I1	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3)	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3)	0	0	0
lambdaT3GeneralQ020101I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ020101I1Tr[y2.Trans(Bar(y2))]	0	0	0
Bar(lambdaT3)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ020101I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ020101I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ020101I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ020101I1Tr[yu.Trans(Bar(yu))]	0	0	0
lambdaH1AdjlambdaT3GeneralQ020101I1	0	0	0
$_\frac{33g2^2\text{lambdaT3GeneralQ020101I1}}{2}$	0	0	0
lambda12AdjlambdaT3GeneralQ020101I1	0	0	0
lambda12CrosslambdaT3GeneralQ020101I1	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3)	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$_\frac{21g2^2\text{lambdaT3GeneralQ020101I1}}{2}$	0	0	0
$_\frac{9gY^2\text{lambdaT3GeneralQ020101I1}}{2}$	0	0	0
lambda12lambdaT3GeneralQ020101I1	0	0	0
lambdaHlambdaT3GeneralQ020101I1	0	0	0
lambdaH1lambdaT3GeneralQ020101I1	0	0	0
lambdaH2lambdaT3GeneralQ020101I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ020101I1	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3)	0	0	0
lambdaT3GeneralQ110002I1Bar(lambdaT3)	0	0	0
lambdaT3GeneralQ020101I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ020101I1Tr[y2.Trans(Bar(y2))]	0	0	0
Bar(lambdaT3)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ020101I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ020101I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ020101I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ020101I1Tr[yu.Trans(Bar(yu))]	0	0	0
lambdaH1AdjlambdaT3GeneralQ020101I1	0	0	0
$_\frac{33g2^2\text{lambdaT3GeneralQ020101I1}}{2}$	0	0	0
lambda12AdjlambdaT3GeneralQ020101I1	0	0	0
lambda12CrosslambdaT3GeneralQ020101I1	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3)	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$_\frac{21g2^2\text{lambdaT3GeneralQ020101I1}}{2}$	0	0	1

$_\frac{9gY^2 lambdaT3GeneralQ020101I1}{2}$	0	0	1
lambda12lambdaT3GeneralQ020101I1	0	0	2
lambdaHlambdaT3GeneralQ020101I1	0	0	2
lambdaH1lambdaT3GeneralQ020101I1	0	0	4
lambdaH2lambdaT3GeneralQ020101I1	0	0	4
lambdaH2AdjlambdaT3GeneralQ020101I1	0	0	0
lambdaT3GeneralQ001102I1Bar(lambdaT3)	0	0	4
lambdaT3GeneralQ110002I1Bar(lambdaT3)	0	0	8
lambdaT3GeneralQ020101I1Tr[y1.Trans(Bar(y1))]	0	0	1
lambdaT3GeneralQ020101I1Tr[y2.Trans(Bar(y2))]	0	0	2
Bar(lambdaT3)Tr[y2.Trans(Bar(y2Conj))]	0	0	4
lambdaT3GeneralQ020101I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	2
lambdaT3GeneralQ020101I1Tr[yd.Trans(Bar(yd))]	0	0	6
lambdaT3GeneralQ020101I1Tr[ye.Trans(Bar(ye))]	0	0	2
lambdaT3GeneralQ020101I1Tr[yu.Trans(Bar(yu))]	0	0	6
lambdaH1AdjlambdaT3GeneralQ020101I1	0	0	-4
$_\frac{33g2^2 lambdaT3GeneralQ020101I1}{2}$	0	0	0
lambda12AdjlambdaT3GeneralQ020101I1	0	0	0
lambda12CrosslambdaT3GeneralQ020101I1	0	0	0
lambdaT3GeneralQ001102I2Bar(lambdaT3)	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$_\frac{21g2^2 lambdaT3GeneralQ020101I1}{2}$	0	0	0
$_\frac{9gY^2 lambdaT3GeneralQ020101I1}{2}$	0	0	1
lambda12lambdaT3GeneralQ020101I1	0	0	2
lambdaHlambdaT3GeneralQ020101I1	0	0	2
lambdaH1lambdaT3GeneralQ020101I1	0	0	4
lambdaH2lambdaT3GeneralQ020101I1	0	0	4
lambdaH2AdjlambdaT3GeneralQ020101I1	0	0	-2
lambdaT3GeneralQ001102I1Bar(lambdaT3)	0	0	4
lambdaT3GeneralQ110002I1Bar(lambdaT3)	0	0	8
lambdaT3GeneralQ020101I1Tr[y1.Trans(Bar(y1))]	0	0	1
lambdaT3GeneralQ020101I1Tr[y2.Trans(Bar(y2))]	0	0	1
Bar(lambdaT3)Tr[y2.Trans(Bar(y2Conj))]	0	0	2
lambdaT3GeneralQ020101I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	1
lambdaT3GeneralQ020101I1Tr[yd.Trans(Bar(yd))]	0	0	6
lambdaT3GeneralQ020101I1Tr[ye.Trans(Bar(ye))]	0	0	2
lambdaT3GeneralQ020101I1Tr[yu.Trans(Bar(yu))]	0	0	6
lambdaH1AdjlambdaT3GeneralQ020101I1	0	0	-2
$_\frac{33g2^2 lambdaT3GeneralQ020101I1}{2}$	0	0	1
lambda12AdjlambdaT3GeneralQ020101I1	0	0	-2
lambda12CrosslambdaT3GeneralQ020101I1	0	0	-2
lambdaT3GeneralQ001102I2Bar(lambdaT3)	0	0	-2

$\mathcal{S}$	E ($\alpha = -4$)
$_\frac{21g2^2 lambdaT3GeneralQ020101I1}{2}$	0
$_\frac{9gY^2 lambdaT3GeneralQ020101I1}{2}$	0
lambda12lambdaT3GeneralQ020101I1	0
lambdaHlambdaT3GeneralQ020101I1	0
lambdaH1lambdaT3GeneralQ020101I1	0
lambdaH2lambdaT3GeneralQ020101I1	0
lambdaH2AdjlambdaT3GeneralQ020101I1	0
lambdaT3GeneralQ001102I1Bar(lambdaT3)	0
lambdaT3GeneralQ110002I1Bar(lambdaT3)	0
lambdaT3GeneralQ020101I1Tr[y1.Trans(Bar(y1))]	0
lambdaT3GeneralQ020101I1Tr[y2.Trans(Bar(y2))]	0

Bar(lambdaT3)Tr[y2.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ020101I1Tr[y2Conj.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ020101I1Tr[yd.Trans(Bar(yd))]	0
lambdaT3GeneralQ020101I1Tr[ye.Trans(Bar(ye))]	0
lambdaT3GeneralQ020101I1Tr[yu.Trans(Bar(yu))]	0
lambdaH1AdjlambdaT3GeneralQ020101I1	0
$-\frac{33g2^2 \text{lambdaT3GeneralQ020101I1}}{2}$	0
lambda12AdjlambdaT3GeneralQ020101I1	0
lambda12CrosslambdaT3GeneralQ020101I1	0
lambdaT3GeneralQ001102I2Bar(lambdaT3)	0

$\beta^{(1)}_{lambdaT3GeneralQ021010I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$-\frac{21g2^2 \text{lambdaT3GeneralQ021010I1}}{2}$	1	0	0
$-\frac{9gY^2 \text{lambdaT3GeneralQ021010I1}}{2}$	1	0	0
lambda12lambdaT3GeneralQ021010I1	2	0	0
lambdaHlambdaT3GeneralQ021010I1	2	0	0
lambdaH1lambdaT3GeneralQ021010I1	4	0	0
lambdaH2lambdaT3GeneralQ021010I1	4	0	0
lambdaH2AdjlambdaT3GeneralQ021010I1	4	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ000211I1)	4	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ110200I1)	8	0	0
lambdaT3GeneralQ021010I1Tr[y1.Trans(Bar(y1))]	2	0	0
Bar(lambdaT3)Tr[y1.Trans(Bar(y1Conj))]	4	0	0
lambdaT3GeneralQ021010I1Tr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
lambdaT3GeneralQ021010I1Tr[y2.Trans(Bar(y2))]	1	0	0
lambdaT3GeneralQ021010I1Tr[yd.Trans(Bar(yd))]	6	0	0
lambdaT3GeneralQ021010I1Tr[ye.Trans(Bar(ye))]	2	0	0
lambdaT3GeneralQ021010I1Tr[yu.Trans(Bar(yu))]	6	0	0
lambdaH1AdjlambdaT3GeneralQ021010I1	0	0	0
$-\frac{33g2^2 \text{lambdaT3GeneralQ021010I1}}{2}$	0	0	0
lambda12AdjlambdaT3GeneralQ021010I1	0	0	0
lambda12CrosslambdaT3GeneralQ021010I1	0	0	0
$-\frac{4 \text{lambdaH1AdjlambdaT3GeneralQ021010I1}}{3}$	0	0	0
$-\frac{4 \text{lambdaH2AdjlambdaT3GeneralQ021010I1}}{3}$	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ000211I2)	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$-\frac{21g2^2 \text{lambdaT3GeneralQ021010I1}}{2}$	0	0	0
$-\frac{9gY^2 \text{lambdaT3GeneralQ021010I1}}{2}$	0	0	0
lambda12lambdaT3GeneralQ021010I1	0	0	0
lambdaHlambdaT3GeneralQ021010I1	0	0	0
lambdaH1lambdaT3GeneralQ021010I1	0	0	0
lambdaH2lambdaT3GeneralQ021010I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ021010I1	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ000211I1)	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaT3GeneralQ021010I1Tr[y1.Trans(Bar(y1))]	0	0	0
Bar(lambdaT3)Tr[y1.Trans(Bar(y1Conj))]	0	0	0
lambdaT3GeneralQ021010I1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaT3GeneralQ021010I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ021010I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ021010I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ021010I1Tr[yu.Trans(Bar(yu))]	0	0	0

lambdaH1AdjlambdaT3GeneralQ021010I1	0	0	0
$_\frac{33g2^2\text{lambdaT3GeneralQ021010I1}}{2}$	0	0	0
lambda12AdjlambdaT3GeneralQ021010I1	0	0	0
lambda12CrosslambdaT3GeneralQ021010I1	0	0	0
$_\frac{4\text{lambdaH1AdjlambdaT3GeneralQ021010I1}}{3}$	0	0	0
$_\frac{4\text{lambdaH2AdjlambdaT3GeneralQ021010I1}}{3}$	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ000211I2)	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$_\frac{21g2^2\text{lambdaT3GeneralQ021010I1}}{2}$	0	0	0
$_\frac{9gY^2\text{lambdaT3GeneralQ021010I1}}{2}$	0	0	0
lambda12lambdaT3GeneralQ021010I1	0	0	0
lambdaHlambdaT3GeneralQ021010I1	0	0	0
lambdaH1lambdaT3GeneralQ021010I1	0	0	0
lambdaH2lambdaT3GeneralQ021010I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ021010I1	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ000211I1)	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaT3GeneralQ021010I1Tr[y1.Trans(Bar(y1))]	0	0	0
Bar(lambdaT3)Tr[y1.Trans(Bar(y1Conj))]	0	0	0
lambdaT3GeneralQ021010I1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaT3GeneralQ021010I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ021010I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ021010I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ021010I1Tr[yu.Trans(Bar(yu))]	0	0	0
lambdaH1AdjlambdaT3GeneralQ021010I1	0	0	0
$_\frac{33g2^2\text{lambdaT3GeneralQ021010I1}}{2}$	0	0	0
lambda12AdjlambdaT3GeneralQ021010I1	0	0	0
lambda12CrosslambdaT3GeneralQ021010I1	0	0	0
$_\frac{4\text{lambdaH1AdjlambdaT3GeneralQ021010I1}}{3}$	0	0	0
$_\frac{4\text{lambdaH2AdjlambdaT3GeneralQ021010I1}}{3}$	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ000211I2)	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$_\frac{21g2^2\text{lambdaT3GeneralQ021010I1}}{2}$	1	0	0
$_\frac{9gY^2\text{lambdaT3GeneralQ021010I1}}{2}$	1	0	0
lambda12lambdaT3GeneralQ021010I1	2	0	0
lambdaHlambdaT3GeneralQ021010I1	2	0	0
lambdaH1lambdaT3GeneralQ021010I1	4	0	0
lambdaH2lambdaT3GeneralQ021010I1	4	0	0
lambdaH2AdjlambdaT3GeneralQ021010I1	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ000211I1)	4	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ110200I1)	8	0	0
lambdaT3GeneralQ021010I1Tr[y1.Trans(Bar(y1))]	1	0	0
Bar(lambdaT3)Tr[y1.Trans(Bar(y1Conj))]	2	0	0
lambdaT3GeneralQ021010I1Tr[y1Conj.Trans(Bar(y1Conj))]	1	0	0
lambdaT3GeneralQ021010I1Tr[y2.Trans(Bar(y2))]	2	0	0
lambdaT3GeneralQ021010I1Tr[yd.Trans(Bar(yd))]	6	0	0
lambdaT3GeneralQ021010I1Tr[ye.Trans(Bar(ye))]	2	0	0
lambdaT3GeneralQ021010I1Tr[yu.Trans(Bar(yu))]	6	0	0
lambdaH1AdjlambdaT3GeneralQ021010I1	4	0	0
$_\frac{33g2^2\text{lambdaT3GeneralQ021010I1}}{2}$	0	0	0
lambda12AdjlambdaT3GeneralQ021010I1	0	0	0
lambda12CrosslambdaT3GeneralQ021010I1	0	0	0
$_\frac{4\text{lambdaH1AdjlambdaT3GeneralQ021010I1}}{3}$	0	0	0

$_\frac{4\text{lambdaH2AdjlambdaT3GeneralQ021010I1}}{3}$	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ000211I2)	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$_\frac{21\text{g2}^2\text{lambdaT3GeneralQ021010I1}}{2}$	0	0	0
$_\frac{9\text{gY}^2\text{lambdaT3GeneralQ021010I1}}{2}$	1	0	0
lambda12lambdaT3GeneralQ021010I1	2	0	0
lambdaHlambdaT3GeneralQ021010I1	2	0	0
lambdaH1lambdaT3GeneralQ021010I1	4	0	0
lambdaH2lambdaT3GeneralQ021010I1	4	0	0
lambdaH2AdjlambdaT3GeneralQ021010I1	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ000211I1)	4	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ110200I1)	8	0	0
lambdaT3GeneralQ021010I1Tr[y1.Trans(Bar(y1))]	1	0	0
Bar(lambdaT3)Tr[y1.Trans(Bar(y1Conj))]	2	0	0
lambdaT3GeneralQ021010I1Tr[y1Conj.Trans(Bar(y1Conj))]	1	0	0
lambdaT3GeneralQ021010I1Tr[y2.Trans(Bar(y2))]	1	0	0
lambdaT3GeneralQ021010I1Tr[yd.Trans(Bar(yd))]	6	0	0
lambdaT3GeneralQ021010I1Tr[ye.Trans(Bar(ye))]	2	0	0
lambdaT3GeneralQ021010I1Tr[yu.Trans(Bar(yu))]	6	0	0
lambdaH1AdjlambdaT3GeneralQ021010I1	0	0	0
$_\frac{33\text{g2}^2\text{lambdaT3GeneralQ021010I1}}{2}$	1	0	0
lambda12AdjlambdaT3GeneralQ021010I1	-2	0	0
lambda12CrosslambdaT3GeneralQ021010I1	-2	0	0
$_\frac{4\text{lambdaH1AdjlambdaT3GeneralQ021010I1}}{3}$	1	0	0
$_\frac{4\text{lambdaH2AdjlambdaT3GeneralQ021010I1}}{3}$	1	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ000211I2)	-2	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$_\frac{21\text{g2}^2\text{lambdaT3GeneralQ021010I1}}{2}$	0
$_\frac{9\text{gY}^2\text{lambdaT3GeneralQ021010I1}}{2}$	0
lambda12lambdaT3GeneralQ021010I1	0
lambdaHlambdaT3GeneralQ021010I1	0
lambdaH1lambdaT3GeneralQ021010I1	0
lambdaH2lambdaT3GeneralQ021010I1	0
lambdaH2AdjlambdaT3GeneralQ021010I1	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ000211I1)	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ110200I1)	0
lambdaT3GeneralQ021010I1Tr[y1.Trans(Bar(y1))]	0
Bar(lambdaT3)Tr[y1.Trans(Bar(y1Conj))]	0
lambdaT3GeneralQ021010I1Tr[y1Conj.Trans(Bar(y1Conj))]	0
lambdaT3GeneralQ021010I1Tr[y2.Trans(Bar(y2))]	0
lambdaT3GeneralQ021010I1Tr[yd.Trans(Bar(yd))]	0
lambdaT3GeneralQ021010I1Tr[ye.Trans(Bar(ye))]	0
lambdaT3GeneralQ021010I1Tr[yu.Trans(Bar(yu))]	0
lambdaH1AdjlambdaT3GeneralQ021010I1	0
$_\frac{33\text{g2}^2\text{lambdaT3GeneralQ021010I1}}{2}$	0
lambda12AdjlambdaT3GeneralQ021010I1	0
lambda12CrosslambdaT3GeneralQ021010I1	0
$_\frac{4\text{lambdaH1AdjlambdaT3GeneralQ021010I1}}{3}$	0
$_\frac{4\text{lambdaH2AdjlambdaT3GeneralQ021010I1}}{3}$	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ000211I2)	0

$\beta^{(1)}_{lambdaT3GeneralQ022000I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
g2^{2}lambdaT3GeneralQ022000I1	0	0	0
gY^{2}lambdaT3GeneralQ022000I1	0	0	0
lambdaHlambdaT3GeneralQ022000I1	0	0	0
lambdaH1lambdaT3GeneralQ022000I1	0	0	0
lambdaH1AdjlambdaT3GeneralQ022000I1	0	0	0
lambdaS1lambdaT3GeneralQ022000I1	0	0	0
lambdaT3GeneralQ022000I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ022000I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ022000I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ022000I1Tr[yu.Trans(Bar(yu))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
g2^{2}lambdaT3GeneralQ022000I1	-9	0	0
gY^{2}lambdaT3GeneralQ022000I1	-3	0	0
lambdaHlambdaT3GeneralQ022000I1	2	0	0
lambdaH1lambdaT3GeneralQ022000I1	8	0	0
lambdaH1AdjlambdaT3GeneralQ022000I1	4	0	0
lambdaS1lambdaT3GeneralQ022000I1	2	0	0
lambdaT3GeneralQ022000I1Tr[y1.Trans(Bar(y1))]	2	0	0
lambdaT3GeneralQ022000I1Tr[yd.Trans(Bar(yd))]	6	0	0
lambdaT3GeneralQ022000I1Tr[ye.Trans(Bar(ye))]	2	0	0
lambdaT3GeneralQ022000I1Tr[yu.Trans(Bar(yu))]	6	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
g2^{2}lambdaT3GeneralQ022000I1	-9	0	0
gY^{2}lambdaT3GeneralQ022000I1	-3	0	0
lambdaHlambdaT3GeneralQ022000I1	2	0	0
lambdaH1lambdaT3GeneralQ022000I1	8	0	0
lambdaH1AdjlambdaT3GeneralQ022000I1	4	0	0
lambdaS1lambdaT3GeneralQ022000I1	2	0	0
lambdaT3GeneralQ022000I1Tr[y1.Trans(Bar(y1))]	3	0	0
lambdaT3GeneralQ022000I1Tr[yd.Trans(Bar(yd))]	6	0	0
lambdaT3GeneralQ022000I1Tr[ye.Trans(Bar(ye))]	2	0	0
lambdaT3GeneralQ022000I1Tr[yu.Trans(Bar(yu))]	6	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
g2^{2}lambdaT3GeneralQ022000I1	0	0	0
gY^{2}lambdaT3GeneralQ022000I1	0	0	0
lambdaHlambdaT3GeneralQ022000I1	0	0	0
lambdaH1lambdaT3GeneralQ022000I1	0	0	0
lambdaH1AdjlambdaT3GeneralQ022000I1	0	0	0
lambdaS1lambdaT3GeneralQ022000I1	0	0	0
lambdaT3GeneralQ022000I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ022000I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ022000I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ022000I1Tr[yu.Trans(Bar(yu))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
g2^{2}lambdaT3GeneralQ022000I1	0	0	0
gY^{2}lambdaT3GeneralQ022000I1	0	0	0

lambdaHlambdaT3GeneralQ022000I1	0	0	0
lambdaH1lambdaT3GeneralQ022000I1	0	0	0
lambdaH1AdjlambdaT3GeneralQ022000I1	0	0	0
lambdaS1lambdaT3GeneralQ022000I1	0	0	0
lambdaT3GeneralQ022000I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ022000I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ022000I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ022000I1Tr[yu.Trans(Bar(yu))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$g2^2$lambdaT3GeneralQ022000I1	0
gY^2lambdaT3GeneralQ022000I1	0
lambdaHlambdaT3GeneralQ022000I1	0
lambdaH1lambdaT3GeneralQ022000I1	0
lambdaH1AdjlambdaT3GeneralQ022000I1	0
lambdaS1lambdaT3GeneralQ022000I1	0
lambdaT3GeneralQ022000I1Tr[y1.Trans(Bar(y1))]	0
lambdaT3GeneralQ022000I1Tr[yd.Trans(Bar(yd))]	0
lambdaT3GeneralQ022000I1Tr[ye.Trans(Bar(ye))]	0
lambdaT3GeneralQ022000I1Tr[yu.Trans(Bar(yu))]	0

$\beta^{(1)}_{lambdaT3GeneralQ110002I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambdaH2lambdaT3GeneralQ000013I1	0	10	0
lambdaH1lambdaT3GeneralQ001102I1	0	2	0
lambdaT3lambdaT3GeneralQ020101I1	0	4	0
$-\frac{33g2^2\text{lambdaT3GeneralQ110002I1}}{2}$	0	1	0
$-\frac{3gY^2\text{lambdaT3GeneralQ110002I1}}{2}$	0	1	0
lambdaHlambdaT3GeneralQ110002I1	0	6	0
lambdaH2lambdaT3GeneralQ110002I1	0	8	0
lambdaS2lambdaT3GeneralQ110002I1	0	2	0
lambdaS2AdjlambdaT3GeneralQ110002I1	0	-4	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ110002I1)	0	40	0
lambdaT3GeneralQ110002I1Tr[y2.Trans(Bar(y2))]	0	2	0
lambdaH2Tr[y2.Trans(Bar(y2Conj))]	0	2	0
lambdaT3GeneralQ110002I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	2	0
lambdaT3GeneralQ110002I1Tr[yd.Trans(Bar(yd))]	0	6	0
lambdaT3GeneralQ110002I1Tr[ye.Trans(Bar(ye))]	0	2	0
lambdaT3GeneralQ110002I1Tr[yu.Trans(Bar(yu))]	0	6	0
Tr[y2.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	-2	0
$-\frac{9g2^2\text{lambdaT3GeneralQ110002I1}}{2}$	0	0	0
lambdaH1lambdaT3GeneralQ001102I2	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambdaH2lambdaT3GeneralQ000013I1	0	0	0
lambdaH1lambdaT3GeneralQ001102I1	0	0	0
lambdaT3lambdaT3GeneralQ020101I1	0	0	0
$-\frac{33g2^2\text{lambdaT3GeneralQ110002I1}}{2}$	0	0	0
$-\frac{3gY^2\text{lambdaT3GeneralQ110002I1}}{2}$	0	0	0
lambdaHlambdaT3GeneralQ110002I1	0	0	0
lambdaH2lambdaT3GeneralQ110002I1	0	0	0
lambdaS2lambdaT3GeneralQ110002I1	0	0	0
lambdaS2AdjlambdaT3GeneralQ110002I1	0	0	0

lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
lambdaT3GeneralQ110002I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaH2Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ110002I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ110002I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ110002I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ110002I1Tr[yu.Trans(Bar(yu))]	0	0	0
Tr[y2.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	0	0
$-\frac{9g2^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0	0	0
lambdaH1lambdaT3GeneralQ001102I2	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambdaH2lambdaT3GeneralQ000013I1	0	0	0
lambdaH1lambdaT3GeneralQ001102I1	0	0	0
lambdaT3lambdaT3GeneralQ020101I1	0	0	0
$-\frac{33g2^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0	0	0
$-\frac{3gY^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0	0	0
lambdaHlambdaT3GeneralQ110002I1	0	0	0
lambdaH2lambdaT3GeneralQ110002I1	0	0	0
lambdaS2lambdaT3GeneralQ110002I1	0	0	0
lambdaS2AdjlambdaT3GeneralQ110002I1	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ110002I1)	0	0	0
lambdaT3GeneralQ110002I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaH2Tr[y2.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ110002I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
lambdaT3GeneralQ110002I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ110002I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ110002I1Tr[yu.Trans(Bar(yu))]	0	0	0
Tr[y2.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	0	0
$-\frac{9g2^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0	0	0
lambdaH1lambdaT3GeneralQ001102I2	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambdaH2lambdaT3GeneralQ000013I1	0	0	6
lambdaH1lambdaT3GeneralQ001102I1	0	0	6
lambdaT3lambdaT3GeneralQ020101I1	0	0	12
$-\frac{33g2^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0	0	0
$-\frac{3gY^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0	0	1
lambdaHlambdaT3GeneralQ110002I1	0	0	6
lambdaH2lambdaT3GeneralQ110002I1	0	0	8
lambdaS2lambdaT3GeneralQ110002I1	0	0	2
lambdaS2AdjlambdaT3GeneralQ110002I1	0	0	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ110002I1)	0	0	24
lambdaT3GeneralQ110002I1Tr[y2.Trans(Bar(y2))]	0	0	4
lambdaH2Tr[y2.Trans(Bar(y2Conj))]	0	0	4
lambdaT3GeneralQ110002I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	4
lambdaT3GeneralQ110002I1Tr[yd.Trans(Bar(yd))]	0	0	6
lambdaT3GeneralQ110002I1Tr[ye.Trans(Bar(ye))]	0	0	2
lambdaT3GeneralQ110002I1Tr[yu.Trans(Bar(yu))]	0	0	6
Tr[y2.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	0	-4
$-\frac{9g2^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0	0	1
lambdaH1lambdaT3GeneralQ001102I2	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambdaH2lambdaT3GeneralQ000013I1	0	0	10
lambdaH1lambdaT3GeneralQ001102I1	0	0	6
lambdaT3lambdaT3GeneralQ020101I1	0	0	4
$-\frac{33g2^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0	0	1
$-\frac{3gY^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0	0	1
lambdaHlambdaT3GeneralQ110002I1	0	0	6
lambdaH2lambdaT3GeneralQ110002I1	0	0	8
lambdaS2lambdaT3GeneralQ110002I1	0	0	2
lambdaS2AdjlambdaT3GeneralQ110002I1	0	0	-4
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ110002I1)	0	0	40
lambdaT3GeneralQ110002I1Tr[y2.Trans(Bar(y2))]	0	0	2
lambdaH2Tr[y2.Trans(Bar(y2Conj))]	0	0	2
lambdaT3GeneralQ110002I1Tr[y2Conj.Trans(Bar(y2Conj))]	0	0	2
lambdaT3GeneralQ110002I1Tr[yd.Trans(Bar(yd))]	0	0	6
lambdaT3GeneralQ110002I1Tr[ye.Trans(Bar(ye))]	0	0	2
lambdaT3GeneralQ110002I1Tr[yu.Trans(Bar(yu))]	0	0	6
Tr[y2.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0	0	-2
$-\frac{9g2^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0	0	0
lambdaH1lambdaT3GeneralQ001102I2	0	0	2

$\mathcal{S}$	E ($\alpha = -4$)
lambdaH2lambdaT3GeneralQ000013I1	0
lambdaH1lambdaT3GeneralQ001102I1	0
lambdaT3lambdaT3GeneralQ020101I1	0
$-\frac{33g2^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0
$-\frac{3gY^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0
lambdaHlambdaT3GeneralQ110002I1	0
lambdaH2lambdaT3GeneralQ110002I1	0
lambdaS2lambdaT3GeneralQ110002I1	0
lambdaS2AdjlambdaT3GeneralQ110002I1	0
lambdaT3GeneralQ000004I1Bar(lambdaT3GeneralQ110002I1)	0
lambdaT3GeneralQ110002I1Tr[y2.Trans(Bar(y2))]	0
lambdaH2Tr[y2.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ110002I1Tr[y2Conj.Trans(Bar(y2Conj))]	0
lambdaT3GeneralQ110002I1Tr[yd.Trans(Bar(yd))]	0
lambdaT3GeneralQ110002I1Tr[ye.Trans(Bar(ye))]	0
lambdaT3GeneralQ110002I1Tr[yu.Trans(Bar(yu))]	0
Tr[y2.Trans(Bar(y2Conj)).ye.Trans(Bar(ye))]	0
$-\frac{9g2^2 \text{lambdaT3GeneralQ110002I1}}{2}$	0
lambdaH1lambdaT3GeneralQ001102I2	0

$\beta^{(1)}_{lambdaT3GeneralQ110101I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambda12AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaHlambdaHHdagS1barS2barCross	0	0	0
lambdaH1AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaH2AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaH2lambdaS1barS2S2bar2	0	0	0
$-\frac{\text{lambdaH2AdjlambdaS1barS2S2bar2}}{2}$	0	0	0
lambdaH1lambdaS1S1bar2S2bar	0	0	0
$\frac{\text{lambdaH1AdjlambdaS1S1bar2S2bar}}{2}$	0	0	0
$g2^2$lambdaT3GeneralQ110101I1	0	0	0
gY^2lambdaT3GeneralQ110101I1	0	0	0

lambda12lambdaT3GeneralQ110101I1	0	0	0
$_\frac{\text{lambda12AdjlambdaT3GeneralQ110101I1}}{2}$	0	0	0
lambdaHlambdaT3GeneralQ110101I1	0	0	0
lambdaH1lambdaT3GeneralQ110101I1	0	0	0
lambdaH1AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaH2lambdaT3GeneralQ110101I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaS1bar2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ110101I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ110101I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ110101I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ110101I1Tr[yu.Trans(Bar(yu))]	0	0	0
$\frac{3}{2}$lambdaT3GeneralQ110101I1Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaT3GeneralQ110101I1Tr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambda12AdjlambdaHHdagS1barS2barCross	0	1	0
lambdaHlambdaHHdagS1barS2barCross	0	-2	0
lambdaH1AdjlambdaHHdagS1barS2barCross	0	-1	0
lambdaH2AdjlambdaHHdagS1barS2barCross	0	1	0
lambdaH2lambdaS1barS2S2bar2	0	-6	0
$_\frac{\text{lambdaH2AdjlambdaS1barS2S2bar2}}{2}$	0	1	0
lambdaH1lambdaS1S1bar2S2bar	0	-6	0
$\frac{\text{lambdaH1AdjlambdaS1S1bar2S2bar}}{2}$	0	1	0
$g2^2$lambdaT3GeneralQ110101I1	0	-9	0
gY^2lambdaT3GeneralQ110101I1	0	-3	0
lambda12lambdaT3GeneralQ110101I1	0	2	0
$_\frac{\text{lambda12AdjlambdaT3GeneralQ110101I1}}{2}$	0	1	0
lambdaHlambdaT3GeneralQ110101I1	0	4	0
lambdaH1lambdaT3GeneralQ110101I1	0	4	0
lambdaH1AdjlambdaT3GeneralQ110101I1	0	-2	0
lambdaH2lambdaT3GeneralQ110101I1	0	4	0
lambdaH2AdjlambdaT3GeneralQ110101I1	0	2	0
lambdaS1bar2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	-4	0
lambdaS1bar2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	8	0
lambdaT3GeneralQ110101I1Tr[y1.Trans(Bar(y1))]	0	1	0
lambdaT3GeneralQ110101I1Tr[y2.Trans(Bar(y2))]	0	1	0
lambdaT3GeneralQ110101I1Tr[yd.Trans(Bar(yd))]	0	6	0
lambdaT3GeneralQ110101I1Tr[ye.Trans(Bar(ye))]	0	2	0
lambdaT3GeneralQ110101I1Tr[yu.Trans(Bar(yu))]	0	6	0
$\frac{3}{2}$lambdaT3GeneralQ110101I1Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaT3GeneralQ110101I1Tr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambda12AdjlambdaHHdagS1barS2barCross	0	1	0
lambdaHlambdaHHdagS1barS2barCross	0	-2	0
lambdaH1AdjlambdaHHdagS1barS2barCross	0	-1	0
lambdaH2AdjlambdaHHdagS1barS2barCross	0	1	0
lambdaH2lambdaS1barS2S2bar2	0	-6	0
$_\frac{\text{lambdaH2AdjlambdaS1barS2S2bar2}}{2}$	0	1	0
lambdaH1lambdaS1S1bar2S2bar	0	-6	0
$\frac{\text{lambdaH1AdjlambdaS1S1bar2S2bar}}{2}$	0	1	0
$g2^2$lambdaT3GeneralQ110101I1	0	-9	0
gY^2lambdaT3GeneralQ110101I1	0	-3	0

lambda12lambdaT3GeneralQ110101I1	0	2	0
$-\frac{\text{lambda12AdjlambdaT3GeneralQ110101I1}}{2}$	0	1	0
lambdaHlambdaT3GeneralQ110101I1	0	4	0
lambdaH1lambdaT3GeneralQ110101I1	0	4	0
lambdaH1AdjlambdaT3GeneralQ110101I1	0	-2	0
lambdaH2lambdaT3GeneralQ110101I1	0	4	0
lambdaH2AdjlambdaT3GeneralQ110101I1	0	2	0
lambdaS1bar2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	-4	0
lambdaS1bar2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	8	0
lambdaT3GeneralQ110101I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ110101I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ110101I1Tr[yd.Trans(Bar(yd))]	0	6	0
lambdaT3GeneralQ110101I1Tr[ye.Trans(Bar(ye))]	0	2	0
lambdaT3GeneralQ110101I1Tr[yu.Trans(Bar(yu))]	0	6	0
$\frac{3}{2}$lambdaT3GeneralQ110101I1Tr[y1.Trans(Bar(y1))]	0	1	0
$\frac{3}{2}$lambdaT3GeneralQ110101I1Tr[y2.Trans(Bar(y2))]	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambda12AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaHlambdaHHdagS1barS2barCross	0	0	0
lambdaH1AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaH2AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaH2lambdaS1barS2S2bar2	0	0	0
$-\frac{\text{lambdaH2AdjlambdaS1barS2S2bar2}}{2}$	0	0	0
lambdaH1lambdaS1S1bar2S2bar	0	0	0
$\frac{\text{lambdaH1AdjlambdaS1S1bar2S2bar}}{2}$	0	0	0
$g2^2$lambdaT3GeneralQ110101I1	0	0	0
gY^2lambdaT3GeneralQ110101I1	0	0	0
lambda12lambdaT3GeneralQ110101I1	0	0	0
$-\frac{\text{lambda12AdjlambdaT3GeneralQ110101I1}}{2}$	0	0	0
lambdaHlambdaT3GeneralQ110101I1	0	0	0
lambdaH1lambdaT3GeneralQ110101I1	0	0	0
lambdaH1AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaH2lambdaT3GeneralQ110101I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaS1bar2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ110101I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ110101I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ110101I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ110101I1Tr[yu.Trans(Bar(yu))]	0	0	0
$\frac{3}{2}$lambdaT3GeneralQ110101I1Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaT3GeneralQ110101I1Tr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambda12AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaHlambdaHHdagS1barS2barCross	0	0	0
lambdaH1AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaH2AdjlambdaHHdagS1barS2barCross	0	0	0
lambdaH2lambdaS1barS2S2bar2	0	0	0
$-\frac{\text{lambdaH2AdjlambdaS1barS2S2bar2}}{2}$	0	0	0
lambdaH1lambdaS1S1bar2S2bar	0	0	0
$\frac{\text{lambdaH1AdjlambdaS1S1bar2S2bar}}{2}$	0	0	0
$g2^2$lambdaT3GeneralQ110101I1	0	0	0
gY^2lambdaT3GeneralQ110101I1	0	0	0

lambda12lambdaT3GeneralQ110101I1	0	0	0
$_\frac{\text{lambda12AdjlambdaT3GeneralQ110101I1}}{2}$	0	0	0
lambdaHlambdaT3GeneralQ110101I1	0	0	0
lambdaH1lambdaT3GeneralQ110101I1	0	0	0
lambdaH1AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaH2lambdaT3GeneralQ110101I1	0	0	0
lambdaH2AdjlambdaT3GeneralQ110101I1	0	0	0
lambdaS1bar2S2bar2Bar(lambdaHHdagS1barS2barCross)	0	0	0
lambdaS1bar2S2bar2Bar(lambdaT3GeneralQ110101I1)	0	0	0
lambdaT3GeneralQ110101I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ110101I1Tr[y2.Trans(Bar(y2))]	0	0	0
lambdaT3GeneralQ110101I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ110101I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ110101I1Tr[yu.Trans(Bar(yu))]	0	0	0
$\frac{3}{2}$lambdaT3GeneralQ110101I1Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$lambdaT3GeneralQ110101I1Tr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
lambda12AdjlambdaHHdagS1barS2barCross	0
lambdaHlambdaHHdagS1barS2barCross	0
lambdaH1AdjlambdaHHdagS1barS2barCross	0
lambdaH2AdjlambdaHHdagS1barS2barCross	0
lambdaH2lambdaS1barS2S2bar2	0
$_\frac{\text{lambdaH2AdjlambdaS1barS2S2bar2}}{2}$	0
lambdaH1lambdaS1S1bar2S2bar	0
$\frac{\text{lambdaH1AdjlambdaS1S1bar2S2bar}}{2}$	0
$g2^2$lambdaT3GeneralQ110101I1	0
gY^2lambdaT3GeneralQ110101I1	0
lambda12lambdaT3GeneralQ110101I1	0
$_\frac{\text{lambda12AdjlambdaT3GeneralQ110101I1}}{2}$	0
lambdaHlambdaT3GeneralQ110101I1	0
lambdaH1lambdaT3GeneralQ110101I1	0
lambdaH1AdjlambdaT3GeneralQ110101I1	0
lambdaH2lambdaT3GeneralQ110101I1	0
lambdaH2AdjlambdaT3GeneralQ110101I1	0
lambdaS1bar2S2bar2Bar(lambdaHHdagS1barS2barCross)	0
lambdaS1bar2S2bar2Bar(lambdaT3GeneralQ110101I1)	0
lambdaT3GeneralQ110101I1Tr[y1.Trans(Bar(y1))]	0
lambdaT3GeneralQ110101I1Tr[y2.Trans(Bar(y2))]	0
lambdaT3GeneralQ110101I1Tr[yd.Trans(Bar(yd))]	0
lambdaT3GeneralQ110101I1Tr[ye.Trans(Bar(ye))]	0
lambdaT3GeneralQ110101I1Tr[yu.Trans(Bar(yu))]	0
$\frac{3}{2}$lambdaT3GeneralQ110101I1Tr[y1.Trans(Bar(y1))]	0
$\frac{3}{2}$lambdaT3GeneralQ110101I1Tr[y2.Trans(Bar(y2))]	0

$\beta^{(1)}_{lambdaT3GeneralQ110200I1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambdaH2lambdaT3GeneralQ000211I1	6	0	0
lambdaH1lambdaT3GeneralQ001300I1	6	0	0
$_\frac{9g2^2\text{lambdaT3GeneralQ110200I1}}{2}$	1	0	0
$_\frac{3gY^2\text{lambdaT3GeneralQ110200I1}}{2}$	1	0	0
lambdaHlambdaT3GeneralQ110200I1	6	0	0
lambdaH1lambdaT3GeneralQ110200I1	8	0	0
lambdaS1lambdaT3GeneralQ110200I1	2	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	12	0	0

lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ110200I1)	24	0	0
lambdaT3GeneralQ110200I1Tr[y1.Trans(Bar(y1))]	4	0	0
lambdaH1Tr[y1Conj.Trans(Bar(y1))]	4	0	0
lambdaT3GeneralQ110200I1Tr[y1Conj.Trans(Bar(y1Conj))]	4	0	0
lambdaT3GeneralQ110200I1Tr[yd.Trans(Bar(yd))]	6	0	0
lambdaT3GeneralQ110200I1Tr[ye.Trans(Bar(ye))]	2	0	0
lambdaT3GeneralQ110200I1Tr[yu.Trans(Bar(yu))]	6	0	0
Tr[y1Conj.Trans(Bar(y1)).ye.Trans(Bar(ye))]	-4	0	0
$_\frac{33g2^2\text{lambdaT3GeneralQ110200I1}}{2}$	0	0	0
lambdaS1AdjlambdaT3GeneralQ110200I1	0	0	0
lambdaH2lambdaT3GeneralQ000211I2	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambdaH2lambdaT3GeneralQ000211I1	0	0	0
lambdaH1lambdaT3GeneralQ001300I1	0	0	0
$_\frac{9g2^2\text{lambdaT3GeneralQ110200I1}}{2}$	0	0	0
$_\frac{3gY^2\text{lambdaT3GeneralQ110200I1}}{2}$	0	0	0
lambdaHlambdaT3GeneralQ110200I1	0	0	0
lambdaH1lambdaT3GeneralQ110200I1	0	0	0
lambdaS1lambdaT3GeneralQ110200I1	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaT3GeneralQ110200I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaH1Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ110200I1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaT3GeneralQ110200I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ110200I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ110200I1Tr[yu.Trans(Bar(yu))]	0	0	0
Tr[y1Conj.Trans(Bar(y1)).ye.Trans(Bar(ye))]	0	0	0
$_\frac{33g2^2\text{lambdaT3GeneralQ110200I1}}{2}$	0	0	0
lambdaS1AdjlambdaT3GeneralQ110200I1	0	0	0
lambdaH2lambdaT3GeneralQ000211I2	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambdaH2lambdaT3GeneralQ000211I1	0	0	0
lambdaH1lambdaT3GeneralQ001300I1	0	0	0
$_\frac{9g2^2\text{lambdaT3GeneralQ110200I1}}{2}$	0	0	0
$_\frac{3gY^2\text{lambdaT3GeneralQ110200I1}}{2}$	0	0	0
lambdaHlambdaT3GeneralQ110200I1	0	0	0
lambdaH1lambdaT3GeneralQ110200I1	0	0	0
lambdaS1lambdaT3GeneralQ110200I1	0	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	0	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ110200I1)	0	0	0
lambdaT3GeneralQ110200I1Tr[y1.Trans(Bar(y1))]	0	0	0
lambdaH1Tr[y1Conj.Trans(Bar(y1))]	0	0	0
lambdaT3GeneralQ110200I1Tr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
lambdaT3GeneralQ110200I1Tr[yd.Trans(Bar(yd))]	0	0	0
lambdaT3GeneralQ110200I1Tr[ye.Trans(Bar(ye))]	0	0	0
lambdaT3GeneralQ110200I1Tr[yu.Trans(Bar(yu))]	0	0	0
Tr[y1Conj.Trans(Bar(y1)).ye.Trans(Bar(ye))]	0	0	0
$_\frac{33g2^2\text{lambdaT3GeneralQ110200I1}}{2}$	0	0	0
lambdaS1AdjlambdaT3GeneralQ110200I1	0	0	0
lambdaH2lambdaT3GeneralQ000211I2	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambdaH2lambdaT3GeneralQ000211I1	2	0	0
lambdaH1lambdaT3GeneralQ001300I1	10	0	0
$-\frac{9g2^2\text{lambdaT3GeneralQ110200I1}}{2}$	0	0	0
$-\frac{3gY^2\text{lambdaT3GeneralQ110200I1}}{2}$	1	0	0
lambdaHlambdaT3GeneralQ110200I1	6	0	0
lambdaH1lambdaT3GeneralQ110200I1	8	0	0
lambdaS1lambdaT3GeneralQ110200I1	2	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ110200I1)	40	0	0
lambdaT3GeneralQ110200I1Tr[y1.Trans(Bar(y1))]	2	0	0
lambdaH1Tr[y1Conj.Trans(Bar(y1))]	2	0	0
lambdaT3GeneralQ110200I1Tr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
lambdaT3GeneralQ110200I1Tr[yd.Trans(Bar(yd))]	6	0	0
lambdaT3GeneralQ110200I1Tr[ye.Trans(Bar(ye))]	2	0	0
lambdaT3GeneralQ110200I1Tr[yu.Trans(Bar(yu))]	6	0	0
Tr[y1Conj.Trans(Bar(y1)).ye.Trans(Bar(ye))]	-2	0	0
$-\frac{33g2^2\text{lambdaT3GeneralQ110200I1}}{2}$	1	0	0
lambdaS1AdjlambdaT3GeneralQ110200I1	-4	0	0
lambdaH2lambdaT3GeneralQ000211I2	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambdaH2lambdaT3GeneralQ000211I1	6	0	0
lambdaH1lambdaT3GeneralQ001300I1	10	0	0
$-\frac{9g2^2\text{lambdaT3GeneralQ110200I1}}{2}$	0	0	0
$-\frac{3gY^2\text{lambdaT3GeneralQ110200I1}}{2}$	1	0	0
lambdaHlambdaT3GeneralQ110200I1	6	0	0
lambdaH1lambdaT3GeneralQ110200I1	8	0	0
lambdaS1lambdaT3GeneralQ110200I1	2	0	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	4	0	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ110200I1)	40	0	0
lambdaT3GeneralQ110200I1Tr[y1.Trans(Bar(y1))]	2	0	0
lambdaH1Tr[y1Conj.Trans(Bar(y1))]	2	0	0
lambdaT3GeneralQ110200I1Tr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
lambdaT3GeneralQ110200I1Tr[yd.Trans(Bar(yd))]	6	0	0
lambdaT3GeneralQ110200I1Tr[ye.Trans(Bar(ye))]	2	0	0
lambdaT3GeneralQ110200I1Tr[yu.Trans(Bar(yu))]	6	0	0
Tr[y1Conj.Trans(Bar(y1)).ye.Trans(Bar(ye))]	-2	0	0
$-\frac{33g2^2\text{lambdaT3GeneralQ110200I1}}{2}$	1	0	0
lambdaS1AdjlambdaT3GeneralQ110200I1	-4	0	0
lambdaH2lambdaT3GeneralQ000211I2	2	0	0

$\mathcal{S}$	E ($\alpha = -4$)
lambdaH2lambdaT3GeneralQ000211I1	0
lambdaH1lambdaT3GeneralQ001300I1	0
$-\frac{9g2^2\text{lambdaT3GeneralQ110200I1}}{2}$	0
$-\frac{3gY^2\text{lambdaT3GeneralQ110200I1}}{2}$	0
lambdaHlambdaT3GeneralQ110200I1	0
lambdaH1lambdaT3GeneralQ110200I1	0
lambdaS1lambdaT3GeneralQ110200I1	0
Bar(lambdaT3)Bar(lambdaT3GeneralQ021010I1)	0
lambdaT3GeneralQ000400I1Bar(lambdaT3GeneralQ110200I1)	0
lambdaT3GeneralQ110200I1Tr[y1.Trans(Bar(y1))]	0
lambdaH1Tr[y1Conj.Trans(Bar(y1))]	0

lambdaT3GeneralQ110200I1Tr[y1Conj.Trans(Bar(y1Conj))]	0
lambdaT3GeneralQ110200I1Tr[yd.Trans(Bar(yd))]	0
lambdaT3GeneralQ110200I1Tr[ye.Trans(Bar(ye))]	0
lambdaT3GeneralQ110200I1Tr[yu.Trans(Bar(yu))]	0
Tr[y1Conj.Trans(Bar(y1)).ye.Trans(Bar(ye))]	0
$-\frac{33g2^2\text{lambdaT3GeneralQ110200I1}}{2}$	0
lambdaS1AdjlambdaT3GeneralQ110200I1	0
lambdaH2lambdaT3GeneralQ000211I2	0

$\beta^{(1)}_{m_1^2}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambdaS1mS1Sq	4	4	4
lambda12mS2Sq	6	6	6
muT3GeneralS1SqBar(lambdaT3GeneralQ001300I1)	12	0	0
lambdaT3GeneralQ001300I1Bar(muT3GeneralS1Sq)	12	0	0
mS1SqTr[y1.Trans(Bar(y1))]	4	4	4
muT3GeneralS1SqTr[y1.Trans(Bar(y1Conj))]	8	0	0
Bar(muT3GeneralS1Sq)Tr[y1Conj.Trans(Bar(y1))]	8	0	0
mS1SqTr[y1Conj.Trans(Bar(y1Conj))]	4	0	0
Tr[MF.Trans(Bar(MF)).Trans(y1).Bar(y1)]	-8	-8	-8
Tr[MF.Trans(Bar(MF)).Trans(y1Conj).Bar(y1Conj)]	-8	0	0
gY^2mS1Sq	0	-6	-24
muT3GeneralS2SqBar(lambdaT3GeneralQ001102I1)	0	12	0
lambdaT3GeneralQ001102I1Bar(muT3GeneralS2Sq)	0	12	0
$-\frac{9g2^2\text{mS1Sq}}{2}$	0	0	0
$-\frac{3gY^2\text{mS1Sq}}{2}$	0	0	0
muT3GeneralS1S2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barBar(muT3GeneralS1S2)	0	0	0
Tr[MF.Bar(MF).Trans(y1).Bar(y1)]	0	0	0
Tr[MF.Trans(Bar(y1)).y1.Bar(MF)]	0	0	0
$-\frac{27gY^2\text{mS1Sq}}{2}$	0	0	0
$g2^2$mS1Sq	0	0	0
lambdaS1AdjmS1Sq	0	0	0
lambda12CrossmS2Sq	0	0	0
muT3GeneralS2SqBar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(muT3GeneralS2Sq)	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambdaS1mS1Sq	6	6	6
lambda12mS2Sq	4	4	4
muT3GeneralS1SqBar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ001300I1Bar(muT3GeneralS1Sq)	0	0	0
mS1SqTr[y1.Trans(Bar(y1))]	2	2	2
muT3GeneralS1SqTr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(muT3GeneralS1Sq)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
mS1SqTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[MF.Trans(Bar(MF)).Trans(y1).Bar(y1)]	-4	0	-4
Tr[MF.Trans(Bar(MF)).Trans(y1Conj).Bar(y1Conj)]	0	0	0
gY^2mS1Sq	0	0	0
muT3GeneralS2SqBar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ001102I1Bar(muT3GeneralS2Sq)	0	0	0
$-\frac{9g2^2\text{mS1Sq}}{2}$	1	1	1
$-\frac{3gY^2\text{mS1Sq}}{2}$	1	1	0
muT3GeneralS1S2Bar(lambdaS1S1bar2S2bar)	0	6	0

lambdaS1S1bar2S2barBar(muT3GeneralS1S2)	0	6	0
Tr[MF.Bar(MF).Trans(y1).Bar(y1)]	0	−8	0
Tr[MF.Trans(Bar(y1)).y1.Bar(MF)]	0	−8	0
$-\frac{27gY^2mS1Sq}{2}$	0	0	1
$g2^2mS1Sq$	0	0	0
lambdaS1AdjmS1Sq	0	0	0
lambda12CrossmS2Sq	0	0	0
muT3GeneralS2SqBar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(muT3GeneralS2Sq)	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambdaS1mS1Sq	6	6	6
lambda12mS2Sq	4	4	4
muT3GeneralS1SqBar(lambdaT3GeneralQ001300I1)	0	0	0
lambdaT3GeneralQ001300I1Bar(muT3GeneralS1Sq)	0	0	0
mS1SqTr[y1.Trans(Bar(y1))]	3	3	3
muT3GeneralS1SqTr[y1.Trans(Bar(y1Conj))]	0	0	0
Bar(muT3GeneralS1Sq)Tr[y1Conj.Trans(Bar(y1))]	0	0	0
mS1SqTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[MF.Trans(Bar(MF)).Trans(y1).Bar(y1)]	−6	0	−6
Tr[MF.Trans(Bar(MF)).Trans(y1Conj).Bar(y1Conj)]	0	0	0
gY^2mS1Sq	0	0	0
muT3GeneralS2SqBar(lambdaT3GeneralQ001102I1)	0	0	0
lambdaT3GeneralQ001102I1Bar(muT3GeneralS2Sq)	0	0	0
$-\frac{9g2^2mS1Sq}{2}$	1	1	1
$-\frac{3gY^2mS1Sq}{2}$	1	1	0
muT3GeneralS1S2Bar(lambdaS1S1bar2S2bar)	0	6	0
lambdaS1S1bar2S2barBar(muT3GeneralS1S2)	0	6	0
Tr[MF.Bar(MF).Trans(y1).Bar(y1)]	0	−12	0
Tr[MF.Trans(Bar(y1)).y1.Bar(MF)]	0	−12	0
$-\frac{27gY^2mS1Sq}{2}$	0	0	1
$g2^2mS1Sq$	0	0	0
lambdaS1AdjmS1Sq	0	0	0
lambda12CrossmS2Sq	0	0	0
muT3GeneralS2SqBar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(muT3GeneralS2Sq)	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambdaS1mS1Sq	8	8	8
lambda12mS2Sq	2	2	2
muT3GeneralS1SqBar(lambdaT3GeneralQ001300I1)	20	0	0
lambdaT3GeneralQ001300I1Bar(muT3GeneralS1Sq)	20	0	0
mS1SqTr[y1.Trans(Bar(y1))]	2	2	2
muT3GeneralS1SqTr[y1.Trans(Bar(y1Conj))]	4	0	0
Bar(muT3GeneralS1Sq)Tr[y1Conj.Trans(Bar(y1))]	4	0	0
mS1SqTr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
Tr[MF.Trans(Bar(MF)).Trans(y1).Bar(y1)]	−4	−4	−4
Tr[MF.Trans(Bar(MF)).Trans(y1Conj).Bar(y1Conj)]	−4	0	0
gY^2mS1Sq	0	−6	−6
muT3GeneralS2SqBar(lambdaT3GeneralQ001102I1)	0	0	4
lambdaT3GeneralQ001102I1Bar(muT3GeneralS2Sq)	0	0	4
$-\frac{9g2^2mS1Sq}{2}$	0	0	0
$-\frac{3gY^2mS1Sq}{2}$	0	0	0
muT3GeneralS1S2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barBar(muT3GeneralS1S2)	0	0	0

Tr[MF.Bar(MF).Trans(y1).Bar(y1)]	0	0	0
Tr[MF.Trans(Bar(y1)).y1.Bar(MF)]	0	0	0
$-\frac{27gY^2mS1Sq}{2}$	0	0	0
$g2^2mS1Sq$	-12	-12	-12
lambdaS1AdjmS1Sq	4	4	4
lambda12CrossmS2Sq	0	0	0
muT3GeneralS2SqBar(lambdaT3GeneralQ001102I2)	0	0	0
lambdaT3GeneralQ001102I2Bar(muT3GeneralS2Sq)	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambdaS1mS1Sq	8	8	8
lambda12mS2Sq	6	6	6
muT3GeneralS1SqBar(lambdaT3GeneralQ001300I1)	20	0	0
lambdaT3GeneralQ001300I1Bar(muT3GeneralS1Sq)	20	0	0
mS1SqTr[y1.Trans(Bar(y1))]	2	2	2
muT3GeneralS1SqTr[y1.Trans(Bar(y1Conj))]	4	0	0
Bar(muT3GeneralS1Sq)Tr[y1Conj.Trans(Bar(y1))]	4	0	0
mS1SqTr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
Tr[MF.Trans(Bar(MF)).Trans(y1).Bar(y1)]	-4	-4	-4
Tr[MF.Trans(Bar(MF)).Trans(y1Conj).Bar(y1Conj)]	-4	0	0
gY^2mS1Sq	0	-6	-6
muT3GeneralS2SqBar(lambdaT3GeneralQ001102I1)	0	0	12
lambdaT3GeneralQ001102I1Bar(muT3GeneralS2Sq)	0	0	12
$-\frac{9g2^2mS1Sq}{2}$	0	0	0
$-\frac{3gY^2mS1Sq}{2}$	0	0	0
muT3GeneralS1S2Bar(lambdaS1S1bar2S2bar)	0	0	0
lambdaS1S1bar2S2barBar(muT3GeneralS1S2)	0	0	0
Tr[MF.Bar(MF).Trans(y1).Bar(y1)]	0	0	0
Tr[MF.Trans(Bar(y1)).y1.Bar(MF)]	0	0	0
$-\frac{27gY^2mS1Sq}{2}$	0	0	0
$g2^2mS1Sq$	-12	-12	-12
lambdaS1AdjmS1Sq	4	4	4
lambda12CrossmS2Sq	4	4	4
muT3GeneralS2SqBar(lambdaT3GeneralQ001102I2)	0	0	4
lambdaT3GeneralQ001102I2Bar(muT3GeneralS2Sq)	0	0	4

$\mathcal{S}$	E ($\alpha = -4$)
lambdaS1mS1Sq	8
lambda12mS2Sq	6
muT3GeneralS1SqBar(lambdaT3GeneralQ001300I1)	0
lambdaT3GeneralQ001300I1Bar(muT3GeneralS1Sq)	0
mS1SqTr[y1.Trans(Bar(y1))]	2
muT3GeneralS1SqTr[y1.Trans(Bar(y1Conj))]	0
Bar(muT3GeneralS1Sq)Tr[y1Conj.Trans(Bar(y1))]	0
mS1SqTr[y1Conj.Trans(Bar(y1Conj))]	0
Tr[MF.Trans(Bar(MF)).Trans(y1).Bar(y1)]	-4
Tr[MF.Trans(Bar(MF)).Trans(y1Conj).Bar(y1Conj)]	0
gY^2mS1Sq	-24
muT3GeneralS2SqBar(lambdaT3GeneralQ001102I1)	0
lambdaT3GeneralQ001102I1Bar(muT3GeneralS2Sq)	0
$-\frac{9g2^2mS1Sq}{2}$	0
$-\frac{3gY^2mS1Sq}{2}$	0
muT3GeneralS1S2Bar(lambdaS1S1bar2S2bar)	0
lambdaS1S1bar2S2barBar(muT3GeneralS1S2)	0
Tr[MF.Bar(MF).Trans(y1).Bar(y1)]	0

Tr[MF.Trans(Bar(y1)).y1.Bar(MF)]	0
$-\frac{27 g_Y^2 mS1Sq}{2}$	0
$g_2^2 mS1Sq$	-12
lambdaS1AdjmS1Sq	4
lambda12CrossmS2Sq	4
muT3GeneralS2SqBar(lambdaT3GeneralQ001102I2)	0
lambdaT3GeneralQ001102I2Bar(muT3GeneralS2Sq)	0

$\beta^{(1)}_{m_2^2}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambda12mS1Sq	2	2	2
$g_2^2 mS2Sq$	-12	-12	-12
$g_Y^2 mS2Sq$	-6	0	-6
lambdaS2mS2Sq	8	8	8
lambdaS2AdjmS2Sq	4	4	4
muT3GeneralS1SqBar(lambdaT3GeneralQ000211I1)	4	0	0
lambdaT3GeneralQ000211I1Bar(muT3GeneralS1Sq)	4	0	0
mS2SqTr[y2.Trans(Bar(y2))]	2	2	2
Tr[MF.Trans(Bar(y2)).y2.Trans(Bar(MF))]	-4	-4	-4
muT3GeneralS2SqBar(lambdaT3GeneralQ000013I1)	0	20	0
lambdaT3GeneralQ000013I1Bar(muT3GeneralS2Sq)	0	20	0
Bar(muT3GeneralS2Sq)Tr[y2.Trans(Bar(y2Conj))]	0	4	0
muT3GeneralS2SqTr[y2Conj.Trans(Bar(y2))]	0	4	0
mS2SqTr[y2Conj.Trans(Bar(y2Conj))]	0	2	0
Tr[MF.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(MF))]	0	-4	0
$-\frac{9 g_2^2 mS2Sq}{2}$	0	0	0
$-\frac{27 g_Y^2 mS2Sq}{2}$	0	0	0
$-\frac{3 g_Y^2 mS2Sq}{2}$	0	0	0
muT3GeneralS1S2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(muT3GeneralS1S2)	0	0	0
Tr[MF.Bar(MF).Trans(y2).Bar(y2)]	0	0	0
Tr[MF.Trans(Bar(y2)).y2.Bar(MF)]	0	0	0
lambda12CrossmS1Sq	0	0	0
muT3GeneralS1SqBar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(muT3GeneralS1Sq)	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambda12mS1Sq	4	4	4
$g_2^2 mS2Sq$	0	0	0
$g_Y^2 mS2Sq$	0	0	0
lambdaS2mS2Sq	6	6	6
lambdaS2AdjmS2Sq	0	0	0
muT3GeneralS1SqBar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(muT3GeneralS1Sq)	0	0	0
mS2SqTr[y2.Trans(Bar(y2))]	2	2	2
Tr[MF.Trans(Bar(y2)).y2.Trans(Bar(MF))]	-4	0	-4
muT3GeneralS2SqBar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(muT3GeneralS2Sq)	0	0	0
Bar(muT3GeneralS2Sq)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
muT3GeneralS2SqTr[y2Conj.Trans(Bar(y2))]	0	0	0
mS2SqTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[MF.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(MF))]	0	0	0
$-\frac{9 g_2^2 mS2Sq}{2}$	1	1	1
$-\frac{27 g_Y^2 mS2Sq}{2}$	1	0	0

$-\frac{3gY^2mS2Sq}{2}$	0	1	1
muT3GeneralS1S2Bar(lambdaS1barS2S2bar2)	0	6	0
lambdaS1barS2S2bar2Bar(muT3GeneralS1S2)	0	6	0
Tr[MF.Bar(MF).Trans(y2).Bar(y2)]	0	−8	0
Tr[MF.Trans(Bar(y2)).y2.Bar(MF)]	0	−8	0
lambda12CrossmS1Sq	0	0	0
muT3GeneralS1SqBar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(muT3GeneralS1Sq)	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambda12mS1Sq	4	4	4
$g2^2$mS2Sq	0	0	0
gY^2mS2Sq	0	0	0
lambdaS2mS2Sq	6	6	6
lambdaS2AdjmS2Sq	0	0	0
muT3GeneralS1SqBar(lambdaT3GeneralQ000211I1)	0	0	0
lambdaT3GeneralQ000211I1Bar(muT3GeneralS1Sq)	0	0	0
mS2SqTr[y2.Trans(Bar(y2))]	3	3	3
Tr[MF.Trans(Bar(y2)).y2.Trans(Bar(MF))]	−6	0	−6
muT3GeneralS2SqBar(lambdaT3GeneralQ000013I1)	0	0	0
lambdaT3GeneralQ000013I1Bar(muT3GeneralS2Sq)	0	0	0
Bar(muT3GeneralS2Sq)Tr[y2.Trans(Bar(y2Conj))]	0	0	0
muT3GeneralS2SqTr[y2Conj.Trans(Bar(y2))]	0	0	0
mS2SqTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[MF.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(MF))]	0	0	0
$-\frac{9g2^2mS2Sq}{2}$	1	1	1
$-\frac{27gY^2mS2Sq}{2}$	1	0	0
$-\frac{3gY^2mS2Sq}{2}$	0	1	1
muT3GeneralS1S2Bar(lambdaS1barS2S2bar2)	0	6	0
lambdaS1barS2S2bar2Bar(muT3GeneralS1S2)	0	6	0
Tr[MF.Bar(MF).Trans(y2).Bar(y2)]	0	−12	0
Tr[MF.Trans(Bar(y2)).y2.Bar(MF)]	0	−12	0
lambda12CrossmS1Sq	0	0	0
muT3GeneralS1SqBar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(muT3GeneralS1Sq)	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambda12mS1Sq	6	6	6
$g2^2$mS2Sq	0	0	0
gY^2mS2Sq	−6	−24	0
lambdaS2mS2Sq	4	4	4
lambdaS2AdjmS2Sq	0	0	0
muT3GeneralS1SqBar(lambdaT3GeneralQ000211I1)	12	0	0
lambdaT3GeneralQ000211I1Bar(muT3GeneralS1Sq)	12	0	0
mS2SqTr[y2.Trans(Bar(y2))]	4	4	4
Tr[MF.Trans(Bar(y2)).y2.Trans(Bar(MF))]	−8	−8	−8
muT3GeneralS2SqBar(lambdaT3GeneralQ000013I1)	0	0	12
lambdaT3GeneralQ000013I1Bar(muT3GeneralS2Sq)	0	0	12
Bar(muT3GeneralS2Sq)Tr[y2.Trans(Bar(y2Conj))]	0	0	8
muT3GeneralS2SqTr[y2Conj.Trans(Bar(y2))]	0	0	8
mS2SqTr[y2Conj.Trans(Bar(y2Conj))]	0	0	4
Tr[MF.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(MF))]	0	0	−8
$-\frac{9g2^2mS2Sq}{2}$	0	0	0
$-\frac{27gY^2mS2Sq}{2}$	0	0	0
$-\frac{3gY^2mS2Sq}{2}$	0	0	0

muT3GeneralS1S2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(muT3GeneralS1S2)	0	0	0
Tr[MF.Bar(MF).Trans(y2).Bar(y2)]	0	0	0
Tr[MF.Trans(Bar(y2)).y2.Bar(MF)]	0	0	0
lambda12CrossmS1Sq	0	0	0
muT3GeneralS1SqBar(lambdaT3GeneralQ000211I2)	0	0	0
lambdaT3GeneralQ000211I2Bar(muT3GeneralS1Sq)	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambda12mS1Sq	6	6	6
$g2^2$mS2Sq	-12	-12	-12
gY^2mS2Sq	-6	-24	0
lambdaS2mS2Sq	8	8	8
lambdaS2AdjmS2Sq	4	4	4
muT3GeneralS1SqBar(lambdaT3GeneralQ000211I1)	12	0	0
lambdaT3GeneralQ000211I1Bar(muT3GeneralS1Sq)	12	0	0
mS2SqTr[y2.Trans(Bar(y2))]	2	2	2
Tr[MF.Trans(Bar(y2)).y2.Trans(Bar(MF))]	-4	-4	-4
muT3GeneralS2SqBar(lambdaT3GeneralQ000013I1)	0	0	20
lambdaT3GeneralQ000013I1Bar(muT3GeneralS2Sq)	0	0	20
Bar(muT3GeneralS2Sq)Tr[y2.Trans(Bar(y2Conj))]	0	0	4
muT3GeneralS2SqTr[y2Conj.Trans(Bar(y2))]	0	0	4
mS2SqTr[y2Conj.Trans(Bar(y2Conj))]	0	0	2
Tr[MF.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(MF))]	0	0	-4
$-\frac{9g2^2mS2Sq}{2}$	0	0	0
$-\frac{27gY^2mS2Sq}{2}$	0	0	0
$-\frac{3gY^2mS2Sq}{2}$	0	0	0
muT3GeneralS1S2Bar(lambdaS1barS2S2bar2)	0	0	0
lambdaS1barS2S2bar2Bar(muT3GeneralS1S2)	0	0	0
Tr[MF.Bar(MF).Trans(y2).Bar(y2)]	0	0	0
Tr[MF.Trans(Bar(y2)).y2.Bar(MF)]	0	0	0
lambda12CrossmS1Sq	4	4	4
muT3GeneralS1SqBar(lambdaT3GeneralQ000211I2)	4	0	0
lambdaT3GeneralQ000211I2Bar(muT3GeneralS1Sq)	4	0	0

$\mathcal{S}$	E ($\alpha = -4$)
lambda12mS1Sq	6
$g2^2$mS2Sq	-12
gY^2mS2Sq	-6
lambdaS2mS2Sq	8
lambdaS2AdjmS2Sq	4
muT3GeneralS1SqBar(lambdaT3GeneralQ000211I1)	0
lambdaT3GeneralQ000211I1Bar(muT3GeneralS1Sq)	0
mS2SqTr[y2.Trans(Bar(y2))]	2
Tr[MF.Trans(Bar(y2)).y2.Trans(Bar(MF))]	-4
muT3GeneralS2SqBar(lambdaT3GeneralQ000013I1)	0
lambdaT3GeneralQ000013I1Bar(muT3GeneralS2Sq)	0
Bar(muT3GeneralS2Sq)Tr[y2.Trans(Bar(y2Conj))]	0
muT3GeneralS2SqTr[y2Conj.Trans(Bar(y2))]	0
mS2SqTr[y2Conj.Trans(Bar(y2Conj))]	0
Tr[MF.Trans(Bar(y2Conj)).y2Conj.Trans(Bar(MF))]	0
$-\frac{9g2^2mS2Sq}{2}$	0
$-\frac{27gY^2mS2Sq}{2}$	0
$-\frac{3gY^2mS2Sq}{2}$	0
muT3GeneralS1S2Bar(lambdaS1barS2S2bar2)	0

lambdaS1barS2S2bar2Bar(muT3GeneralS1S2)	0
Tr[MF.Bar(MF).Trans(y2).Bar(y2)]	0
Tr[MF.Trans(Bar(y2)).y2.Bar(MF)]	0
lambda12CrossmS1Sq	4
muT3GeneralS1SqBar(lambdaT3GeneralQ000211I2)	0
lambdaT3GeneralQ000211I2Bar(muT3GeneralS1Sq)	0

$\beta^{(1)}_{muT3GeneralS1S2}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambdaS1S1bar2S2barmS1Sq	0	0	0
lambdaS1barS2S2bar2mS2Sq	0	0	0
$-\frac{9g2^2 muT3GeneralS1S2}{2}$	0	0	0
$-\frac{3gY^2 muT3GeneralS1S2}{2}$	0	0	0
lambda12muT3GeneralS1S2	0	0	0
$-\frac{3lambda12AdjmuT3GeneralS1S2}{2}$	0	0	0
lambdaS1bar2S2bar2Bar(muT3GeneralS1S2)	0	0	0
muT3GeneralS1S2Tr[y1.Trans(Bar(y1))]	0	0	0
muT3GeneralS1S2Tr[y2.Trans(Bar(y2))]	0	0	0
$\frac{3}{2}$muT3GeneralS1S2Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$muT3GeneralS1S2Tr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambdaS1S1bar2S2barmS1Sq	0	6	0
lambdaS1barS2S2bar2mS2Sq	0	6	0
$-\frac{9g2^2 muT3GeneralS1S2}{2}$	0	1	0
$-\frac{3gY^2 muT3GeneralS1S2}{2}$	0	1	0
lambda12muT3GeneralS1S2	0	2	0
$-\frac{3lambda12AdjmuT3GeneralS1S2}{2}$	0	1	0
lambdaS1bar2S2bar2Bar(muT3GeneralS1S2)	0	12	0
muT3GeneralS1S2Tr[y1.Trans(Bar(y1))]	0	1	0
muT3GeneralS1S2Tr[y2.Trans(Bar(y2))]	0	1	0
$\frac{3}{2}$muT3GeneralS1S2Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$muT3GeneralS1S2Tr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambdaS1S1bar2S2barmS1Sq	0	6	0
lambdaS1barS2S2bar2mS2Sq	0	6	0
$-\frac{9g2^2 muT3GeneralS1S2}{2}$	0	1	0
$-\frac{3gY^2 muT3GeneralS1S2}{2}$	0	1	0
lambda12muT3GeneralS1S2	0	2	0
$-\frac{3lambda12AdjmuT3GeneralS1S2}{2}$	0	1	0
lambdaS1bar2S2bar2Bar(muT3GeneralS1S2)	0	12	0
muT3GeneralS1S2Tr[y1.Trans(Bar(y1))]	0	0	0
muT3GeneralS1S2Tr[y2.Trans(Bar(y2))]	0	0	0
$\frac{3}{2}$muT3GeneralS1S2Tr[y1.Trans(Bar(y1))]	0	1	0
$\frac{3}{2}$muT3GeneralS1S2Tr[y2.Trans(Bar(y2))]	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambdaS1S1bar2S2barmS1Sq	0	0	0
lambdaS1barS2S2bar2mS2Sq	0	0	0

$-\frac{9g2^2muT3GeneralS1S2}{2}$	0	0	0
$-\frac{3gY^2muT3GeneralS1S2}{2}$	0	0	0
lambda12muT3GeneralS1S2	0	0	0
$-\frac{3lambda12AdjmuT3GeneralS1S2}{2}$	0	0	0
lambdaS1bar2S2bar2Bar(muT3GeneralS1S2)	0	0	0
muT3GeneralS1S2Tr[y1.Trans(Bar(y1))]	0	0	0
muT3GeneralS1S2Tr[y2.Trans(Bar(y2))]	0	0	0
$\frac{3}{2}$muT3GeneralS1S2Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$muT3GeneralS1S2Tr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambdaS1S1bar2S2barmS1Sq	0	0	0
lambdaS1barS2S2bar2mS2Sq	0	0	0
$-\frac{9g2^2muT3GeneralS1S2}{2}$	0	0	0
$-\frac{3gY^2muT3GeneralS1S2}{2}$	0	0	0
lambda12muT3GeneralS1S2	0	0	0
$-\frac{3lambda12AdjmuT3GeneralS1S2}{2}$	0	0	0
lambdaS1bar2S2bar2Bar(muT3GeneralS1S2)	0	0	0
muT3GeneralS1S2Tr[y1.Trans(Bar(y1))]	0	0	0
muT3GeneralS1S2Tr[y2.Trans(Bar(y2))]	0	0	0
$\frac{3}{2}$muT3GeneralS1S2Tr[y1.Trans(Bar(y1))]	0	0	0
$\frac{3}{2}$muT3GeneralS1S2Tr[y2.Trans(Bar(y2))]	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
lambdaS1S1bar2S2barmS1Sq	0
lambdaS1barS2S2bar2mS2Sq	0
$-\frac{9g2^2muT3GeneralS1S2}{2}$	0
$-\frac{3gY^2muT3GeneralS1S2}{2}$	0
lambda12muT3GeneralS1S2	0
$-\frac{3lambda12AdjmuT3GeneralS1S2}{2}$	0
lambdaS1bar2S2bar2Bar(muT3GeneralS1S2)	0
muT3GeneralS1S2Tr[y1.Trans(Bar(y1))]	0
muT3GeneralS1S2Tr[y2.Trans(Bar(y2))]	0
$\frac{3}{2}$muT3GeneralS1S2Tr[y1.Trans(Bar(y1))]	0
$\frac{3}{2}$muT3GeneralS1S2Tr[y2.Trans(Bar(y2))]	0

$\beta^{(1)}_{muT3GeneralS1Sq}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambdaT3GeneralQ001300I1mS1Sq	6	0	0
lambdaT3GeneralQ000211I1mS2Sq	6	0	0
lambdaS1muT3GeneralS1Sq	2	0	0
lambdaT3GeneralQ000400I1Bar(muT3GeneralS1Sq)	24	0	0
muT3GeneralS1SqTr[y1.Trans(Bar(y1))]	4	0	0
mS1SqTr[y1Conj.Trans(Bar(y1))]	4	0	0
muT3GeneralS1SqTr[y1Conj.Trans(Bar(y1Conj))]	4	0	0
Tr[MF.Trans(Bar(MF)).Trans(y1Conj).Bar(y1)]	-8	0	0
$g2^2$muT3GeneralS1Sq	0	0	0
lambdaS1AdjmuT3GeneralS1Sq	0	0	0
lambdaT3GeneralQ000211I2mS2Sq	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambdaT3GeneralQ001300I1mS1Sq	0	0	0
lambdaT3GeneralQ000211I1mS2Sq	0	0	0
lambdaS1muT3GeneralS1Sq	0	0	0
lambdaT3GeneralQ000400I1Bar(muT3GeneralS1Sq)	0	0	0
muT3GeneralS1SqTr[y1.Trans(Bar(y1))]	0	0	0
mS1SqTr[y1Conj.Trans(Bar(y1))]	0	0	0
muT3GeneralS1SqTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[MF.Trans(Bar(MF)).Trans(y1Conj).Bar(y1)]	0	0	0
$g2^2$muT3GeneralS1Sq	0	0	0
lambdaS1AdjmuT3GeneralS1Sq	0	0	0
lambdaT3GeneralQ000211I2mS2Sq	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambdaT3GeneralQ001300I1mS1Sq	0	0	0
lambdaT3GeneralQ000211I1mS2Sq	0	0	0
lambdaS1muT3GeneralS1Sq	0	0	0
lambdaT3GeneralQ000400I1Bar(muT3GeneralS1Sq)	0	0	0
muT3GeneralS1SqTr[y1.Trans(Bar(y1))]	0	0	0
mS1SqTr[y1Conj.Trans(Bar(y1))]	0	0	0
muT3GeneralS1SqTr[y1Conj.Trans(Bar(y1Conj))]	0	0	0
Tr[MF.Trans(Bar(MF)).Trans(y1Conj).Bar(y1)]	0	0	0
$g2^2$muT3GeneralS1Sq	0	0	0
lambdaS1AdjmuT3GeneralS1Sq	0	0	0
lambdaT3GeneralQ000211I2mS2Sq	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambdaT3GeneralQ001300I1mS1Sq	10	0	0
lambdaT3GeneralQ000211I1mS2Sq	2	0	0
lambdaS1muT3GeneralS1Sq	2	0	0
lambdaT3GeneralQ000400I1Bar(muT3GeneralS1Sq)	40	0	0
muT3GeneralS1SqTr[y1.Trans(Bar(y1))]	2	0	0
mS1SqTr[y1Conj.Trans(Bar(y1))]	2	0	0
muT3GeneralS1SqTr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
Tr[MF.Trans(Bar(MF)).Trans(y1Conj).Bar(y1)]	-4	0	0
$g2^2$muT3GeneralS1Sq	-12	0	0
lambdaS1AdjmuT3GeneralS1Sq	-4	0	0
lambdaT3GeneralQ000211I2mS2Sq	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambdaT3GeneralQ001300I1mS1Sq	10	0	0
lambdaT3GeneralQ000211I1mS2Sq	6	0	0
lambdaS1muT3GeneralS1Sq	2	0	0
lambdaT3GeneralQ000400I1Bar(muT3GeneralS1Sq)	40	0	0
muT3GeneralS1SqTr[y1.Trans(Bar(y1))]	2	0	0
mS1SqTr[y1Conj.Trans(Bar(y1))]	2	0	0
muT3GeneralS1SqTr[y1Conj.Trans(Bar(y1Conj))]	2	0	0
Tr[MF.Trans(Bar(MF)).Trans(y1Conj).Bar(y1)]	-4	0	0
$g2^2$muT3GeneralS1Sq	-12	0	0
lambdaS1AdjmuT3GeneralS1Sq	-4	0	0
lambdaT3GeneralQ000211I2mS2Sq	2	0	0

$\mathcal{S}$	E ($\alpha = -4$)
lambdaT3GeneralQ001300I1mS1Sq	0
lambdaT3GeneralQ000211I1mS2Sq	0
lambdaS1muT3GeneralS1Sq	0
lambdaT3GeneralQ000400I1Bar(muT3GeneralS1Sq)	0
muT3GeneralS1SqTr[y1.Trans(Bar(y1))]	0
mS1SqTr[y1Conj.Trans(Bar(y1))]	0
muT3GeneralS1SqTr[y1Conj.Trans(Bar(y1Conj))]	0
Tr[MF.Trans(Bar(MF)).Trans(y1Conj).Bar(y1)]	0
g2^{2}muT3GeneralS1Sq	0
lambdaS1AdjmuT3GeneralS1Sq	0
lambdaT3GeneralQ000211I2mS2Sq	0

$\beta^{(1)}_{muT3GeneralS2Sq}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
lambdaT3GeneralQ001102I1mS1Sq	0	2	0
lambdaT3GeneralQ000013I1mS2Sq	0	10	0
g2^{2}muT3GeneralS2Sq	0	-12	0
lambdaS2muT3GeneralS2Sq	0	2	0
lambdaS2AdjmuT3GeneralS2Sq	0	-4	0
lambdaT3GeneralQ000004I1Bar(muT3GeneralS2Sq)	0	40	0
muT3GeneralS2SqTr[y2.Trans(Bar(y2))]	0	2	0
mS2SqTr[y2.Trans(Bar(y2Conj))]	0	2	0
muT3GeneralS2SqTr[y2Conj.Trans(Bar(y2Conj))]	0	2	0
Tr[MF.Trans(Bar(y2Conj)).y2.Trans(Bar(MF))]	0	-4	0
lambdaT3GeneralQ001102I2mS1Sq	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
lambdaT3GeneralQ001102I1mS1Sq	0	0	0
lambdaT3GeneralQ000013I1mS2Sq	0	0	0
g2^{2}muT3GeneralS2Sq	0	0	0
lambdaS2muT3GeneralS2Sq	0	0	0
lambdaS2AdjmuT3GeneralS2Sq	0	0	0
lambdaT3GeneralQ000004I1Bar(muT3GeneralS2Sq)	0	0	0
muT3GeneralS2SqTr[y2.Trans(Bar(y2))]	0	0	0
mS2SqTr[y2.Trans(Bar(y2Conj))]	0	0	0
muT3GeneralS2SqTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[MF.Trans(Bar(y2Conj)).y2.Trans(Bar(MF))]	0	0	0
lambdaT3GeneralQ001102I2mS1Sq	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
lambdaT3GeneralQ001102I1mS1Sq	0	0	0
lambdaT3GeneralQ000013I1mS2Sq	0	0	0
g2^{2}muT3GeneralS2Sq	0	0	0
lambdaS2muT3GeneralS2Sq	0	0	0
lambdaS2AdjmuT3GeneralS2Sq	0	0	0
lambdaT3GeneralQ000004I1Bar(muT3GeneralS2Sq)	0	0	0
muT3GeneralS2SqTr[y2.Trans(Bar(y2))]	0	0	0
mS2SqTr[y2.Trans(Bar(y2Conj))]	0	0	0
muT3GeneralS2SqTr[y2Conj.Trans(Bar(y2Conj))]	0	0	0
Tr[MF.Trans(Bar(y2Conj)).y2.Trans(Bar(MF))]	0	0	0
lambdaT3GeneralQ001102I2mS1Sq	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
lambdaT3GeneralQ001102I1mS1Sq	0	0	6
lambdaT3GeneralQ000013I1mS2Sq	0	0	6
g2^{2}muT3GeneralS2Sq	0	0	0
lambdaS2muT3GeneralS2Sq	0	0	2
lambdaS2AdjmuT3GeneralS2Sq	0	0	0
lambdaT3GeneralQ000004I1Bar(muT3GeneralS2Sq)	0	0	24
muT3GeneralS2SqTr[y2.Trans(Bar(y2))]	0	0	4
mS2SqTr[y2.Trans(Bar(y2Conj))]	0	0	4
muT3GeneralS2SqTr[y2Conj.Trans(Bar(y2Conj))]	0	0	4
Tr[MF.Trans(Bar(y2Conj)).y2.Trans(Bar(MF))]	0	0	-8
lambdaT3GeneralQ001102I2mS1Sq	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
lambdaT3GeneralQ001102I1mS1Sq	0	0	6
lambdaT3GeneralQ000013I1mS2Sq	0	0	10
g2^{2}muT3GeneralS2Sq	0	0	-12
lambdaS2muT3GeneralS2Sq	0	0	2
lambdaS2AdjmuT3GeneralS2Sq	0	0	-4
lambdaT3GeneralQ000004I1Bar(muT3GeneralS2Sq)	0	0	40
muT3GeneralS2SqTr[y2.Trans(Bar(y2))]	0	0	2
mS2SqTr[y2.Trans(Bar(y2Conj))]	0	0	2
muT3GeneralS2SqTr[y2Conj.Trans(Bar(y2Conj))]	0	0	2
Tr[MF.Trans(Bar(y2Conj)).y2.Trans(Bar(MF))]	0	0	-4
lambdaT3GeneralQ001102I2mS1Sq	0	0	2

$\mathcal{S}$	E ($\alpha = -4$)
lambdaT3GeneralQ001102I1mS1Sq	0
lambdaT3GeneralQ000013I1mS2Sq	0
g2^{2}muT3GeneralS2Sq	0
lambdaS2muT3GeneralS2Sq	0
lambdaS2AdjmuT3GeneralS2Sq	0
lambdaT3GeneralQ000004I1Bar(muT3GeneralS2Sq)	0
muT3GeneralS2SqTr[y2.Trans(Bar(y2))]	0
mS2SqTr[y2.Trans(Bar(y2Conj))]	0
muT3GeneralS2SqTr[y2Conj.Trans(Bar(y2Conj))]	0
Tr[MF.Trans(Bar(y2Conj)).y2.Trans(Bar(MF))]	0
lambdaT3GeneralQ001102I2mS1Sq	0

$\beta^{(1)}_{y_1}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$-\frac{9}{2}$g2^{2}Matrix(y1)(gen($i), heavy($j))	1	1	1
$-\frac{3}{2}$gY^{2}Matrix(y1)(gen($i), heavy($j))	1	1	0
Tr[y1.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	2	2	2
Tr[y1Conj.Trans(Bar(y1Conj))]Matrix(y1)(gen($i), heavy($j))	2	0	0
Tr[y1.Trans(Bar(y1Conj))]Matrix(y1Conj)(gen($i), heavy($j))	4	0	0
Matrix(y1, Trans(Bar(y1)), y1)(gen($i), heavy($j))	1	1	1
$\frac{5}{2}$Matrix(y1, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	1	0	0
$\frac{5}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), y1)(gen($i), heavy($j))	1	0	0
$\frac{3}{4}$Matrix(y2, Trans(Bar(y2)), y1)(gen($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y1)(gen($i), heavy($j))	1	1	1
$\frac{3}{4}$Matrix(y2Conj, Trans(Bar(y2Conj)), y1)(gen($i), heavy($j))	0	1	0

$-\frac{15}{2}gY^2Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	1
$\frac{3}{2}Matrix(y1,Trans(Bar(y1)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{1}{2}Matrix(y2,Trans(Bar(y2)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{29}{12}Matrix(y1,Trans(Bar(y1Conj)),y1Conj)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{29}{12}Matrix(y1Conj,Trans(Bar(y1Conj)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{1}{2}Matrix(y2Conj,Trans(Bar(y2Conj)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$-\frac{9}{4}g2^2Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$-\frac{15}{4}gY^2Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$-\frac{3}{4}gY^2Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$Tr[y1.Trans(Bar(y2))]Matrix(y2)(gen(\$i),heavy(\$j))$	0	0	0
$Matrix(y1,Trans(Bar(y2)),y2)(gen(\$i),heavy(\$j))$	0	0	0
$-\frac{33}{4}g2^2Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{3}{2}Tr[y1.Trans(Bar(y1))]Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{5}{4}Matrix(y1,Trans(Bar(y1)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{3}{2}Tr[y1.Trans(Bar(y2))]Matrix(y2)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{1}{2}Matrix(y1,Trans(Bar(y2)),y2)(gen(\$i),heavy(\$j))$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$-\frac{9}{2}g2^2Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$-\frac{3}{2}gY^2Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$Tr[y1.Trans(Bar(y1))]Matrix(y1)(gen(\$i),heavy(\$j))$	1	1	1
$Tr[y1Conj.Trans(Bar(y1Conj))]Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$Tr[y1.Trans(Bar(y1Conj))]Matrix(y1Conj)(gen(\$i),heavy(\$j))$	0	0	0
$Matrix(y1,Trans(Bar(y1)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{5}{2}Matrix(y1,Trans(Bar(y1Conj)),y1Conj)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{5}{2}Matrix(y1Conj,Trans(Bar(y1Conj)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{3}{4}Matrix(y2,Trans(Bar(y2)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{1}{2}Matrix(ye,Trans(Bar(ye)),y1)(gen(\$i),heavy(\$j))$	1	1	1
$\frac{3}{4}Matrix(y2Conj,Trans(Bar(y2Conj)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$-\frac{15}{2}gY^2Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{3}{2}Matrix(y1,Trans(Bar(y1)),y1)(gen(\$i),heavy(\$j))$	1	1	1
$\frac{1}{2}Matrix(y2,Trans(Bar(y2)),y1)(gen(\$i),heavy(\$j))$	1	1	1
$\frac{29}{12}Matrix(y1,Trans(Bar(y1Conj)),y1Conj)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{29}{12}Matrix(y1Conj,Trans(Bar(y1Conj)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{1}{2}Matrix(y2Conj,Trans(Bar(y2Conj)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$-\frac{9}{4}g2^2Matrix(y1)(gen(\$i),heavy(\$j))$	1	1	1
$-\frac{15}{4}gY^2Matrix(y1)(gen(\$i),heavy(\$j))$	1	0	1
$-\frac{3}{4}gY^2Matrix(y1)(gen(\$i),heavy(\$j))$	0	1	0
$Tr[y1.Trans(Bar(y2))]Matrix(y2)(gen(\$i),heavy(\$j))$	0	1	0
$Matrix(y1,Trans(Bar(y2)),y2)(gen(\$i),heavy(\$j))$	0	1	0
$-\frac{33}{4}g2^2Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{3}{2}Tr[y1.Trans(Bar(y1))]Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{5}{4}Matrix(y1,Trans(Bar(y1)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{3}{2}Tr[y1.Trans(Bar(y2))]Matrix(y2)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{1}{2}Matrix(y1,Trans(Bar(y2)),y2)(gen(\$i),heavy(\$j))$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$-\frac{9}{2}g2^2Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$-\frac{3}{2}gY^2Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$Tr[y1.Trans(Bar(y1))]Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$Tr[y1Conj.Trans(Bar(y1Conj))]Matrix(y1)(gen(\$i),heavy(\$j))$	0	0	0
$Tr[y1.Trans(Bar(y1Conj))]Matrix(y1Conj)(gen(\$i),heavy(\$j))$	0	0	0
$Matrix(y1,Trans(Bar(y1)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{5}{2}Matrix(y1,Trans(Bar(y1Conj)),y1Conj)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{5}{2}Matrix(y1Conj,Trans(Bar(y1Conj)),y1)(gen(\$i),heavy(\$j))$	0	0	0
$\frac{3}{4}Matrix(y2,Trans(Bar(y2)),y1)(gen(\$i),heavy(\$j))$	1	1	1

$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y1)(gen($i), heavy($j))	1	1	1
$\frac{3}{4}$Matrix(y2Conj, Trans(Bar(y2Conj)), y1)(gen($i), heavy($j))	0	0	0
$-\frac{15}{2}$gY^{2}Matrix(y1)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Matrix(y1, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), y1)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y1, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y1Conj, Trans(Bar(y1Conj)), y1)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y1)(gen($i), heavy($j))	0	0	0
$-\frac{9}{4}$g2^{2}Matrix(y1)(gen($i), heavy($j))	0	0	0
$-\frac{15}{4}$gY^{2}Matrix(y1)(gen($i), heavy($j))	1	0	1
$-\frac{3}{4}$gY^{2}Matrix(y1)(gen($i), heavy($j))	0	1	0
Tr[y1.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	0
Matrix(y1, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$-\frac{33}{4}$g2^{2}Matrix(y1)(gen($i), heavy($j))	1	1	1
$\frac{3}{2}$Tr[y1.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	1	1	1
$\frac{5}{4}$Matrix(y1, Trans(Bar(y1)), y1)(gen($i), heavy($j))	1	1	1
$\frac{3}{2}$Tr[y1.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	1	0
$\frac{1}{2}$Matrix(y1, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$-\frac{9}{2}$g2^{2}Matrix(y1)(gen($i), heavy($j))	1	1	1
$-\frac{3}{2}$gY^{2}Matrix(y1)(gen($i), heavy($j))	1	0	1
Tr[y1.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	1	1	1
Tr[y1Conj.Trans(Bar(y1Conj))]Matrix(y1)(gen($i), heavy($j))	1	0	0
Tr[y1.Trans(Bar(y1Conj))]Matrix(y1Conj)(gen($i), heavy($j))	2	0	0
Matrix(y1, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y1, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), y1)(gen($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(y2, Trans(Bar(y2)), y1)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y1)(gen($i), heavy($j))	1	1	1
$\frac{3}{4}$Matrix(y2Conj, Trans(Bar(y2Conj)), y1)(gen($i), heavy($j))	0	0	0
$-\frac{15}{2}$gY^{2}Matrix(y1)(gen($i), heavy($j))	0	1	0
$\frac{3}{2}$Matrix(y1, Trans(Bar(y1)), y1)(gen($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), y1)(gen($i), heavy($j))	1	1	1
$\frac{29}{12}$Matrix(y1, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	1	0	0
$\frac{29}{12}$Matrix(y1Conj, Trans(Bar(y1Conj)), y1)(gen($i), heavy($j))	1	0	0
$\frac{1}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y1)(gen($i), heavy($j))	0	0	1
$-\frac{9}{4}$g2^{2}Matrix(y1)(gen($i), heavy($j))	0	0	0
$-\frac{15}{4}$gY^{2}Matrix(y1)(gen($i), heavy($j))	0	0	0
$-\frac{3}{4}$gY^{2}Matrix(y1)(gen($i), heavy($j))	0	0	0
Tr[y1.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	0
Matrix(y1, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$-\frac{33}{4}$g2^{2}Matrix(y1)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Tr[y1.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	0	0	0
$\frac{5}{4}$Matrix(y1, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Tr[y1.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y1, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$-\frac{9}{2}$g2^{2}Matrix(y1)(gen($i), heavy($j))	1	1	1
$-\frac{3}{2}$gY^{2}Matrix(y1)(gen($i), heavy($j))	1	0	1
Tr[y1.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	1	1	1
Tr[y1Conj.Trans(Bar(y1Conj))]Matrix(y1)(gen($i), heavy($j))	1	0	0
Tr[y1.Trans(Bar(y1Conj))]Matrix(y1Conj)(gen($i), heavy($j))	2	0	0
Matrix(y1, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y1, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	0	0	0

$\frac{5}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), y1)(gen($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(y2, Trans(Bar(y2)), y1)(gen($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y1)(gen($i), heavy($j))	1	1	1
$\frac{3}{4}$Matrix(y2Conj, Trans(Bar(y2Conj)), y1)(gen($i), heavy($j))	0	0	1
$-\frac{15}{2}$gY^{2}Matrix(y1)(gen($i), heavy($j))	0	1	0
$\frac{3}{2}$Matrix(y1, Trans(Bar(y1)), y1)(gen($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), y1)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y1, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	1	0	0
$\frac{29}{12}$Matrix(y1Conj, Trans(Bar(y1Conj)), y1)(gen($i), heavy($j))	1	0	0
$\frac{1}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y1)(gen($i), heavy($j))	0	0	0
$-\frac{9}{4}$g2^{2}Matrix(y1)(gen($i), heavy($j))	0	0	0
$-\frac{15}{4}$gY^{2}Matrix(y1)(gen($i), heavy($j))	0	0	0
$-\frac{3}{4}$gY^{2}Matrix(y1)(gen($i), heavy($j))	0	0	0
Tr[y1.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	0
Matrix(y1, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$-\frac{33}{4}$g2^{2}Matrix(y1)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Tr[y1.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	0	0	0
$\frac{5}{4}$Matrix(y1, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Tr[y1.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y1, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$-\frac{9}{2}$g2^{2}Matrix(y1)(gen($i), heavy($j))	1
$-\frac{3}{2}$gY^{2}Matrix(y1)(gen($i), heavy($j))	0
Tr[y1.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	1
Tr[y1Conj.Trans(Bar(y1Conj))]Matrix(y1)(gen($i), heavy($j))	0
Tr[y1.Trans(Bar(y1Conj))]Matrix(y1Conj)(gen($i), heavy($j))	0
Matrix(y1, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0
$\frac{5}{2}$Matrix(y1, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	0
$\frac{5}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), y1)(gen($i), heavy($j))	0
$\frac{3}{4}$Matrix(y2, Trans(Bar(y2)), y1)(gen($i), heavy($j))	1
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y1)(gen($i), heavy($j))	1
$\frac{3}{4}$Matrix(y2Conj, Trans(Bar(y2Conj)), y1)(gen($i), heavy($j))	0
$-\frac{15}{2}$gY^{2}Matrix(y1)(gen($i), heavy($j))	1
$\frac{3}{2}$Matrix(y1, Trans(Bar(y1)), y1)(gen($i), heavy($j))	1
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), y1)(gen($i), heavy($j))	0
$\frac{29}{12}$Matrix(y1, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	0
$\frac{29}{12}$Matrix(y1Conj, Trans(Bar(y1Conj)), y1)(gen($i), heavy($j))	0
$\frac{1}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y1)(gen($i), heavy($j))	0
$-\frac{9}{4}$g2^{2}Matrix(y1)(gen($i), heavy($j))	0
$-\frac{15}{4}$gY^{2}Matrix(y1)(gen($i), heavy($j))	0
$-\frac{3}{4}$gY^{2}Matrix(y1)(gen($i), heavy($j))	0
Tr[y1.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0
Matrix(y1, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0
$-\frac{33}{4}$g2^{2}Matrix(y1)(gen($i), heavy($j))	0
$\frac{3}{2}$Tr[y1.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	0
$\frac{5}{4}$Matrix(y1, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0
$\frac{3}{2}$Tr[y1.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0
$\frac{1}{2}$Matrix(y1, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0

$\beta^{(1)}_{y1Conj}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
Tr[y1Conj.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	4	0	0
$-\frac{9}{2}$g2^{2}Matrix(y1Conj)(gen($i), heavy($j))	1	0	0
$-\frac{3}{2}$gY^{2}Matrix(y1Conj)(gen($i), heavy($j))	1	0	0

$\mathrm{Tr}[\mathrm{y1.Trans(Bar(y1))}]\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	2	0	0
$\mathrm{Tr}[\mathrm{y1Conj.Trans(Bar(y1Conj))}]\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	2	0	0
$\frac{5}{2}\mathrm{Matrix(y1, Trans(Bar(y1)), y1Conj)(gen(\$i), heavy(\$j))}$	1	0	0
$\frac{5}{2}\mathrm{Matrix(y1Conj, Trans(Bar(y1)), y1)(gen(\$i), heavy(\$j))}$	1	0	0
$\mathrm{Matrix(y1Conj, Trans(Bar(y1Conj)), y1Conj)(gen(\$i), heavy(\$j))}$	1	0	0
$\frac{3}{4}\mathrm{Matrix(y2, Trans(Bar(y2)), y1Conj)(gen(\$i), heavy(\$j))}$	1	0	0
$\frac{1}{2}\mathrm{Matrix(ye, Trans(Bar(ye)), y1Conj)(gen(\$i), heavy(\$j))}$	1	0	0
$\frac{29}{12}\mathrm{Matrix(y1, Trans(Bar(y1)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{29}{12}\mathrm{Matrix(y1Conj, Trans(Bar(y1)), y1)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{3}{2}\mathrm{Matrix(y1Conj, Trans(Bar(y1Conj)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{1}{2}\mathrm{Matrix(y2, Trans(Bar(y2)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\mathrm{Tr}[\mathrm{y1Conj.Trans(Bar(y1))}]\mathrm{Matrix(y1)(gen(\$i), heavy(\$j))}$	0	0	0
$-\frac{9}{2}\mathrm{g2}^2\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$-\frac{3}{2}\mathrm{gY}^2\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\mathrm{Tr}[\mathrm{y1.Trans(Bar(y1))}]\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\mathrm{Tr}[\mathrm{y1Conj.Trans(Bar(y1Conj))}]\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{5}{2}\mathrm{Matrix(y1, Trans(Bar(y1)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{5}{2}\mathrm{Matrix(y1Conj, Trans(Bar(y1)), y1)(gen(\$i), heavy(\$j))}$	0	0	0
$\mathrm{Matrix(y1Conj, Trans(Bar(y1Conj)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{3}{4}\mathrm{Matrix(y2, Trans(Bar(y2)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{1}{2}\mathrm{Matrix(ye, Trans(Bar(ye)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{29}{12}\mathrm{Matrix(y1, Trans(Bar(y1)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{29}{12}\mathrm{Matrix(y1Conj, Trans(Bar(y1)), y1)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{3}{2}\mathrm{Matrix(y1Conj, Trans(Bar(y1Conj)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{1}{2}\mathrm{Matrix(y2, Trans(Bar(y2)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\mathrm{Tr}[\mathrm{y1Conj.Trans(Bar(y1))}]\mathrm{Matrix(y1)(gen(\$i), heavy(\$j))}$	0	0	0
$-\frac{9}{2}\mathrm{g2}^2\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$-\frac{3}{2}\mathrm{gY}^2\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\mathrm{Tr}[\mathrm{y1.Trans(Bar(y1))}]\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\mathrm{Tr}[\mathrm{y1Conj.Trans(Bar(y1Conj))}]\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{5}{2}\mathrm{Matrix(y1, Trans(Bar(y1)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{5}{2}\mathrm{Matrix(y1Conj, Trans(Bar(y1)), y1)(gen(\$i), heavy(\$j))}$	0	0	0
$\mathrm{Matrix(y1Conj, Trans(Bar(y1Conj)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{3}{4}\mathrm{Matrix(y2, Trans(Bar(y2)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{1}{2}\mathrm{Matrix(ye, Trans(Bar(ye)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{29}{12}\mathrm{Matrix(y1, Trans(Bar(y1)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{29}{12}\mathrm{Matrix(y1Conj, Trans(Bar(y1)), y1)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{3}{2}\mathrm{Matrix(y1Conj, Trans(Bar(y1Conj)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{1}{2}\mathrm{Matrix(y2, Trans(Bar(y2)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\mathrm{Tr}[\mathrm{y1Conj.Trans(Bar(y1))}]\mathrm{Matrix(y1)(gen(\$i), heavy(\$j))}$	2	0	0
$-\frac{9}{2}\mathrm{g2}^2\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	1	0	0
$-\frac{3}{2}\mathrm{gY}^2\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	1	0	0
$\mathrm{Tr}[\mathrm{y1.Trans(Bar(y1))}]\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	1	0	0
$\mathrm{Tr}[\mathrm{y1Conj.Trans(Bar(y1Conj))}]\mathrm{Matrix(y1Conj)(gen(\$i), heavy(\$j))}$	1	0	0
$\frac{5}{2}\mathrm{Matrix(y1, Trans(Bar(y1)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{5}{2}\mathrm{Matrix(y1Conj, Trans(Bar(y1)), y1)(gen(\$i), heavy(\$j))}$	0	0	0
$\mathrm{Matrix(y1Conj, Trans(Bar(y1Conj)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{3}{4}\mathrm{Matrix(y2, Trans(Bar(y2)), y1Conj)(gen(\$i), heavy(\$j))}$	0	0	0
$\frac{1}{2}\mathrm{Matrix(ye, Trans(Bar(ye)), y1Conj)(gen(\$i), heavy(\$j))}$	1	0	0

$\frac{29}{12}$Matrix(y1, Trans(Bar(y1)), y1Conj)(gen($i), heavy($j))	1	0	0
$\frac{29}{12}$Matrix(y1Conj, Trans(Bar(y1)), y1)(gen($i), heavy($j))	1	0	0
$\frac{3}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	1	0	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), y1Conj)(gen($i), heavy($j))	1	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
Tr[y1Conj.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	2	0	0
$-\frac{9}{2}$g2^{2}Matrix(y1Conj)(gen($i), heavy($j))	1	0	0
$-\frac{3}{2}$gY^{2}Matrix(y1Conj)(gen($i), heavy($j))	1	0	0
Tr[y1.Trans(Bar(y1))]Matrix(y1Conj)(gen($i), heavy($j))	1	0	0
Tr[y1Conj.Trans(Bar(y1Conj))]Matrix(y1Conj)(gen($i), heavy($j))	1	0	0
$\frac{5}{2}$Matrix(y1, Trans(Bar(y1)), y1Conj)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y1Conj, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	0	0
Matrix(y1Conj, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(y2, Trans(Bar(y2)), y1Conj)(gen($i), heavy($j))	1	0	0
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y1Conj)(gen($i), heavy($j))	1	0	0
$\frac{29}{12}$Matrix(y1, Trans(Bar(y1)), y1Conj)(gen($i), heavy($j))	1	0	0
$\frac{29}{12}$Matrix(y1Conj, Trans(Bar(y1)), y1)(gen($i), heavy($j))	1	0	0
$\frac{3}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	1	0	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), y1Conj)(gen($i), heavy($j))	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
Tr[y1Conj.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	0
$-\frac{9}{2}$g2^{2}Matrix(y1Conj)(gen($i), heavy($j))	0
$-\frac{3}{2}$gY^{2}Matrix(y1Conj)(gen($i), heavy($j))	0
Tr[y1.Trans(Bar(y1))]Matrix(y1Conj)(gen($i), heavy($j))	0
Tr[y1Conj.Trans(Bar(y1Conj))]Matrix(y1Conj)(gen($i), heavy($j))	0
$\frac{5}{2}$Matrix(y1, Trans(Bar(y1)), y1Conj)(gen($i), heavy($j))	0
$\frac{5}{2}$Matrix(y1Conj, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0
Matrix(y1Conj, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	0
$\frac{3}{4}$Matrix(y2, Trans(Bar(y2)), y1Conj)(gen($i), heavy($j))	0
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y1Conj)(gen($i), heavy($j))	0
$\frac{29}{12}$Matrix(y1, Trans(Bar(y1)), y1Conj)(gen($i), heavy($j))	0
$\frac{29}{12}$Matrix(y1Conj, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0
$\frac{3}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), y1Conj)(gen($i), heavy($j))	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), y1Conj)(gen($i), heavy($j))	0

$\beta_{y_2}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$-\frac{9}{2}$g2^{2}Matrix(y2)(gen($i), heavy($j))	1	1	1
$-\frac{3}{2}$gY^{2}Matrix(y2)(gen($i), heavy($j))	1	1	0
Tr[y2.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), y2)(gen($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), y2)(gen($i), heavy($j))	1	0	0
$\frac{3}{2}$Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y2)(gen($i), heavy($j))	1	1	1
Tr[y2Conj.Trans(Bar(y2Conj))]Matrix(y2)(gen($i), heavy($j))	0	1	0
Tr[y2.Trans(Bar(y2Conj))]Matrix(y2Conj)(gen($i), heavy($j))	0	2	0
$\frac{29}{12}$Matrix(y2, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	1	0
$\frac{29}{12}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2)(gen($i), heavy($j))	0	1	0
$-\frac{15}{2}$gY^{2}Matrix(y2)(gen($i), heavy($j))	0	0	1
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(y1Conj, Trans(Bar(y1Conj)), y2)(gen($i), heavy($j))	0	0	0
Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0

$\frac{5}{2}$Matrix(y2, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2)(gen($i), heavy($j))	0	0	0
$-\frac{9}{4}$g2^{2}Matrix(y2)(gen($i), heavy($j))	0	0	0
$-\frac{15}{4}$gY^{2}Matrix(y2)(gen($i), heavy($j))	0	0	0
Tr[y2.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	0	0	0
$-\frac{3}{4}$gY^{2}Matrix(y2)(gen($i), heavy($j))	0	0	0
Matrix(y2, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	0	0
$-\frac{33}{4}$g2^{2}Matrix(y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Tr[y2.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	0
$\frac{5}{4}$Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Tr[y2.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$-\frac{9}{2}$g2^{2}Matrix(y2)(gen($i), heavy($j))	0	0	0
$-\frac{3}{2}$gY^{2}Matrix(y2)(gen($i), heavy($j))	0	0	0
Tr[y2.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), y2)(gen($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y2)(gen($i), heavy($j))	1	1	1
Tr[y2Conj.Trans(Bar(y2Conj))]Matrix(y2)(gen($i), heavy($j))	0	0	0
Tr[y2.Trans(Bar(y2Conj))]Matrix(y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y2, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2)(gen($i), heavy($j))	0	0	0
$-\frac{15}{2}$gY^{2}Matrix(y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(y1Conj, Trans(Bar(y1Conj)), y2)(gen($i), heavy($j))	0	0	0
Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y2, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2)(gen($i), heavy($j))	0	0	0
$-\frac{9}{4}$g2^{2}Matrix(y2)(gen($i), heavy($j))	1	1	1
$-\frac{15}{4}$gY^{2}Matrix(y2)(gen($i), heavy($j))	1	0	1
Tr[y2.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	0	1	0
$-\frac{3}{4}$gY^{2}Matrix(y2)(gen($i), heavy($j))	0	1	0
Matrix(y2, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	1	0
$-\frac{33}{4}$g2^{2}Matrix(y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Tr[y2.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	0
$\frac{5}{4}$Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Tr[y2.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$-\frac{9}{2}$g2^{2}Matrix(y2)(gen($i), heavy($j))	0	0	0
$-\frac{3}{2}$gY^{2}Matrix(y2)(gen($i), heavy($j))	0	0	0
Tr[y2.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), y2)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y2)(gen($i), heavy($j))	1	1	1
Tr[y2Conj.Trans(Bar(y2Conj))]Matrix(y2)(gen($i), heavy($j))	0	0	0
Tr[y2.Trans(Bar(y2Conj))]Matrix(y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y2, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2)(gen($i), heavy($j))	0	0	0
$-\frac{15}{2}$gY^{2}Matrix(y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), y2)(gen($i), heavy($j))	1	1	1

$\frac{3}{4}$Matrix(y1Conj, Trans(Bar(y1Conj)), y2)(gen($i), heavy($j))	0	0	0
Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y2, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2)(gen($i), heavy($j))	0	0	0
$-\frac{9}{4}g2^2$Matrix(y2)(gen($i), heavy($j))	0	0	0
$-\frac{15}{4}gY^2$Matrix(y2)(gen($i), heavy($j))	1	0	1
Tr[y2.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	0	0	0
$-\frac{3}{4}gY^2$Matrix(y2)(gen($i), heavy($j))	0	1	0
Matrix(y2, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	0	0
$-\frac{33}{4}g2^2$Matrix(y2)(gen($i), heavy($j))	1	1	1
$\frac{3}{2}$Tr[y2.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	1	1	1
$\frac{5}{4}$Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	1	1	1
$\frac{3}{2}$Tr[y2.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	0	1	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$-\frac{9}{2}g2^2$Matrix(y2)(gen($i), heavy($j))	1	1	1
$-\frac{3}{2}gY^2$Matrix(y2)(gen($i), heavy($j))	1	0	1
Tr[y2.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	2	2	2
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), y2)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y2)(gen($i), heavy($j))	1	1	1
Tr[y2Conj.Trans(Bar(y2Conj))]Matrix(y2)(gen($i), heavy($j))	0	0	2
Tr[y2.Trans(Bar(y2Conj))]Matrix(y2Conj)(gen($i), heavy($j))	0	0	4
$\frac{29}{12}$Matrix(y2, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2)(gen($i), heavy($j))	0	0	0
$-\frac{15}{2}gY^2$Matrix(y2)(gen($i), heavy($j))	0	1	0
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), y2)(gen($i), heavy($j))	1	1	1
$\frac{3}{4}$Matrix(y1Conj, Trans(Bar(y1Conj)), y2)(gen($i), heavy($j))	1	0	0
Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	1	1	1
$\frac{5}{2}$Matrix(y2, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	1
$\frac{5}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2)(gen($i), heavy($j))	0	0	1
$-\frac{9}{4}g2^2$Matrix(y2)(gen($i), heavy($j))	0	0	0
$-\frac{15}{4}gY^2$Matrix(y2)(gen($i), heavy($j))	0	0	0
Tr[y2.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	0	0	0
$-\frac{3}{4}gY^2$Matrix(y2)(gen($i), heavy($j))	0	0	0
Matrix(y2, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	0	0
$-\frac{33}{4}g2^2$Matrix(y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Tr[y2.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	0
$\frac{5}{4}$Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Tr[y2.Trans(Bar(y1))]Matrix(y1)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y1)), y1)(gen($i), heavy($j))	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$-\frac{9}{2}g2^2$Matrix(y2)(gen($i), heavy($j))	1	1	1
$-\frac{3}{2}gY^2$Matrix(y2)(gen($i), heavy($j))	1	0	1
Tr[y2.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), y2)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Matrix(y2, Trans(Bar(y2)), y2)(gen($i), heavy($j))	1	1	1
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y2)(gen($i), heavy($j))	1	1	1
Tr[y2Conj.Trans(Bar(y2Conj))]Matrix(y2)(gen($i), heavy($j))	0	0	1
Tr[y2.Trans(Bar(y2Conj))]Matrix(y2Conj)(gen($i), heavy($j))	0	0	2
$\frac{29}{12}$Matrix(y2, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	1
$\frac{29}{12}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2)(gen($i), heavy($j))	0	0	1

$-\frac{15}{2}\mathrm{gY}^2\mathrm{Matrix(y2)(gen(\$i),heavy(\$j))}$	0	1	0
$\frac{3}{4}\mathrm{Matrix(y1,Trans(Bar(y1)),y2)(gen(\$i),heavy(\$j))}$	1	1	1
$\frac{3}{4}\mathrm{Matrix(y1Conj,Trans(Bar(y1Conj)),y2)(gen(\$i),heavy(\$j))}$	1	0	0
$\mathrm{Matrix(y2,Trans(Bar(y2)),y2)(gen(\$i),heavy(\$j))}$	0	0	0
$\frac{5}{2}\mathrm{Matrix(y2,Trans(Bar(y2Conj)),y2Conj)(gen(\$i),heavy(\$j))}$	0	0	0
$\frac{5}{2}\mathrm{Matrix(y2Conj,Trans(Bar(y2Conj)),y2)(gen(\$i),heavy(\$j))}$	0	0	0
$-\frac{9}{4}\mathrm{g2}^2\mathrm{Matrix(y2)(gen(\$i),heavy(\$j))}$	0	0	0
$-\frac{15}{4}\mathrm{gY}^2\mathrm{Matrix(y2)(gen(\$i),heavy(\$j))}$	0	0	0
$\mathrm{Tr[y2.Trans(Bar(y1))]Matrix(y1)(gen(\$i),heavy(\$j))}$	0	0	0
$-\frac{3}{4}\mathrm{gY}^2\mathrm{Matrix(y2)(gen(\$i),heavy(\$j))}$	0	0	0
$\mathrm{Matrix(y2,Trans(Bar(y1)),y1)(gen(\$i),heavy(\$j))}$	0	0	0
$-\frac{33}{4}\mathrm{g2}^2\mathrm{Matrix(y2)(gen(\$i),heavy(\$j))}$	0	0	0
$\frac{3}{2}\mathrm{Tr[y2.Trans(Bar(y2))]Matrix(y2)(gen(\$i),heavy(\$j))}$	0	0	0
$\frac{5}{4}\mathrm{Matrix(y2,Trans(Bar(y2)),y2)(gen(\$i),heavy(\$j))}$	0	0	0
$\frac{3}{2}\mathrm{Tr[y2.Trans(Bar(y1))]Matrix(y1)(gen(\$i),heavy(\$j))}$	0	0	0
$\frac{1}{2}\mathrm{Matrix(y2,Trans(Bar(y1)),y1)(gen(\$i),heavy(\$j))}$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$-\frac{9}{2}\mathrm{g2}^2\mathrm{Matrix(y2)(gen(\$i),heavy(\$j))}$	1
$-\frac{3}{2}\mathrm{gY}^2\mathrm{Matrix(y2)(gen(\$i),heavy(\$j))}$	0
$\mathrm{Tr[y2.Trans(Bar(y2))]Matrix(y2)(gen(\$i),heavy(\$j))}$	1
$\frac{1}{2}\mathrm{Matrix(y1,Trans(Bar(y1)),y2)(gen(\$i),heavy(\$j))}$	0
$\frac{1}{2}\mathrm{Matrix(y1Conj,Trans(Bar(y1Conj)),y2)(gen(\$i),heavy(\$j))}$	0
$\frac{3}{2}\mathrm{Matrix(y2,Trans(Bar(y2)),y2)(gen(\$i),heavy(\$j))}$	1
$\frac{1}{2}\mathrm{Matrix(ye,Trans(Bar(ye)),y2)(gen(\$i),heavy(\$j))}$	1
$\mathrm{Tr[y2Conj.Trans(Bar(y2Conj))]Matrix(y2)(gen(\$i),heavy(\$j))}$	0
$\mathrm{Tr[y2.Trans(Bar(y2Conj))]Matrix(y2Conj)(gen(\$i),heavy(\$j))}$	0
$\frac{29}{12}\mathrm{Matrix(y2,Trans(Bar(y2Conj)),y2Conj)(gen(\$i),heavy(\$j))}$	0
$\frac{29}{12}\mathrm{Matrix(y2Conj,Trans(Bar(y2Conj)),y2)(gen(\$i),heavy(\$j))}$	0
$-\frac{15}{2}\mathrm{gY}^2\mathrm{Matrix(y2)(gen(\$i),heavy(\$j))}$	1
$\frac{3}{4}\mathrm{Matrix(y1,Trans(Bar(y1)),y2)(gen(\$i),heavy(\$j))}$	1
$\frac{3}{4}\mathrm{Matrix(y1Conj,Trans(Bar(y1Conj)),y2)(gen(\$i),heavy(\$j))}$	0
$\mathrm{Matrix(y2,Trans(Bar(y2)),y2)(gen(\$i),heavy(\$j))}$	0
$\frac{5}{2}\mathrm{Matrix(y2,Trans(Bar(y2Conj)),y2Conj)(gen(\$i),heavy(\$j))}$	0
$\frac{5}{2}\mathrm{Matrix(y2Conj,Trans(Bar(y2Conj)),y2)(gen(\$i),heavy(\$j))}$	0
$-\frac{9}{4}\mathrm{g2}^2\mathrm{Matrix(y2)(gen(\$i),heavy(\$j))}$	0
$-\frac{15}{4}\mathrm{gY}^2\mathrm{Matrix(y2)(gen(\$i),heavy(\$j))}$	0
$\mathrm{Tr[y2.Trans(Bar(y1))]Matrix(y1)(gen(\$i),heavy(\$j))}$	0
$-\frac{3}{4}\mathrm{gY}^2\mathrm{Matrix(y2)(gen(\$i),heavy(\$j))}$	0
$\mathrm{Matrix(y2,Trans(Bar(y1)),y1)(gen(\$i),heavy(\$j))}$	0
$-\frac{33}{4}\mathrm{g2}^2\mathrm{Matrix(y2)(gen(\$i),heavy(\$j))}$	0
$\frac{3}{2}\mathrm{Tr[y2.Trans(Bar(y2))]Matrix(y2)(gen(\$i),heavy(\$j))}$	0
$\frac{5}{4}\mathrm{Matrix(y2,Trans(Bar(y2)),y2)(gen(\$i),heavy(\$j))}$	0
$\frac{3}{2}\mathrm{Tr[y2.Trans(Bar(y1))]Matrix(y1)(gen(\$i),heavy(\$j))}$	0
$\frac{1}{2}\mathrm{Matrix(y2,Trans(Bar(y1)),y1)(gen(\$i),heavy(\$j))}$	0

$\beta^{(1)}_{y2Conj}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\mathrm{Tr[y2Conj.Trans(Bar(y2))]Matrix(y2)(gen(\$i),heavy(\$j))}$	0	2	0
$-\frac{9}{2}\mathrm{g2}^2\mathrm{Matrix(y2Conj)(gen(\$i),heavy(\$j))}$	0	1	0
$-\frac{3}{2}\mathrm{gY}^2\mathrm{Matrix(y2Conj)(gen(\$i),heavy(\$j))}$	0	1	0
$\mathrm{Tr[y2.Trans(Bar(y2))]Matrix(y2Conj)(gen(\$i),heavy(\$j))}$	0	1	0
$\mathrm{Tr[y2Conj.Trans(Bar(y2Conj))]Matrix(y2Conj)(gen(\$i),heavy(\$j))}$	0	1	0
$\frac{1}{2}\mathrm{Matrix(y1,Trans(Bar(y1)),y2Conj)(gen(\$i),heavy(\$j))}$	0	1	0
$\frac{29}{12}\mathrm{Matrix(y2,Trans(Bar(y2)),y2Conj)(gen(\$i),heavy(\$j))}$	0	1	0

$\frac{29}{12}$Matrix(y2Conj, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	1	0
$\frac{3}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	1	0
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y2Conj)(gen($i), heavy($j))	0	1	0
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y2, Trans(Bar(y2)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y2Conj, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
Matrix(y2Conj, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
Tr[y2Conj.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	0
$-\frac{9}{2}$g2^{2}Matrix(y2Conj)(gen($i), heavy($j))	0	0	0
$-\frac{3}{2}$gY^{2}Matrix(y2Conj)(gen($i), heavy($j))	0	0	0
Tr[y2.Trans(Bar(y2))]Matrix(y2Conj)(gen($i), heavy($j))	0	0	0
Tr[y2Conj.Trans(Bar(y2Conj))]Matrix(y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y2, Trans(Bar(y2)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y2Conj, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y2, Trans(Bar(y2)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y2Conj, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
Matrix(y2Conj, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
Tr[y2Conj.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	0
$-\frac{9}{2}$g2^{2}Matrix(y2Conj)(gen($i), heavy($j))	0	0	0
$-\frac{3}{2}$gY^{2}Matrix(y2Conj)(gen($i), heavy($j))	0	0	0
Tr[y2.Trans(Bar(y2))]Matrix(y2Conj)(gen($i), heavy($j))	0	0	0
Tr[y2Conj.Trans(Bar(y2Conj))]Matrix(y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y2, Trans(Bar(y2)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y2Conj, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y2, Trans(Bar(y2)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y2Conj, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
Matrix(y2Conj, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
Tr[y2Conj.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	4
$-\frac{9}{2}$g2^{2}Matrix(y2Conj)(gen($i), heavy($j))	0	0	1
$-\frac{3}{2}$gY^{2}Matrix(y2Conj)(gen($i), heavy($j))	0	0	1
Tr[y2.Trans(Bar(y2))]Matrix(y2Conj)(gen($i), heavy($j))	0	0	2
Tr[y2Conj.Trans(Bar(y2Conj))]Matrix(y2Conj)(gen($i), heavy($j))	0	0	2
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y2, Trans(Bar(y2)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y2Conj, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
$\frac{3}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y2Conj)(gen($i), heavy($j))	0	0	1
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), y2Conj)(gen($i), heavy($j))	0	0	1
$\frac{5}{2}$Matrix(y2, Trans(Bar(y2)), y2Conj)(gen($i), heavy($j))	0	0	1
$\frac{5}{2}$Matrix(y2Conj, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	1

Matrix(y2Conj, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	1

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
Tr[y2Conj.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0	0	2
$-\frac{9}{2}$g2^{2}Matrix(y2Conj)(gen($i), heavy($j))	0	0	1
$-\frac{3}{2}$gY^{2}Matrix(y2Conj)(gen($i), heavy($j))	0	0	1
Tr[y2.Trans(Bar(y2))]Matrix(y2Conj)(gen($i), heavy($j))	0	0	1
Tr[y2Conj.Trans(Bar(y2Conj))]Matrix(y2Conj)(gen($i), heavy($j))	0	0	1
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{29}{12}$Matrix(y2, Trans(Bar(y2)), y2Conj)(gen($i), heavy($j))	0	0	1
$\frac{29}{12}$Matrix(y2Conj, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	1
$\frac{3}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	1
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y2Conj)(gen($i), heavy($j))	0	0	1
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), y2Conj)(gen($i), heavy($j))	0	0	1
$\frac{5}{2}$Matrix(y2, Trans(Bar(y2)), y2Conj)(gen($i), heavy($j))	0	0	0
$\frac{5}{2}$Matrix(y2Conj, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0	0	0
Matrix(y2Conj, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
Tr[y2Conj.Trans(Bar(y2))]Matrix(y2)(gen($i), heavy($j))	0
$-\frac{9}{2}$g2^{2}Matrix(y2Conj)(gen($i), heavy($j))	0
$-\frac{3}{2}$gY^{2}Matrix(y2Conj)(gen($i), heavy($j))	0
Tr[y2.Trans(Bar(y2))]Matrix(y2Conj)(gen($i), heavy($j))	0
Tr[y2Conj.Trans(Bar(y2Conj))]Matrix(y2Conj)(gen($i), heavy($j))	0
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), y2Conj)(gen($i), heavy($j))	0
$\frac{29}{12}$Matrix(y2, Trans(Bar(y2)), y2Conj)(gen($i), heavy($j))	0
$\frac{29}{12}$Matrix(y2Conj, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0
$\frac{3}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0
$\frac{1}{2}$Matrix(ye, Trans(Bar(ye)), y2Conj)(gen($i), heavy($j))	0
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), y2Conj)(gen($i), heavy($j))	0
$\frac{5}{2}$Matrix(y2, Trans(Bar(y2)), y2Conj)(gen($i), heavy($j))	0
$\frac{5}{2}$Matrix(y2Conj, Trans(Bar(y2)), y2)(gen($i), heavy($j))	0
Matrix(y2Conj, Trans(Bar(y2Conj)), y2Conj)(gen($i), heavy($j))	0

$\beta^{(1)}_{Y_d}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$-\frac{9}{4}$g2^{2}Matrix(yd)(gen($i), gen($j))	1	1	1
g3^{2}Matrix(yd)(gen($i), gen($j))	-8	-8	-8
$-\frac{5}{12}$gY^{2}Matrix(yd)(gen($i), gen($j))	1	1	1
Tr[yd.Trans(Bar(yd))]Matrix(yd)(gen($i), gen($j))	3	3	3
Tr[ye.Trans(Bar(ye))]Matrix(yd)(gen($i), gen($j))	1	1	1
Tr[yu.Trans(Bar(yu))]Matrix(yd)(gen($i), gen($j))	3	3	3
$\frac{3}{2}$Matrix(yd, Trans(Bar(yd)), yd)(gen($i), gen($j))	1	1	1
$-\frac{3}{2}$Matrix(yu, Trans(Bar(yu)), yd)(gen($i), gen($j))	1	1	1

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$-\frac{9}{4}$g2^{2}Matrix(yd)(gen($i), gen($j))	1	1	1
g3^{2}Matrix(yd)(gen($i), gen($j))	-8	-8	-8
$-\frac{5}{12}$gY^{2}Matrix(yd)(gen($i), gen($j))	1	1	1
Tr[yd.Trans(Bar(yd))]Matrix(yd)(gen($i), gen($j))	3	3	3
Tr[ye.Trans(Bar(ye))]Matrix(yd)(gen($i), gen($j))	1	1	1
Tr[yu.Trans(Bar(yu))]Matrix(yd)(gen($i), gen($j))	3	3	3

$\frac{3}{2}$Matrix(yd, Trans(Bar(yd)), yd)(gen($i), gen($j))	1	1	1
$-\frac{3}{2}$Matrix(yu, Trans(Bar(yu)), yd)(gen($i), gen($j))	1	1	1

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$-\frac{9}{4}$g2^{2}Matrix(yd)(gen($i), gen($j))	1	1	1
g3^{2}Matrix(yd)(gen($i), gen($j))	-8	-8	-8
$-\frac{5}{12}$gY^{2}Matrix(yd)(gen($i), gen($j))	1	1	1
Tr[yd.Trans(Bar(yd))]Matrix(yd)(gen($i), gen($j))	3	3	3
Tr[ye.Trans(Bar(ye))]Matrix(yd)(gen($i), gen($j))	1	1	1
Tr[yu.Trans(Bar(yu))]Matrix(yd)(gen($i), gen($j))	3	3	3
$\frac{3}{2}$Matrix(yd, Trans(Bar(yd)), yd)(gen($i), gen($j))	1	1	1
$-\frac{3}{2}$Matrix(yu, Trans(Bar(yu)), yd)(gen($i), gen($j))	1	1	1

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$-\frac{9}{4}$g2^{2}Matrix(yd)(gen($i), gen($j))	1	1	1
g3^{2}Matrix(yd)(gen($i), gen($j))	-8	-8	-8
$-\frac{5}{12}$gY^{2}Matrix(yd)(gen($i), gen($j))	1	1	1
Tr[yd.Trans(Bar(yd))]Matrix(yd)(gen($i), gen($j))	3	3	3
Tr[ye.Trans(Bar(ye))]Matrix(yd)(gen($i), gen($j))	1	1	1
Tr[yu.Trans(Bar(yu))]Matrix(yd)(gen($i), gen($j))	3	3	3
$\frac{3}{2}$Matrix(yd, Trans(Bar(yd)), yd)(gen($i), gen($j))	1	1	1
$-\frac{3}{2}$Matrix(yu, Trans(Bar(yu)), yd)(gen($i), gen($j))	1	1	1

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$-\frac{9}{4}$g2^{2}Matrix(yd)(gen($i), gen($j))	1	1	1
g3^{2}Matrix(yd)(gen($i), gen($j))	-8	-8	-8
$-\frac{5}{12}$gY^{2}Matrix(yd)(gen($i), gen($j))	1	1	1
Tr[yd.Trans(Bar(yd))]Matrix(yd)(gen($i), gen($j))	3	3	3
Tr[ye.Trans(Bar(ye))]Matrix(yd)(gen($i), gen($j))	1	1	1
Tr[yu.Trans(Bar(yu))]Matrix(yd)(gen($i), gen($j))	3	3	3
$\frac{3}{2}$Matrix(yd, Trans(Bar(yd)), yd)(gen($i), gen($j))	1	1	1
$-\frac{3}{2}$Matrix(yu, Trans(Bar(yu)), yd)(gen($i), gen($j))	1	1	1

$\mathcal{S}$	E ($\alpha = -4$)
$-\frac{9}{4}$g2^{2}Matrix(yd)(gen($i), gen($j))	1
g3^{2}Matrix(yd)(gen($i), gen($j))	-8
$-\frac{5}{12}$gY^{2}Matrix(yd)(gen($i), gen($j))	1
Tr[yd.Trans(Bar(yd))]Matrix(yd)(gen($i), gen($j))	3
Tr[ye.Trans(Bar(ye))]Matrix(yd)(gen($i), gen($j))	1
Tr[yu.Trans(Bar(yu))]Matrix(yd)(gen($i), gen($j))	3
$\frac{3}{2}$Matrix(yd, Trans(Bar(yd)), yd)(gen($i), gen($j))	1
$-\frac{3}{2}$Matrix(yu, Trans(Bar(yu)), yd)(gen($i), gen($j))	1

$\beta^{(1)}_{Y_e}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$-\frac{9}{4}$g2^{2}Matrix(ye)(gen($i), gen($j))	1	1	1
$-\frac{15}{4}$gY^{2}Matrix(ye)(gen($i), gen($j))	1	1	1
Tr[yd.Trans(Bar(yd))]Matrix(ye)(gen($i), gen($j))	3	3	3
Tr[ye.Trans(Bar(ye))]Matrix(ye)(gen($i), gen($j))	1	1	1
Tr[yu.Trans(Bar(yu))]Matrix(ye)(gen($i), gen($j))	3	3	3

$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), ye)(gen($i), gen($j))	1	1	1
$\frac{1}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), ye)(gen($i), gen($j))	1	0	0
$\frac{3}{4}$Matrix(y2, Trans(Bar(y2)), ye)(gen($i), gen($j))	1	1	1
$\frac{3}{2}$Matrix(ye, Trans(Bar(ye)), ye)(gen($i), gen($j))	1	1	1
$\frac{3}{4}$Matrix(y2Conj, Trans(Bar(y2Conj)), ye)(gen($i), gen($j))	0	1	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), ye)(gen($i), gen($j))	0	0	0
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), ye)(gen($i), gen($j))	0	0	0
$\frac{3}{4}$Matrix(y1Conj, Trans(Bar(y1Conj)), ye)(gen($i), gen($j))	0	0	0
$\frac{1}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), ye)(gen($i), gen($j))	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$-\frac{9}{4}$g2^{2}Matrix(ye)(gen($i), gen($j))	1	1	1
$-\frac{15}{4}$gY^{2}Matrix(ye)(gen($i), gen($j))	1	1	1
Tr[yd.Trans(Bar(yd))]Matrix(ye)(gen($i), gen($j))	3	3	3
Tr[ye.Trans(Bar(ye))]Matrix(ye)(gen($i), gen($j))	1	1	1
Tr[yu.Trans(Bar(yu))]Matrix(ye)(gen($i), gen($j))	3	3	3
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), ye)(gen($i), gen($j))	1	1	1
$\frac{1}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), ye)(gen($i), gen($j))	0	0	0
$\frac{3}{4}$Matrix(y2, Trans(Bar(y2)), ye)(gen($i), gen($j))	0	0	0
$\frac{3}{2}$Matrix(ye, Trans(Bar(ye)), ye)(gen($i), gen($j))	1	1	1
$\frac{3}{4}$Matrix(y2Conj, Trans(Bar(y2Conj)), ye)(gen($i), gen($j))	0	0	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), ye)(gen($i), gen($j))	1	1	1
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), ye)(gen($i), gen($j))	0	0	0
$\frac{3}{4}$Matrix(y1Conj, Trans(Bar(y1Conj)), ye)(gen($i), gen($j))	0	0	0
$\frac{1}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), ye)(gen($i), gen($j))	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$-\frac{9}{4}$g2^{2}Matrix(ye)(gen($i), gen($j))	1	1	1
$-\frac{15}{4}$gY^{2}Matrix(ye)(gen($i), gen($j))	1	1	1
Tr[yd.Trans(Bar(yd))]Matrix(ye)(gen($i), gen($j))	3	3	3
Tr[ye.Trans(Bar(ye))]Matrix(ye)(gen($i), gen($j))	1	1	1
Tr[yu.Trans(Bar(yu))]Matrix(ye)(gen($i), gen($j))	3	3	3
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), ye)(gen($i), gen($j))	0	0	0
$\frac{1}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), ye)(gen($i), gen($j))	0	0	0
$\frac{3}{4}$Matrix(y2, Trans(Bar(y2)), ye)(gen($i), gen($j))	1	1	1
$\frac{3}{2}$Matrix(ye, Trans(Bar(ye)), ye)(gen($i), gen($j))	1	1	1
$\frac{3}{4}$Matrix(y2Conj, Trans(Bar(y2Conj)), ye)(gen($i), gen($j))	0	0	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), ye)(gen($i), gen($j))	0	0	0
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), ye)(gen($i), gen($j))	1	1	1
$\frac{3}{4}$Matrix(y1Conj, Trans(Bar(y1Conj)), ye)(gen($i), gen($j))	0	0	0
$\frac{1}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), ye)(gen($i), gen($j))	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$-\frac{9}{4}$g2^{2}Matrix(ye)(gen($i), gen($j))	1	1	1
$-\frac{15}{4}$gY^{2}Matrix(ye)(gen($i), gen($j))	1	1	1
Tr[yd.Trans(Bar(yd))]Matrix(ye)(gen($i), gen($j))	3	3	3
Tr[ye.Trans(Bar(ye))]Matrix(ye)(gen($i), gen($j))	1	1	1
Tr[yu.Trans(Bar(yu))]Matrix(ye)(gen($i), gen($j))	3	3	3
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), ye)(gen($i), gen($j))	0	0	0
$\frac{1}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), ye)(gen($i), gen($j))	0	0	0
$\frac{3}{4}$Matrix(y2, Trans(Bar(y2)), ye)(gen($i), gen($j))	0	0	0
$\frac{3}{2}$Matrix(ye, Trans(Bar(ye)), ye)(gen($i), gen($j))	1	1	1
$\frac{3}{4}$Matrix(y2Conj, Trans(Bar(y2Conj)), ye)(gen($i), gen($j))	0	0	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), ye)(gen($i), gen($j))	1	1	1
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), ye)(gen($i), gen($j))	1	1	1

$\frac{3}{4}$Matrix(y1Conj, Trans(Bar(y1Conj)), ye)(gen($i), gen($j))	1	0	0
$\frac{1}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), ye)(gen($i), gen($j))	0	0	1

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$-\frac{9}{4}$g2^{2}Matrix(ye)(gen($i), gen($j))	1	1	1
$-\frac{15}{4}$gY^{2}Matrix(ye)(gen($i), gen($j))	1	1	1
Tr[yd.Trans(Bar(yd))]Matrix(ye)(gen($i), gen($j))	3	3	3
Tr[ye.Trans(Bar(ye))]Matrix(ye)(gen($i), gen($j))	1	1	1
Tr[yu.Trans(Bar(yu))]Matrix(ye)(gen($i), gen($j))	3	3	3
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), ye)(gen($i), gen($j))	0	0	0
$\frac{1}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), ye)(gen($i), gen($j))	0	0	0
$\frac{3}{4}$Matrix(y2, Trans(Bar(y2)), ye)(gen($i), gen($j))	1	1	1
$\frac{3}{2}$Matrix(ye, Trans(Bar(ye)), ye)(gen($i), gen($j))	1	1	1
$\frac{3}{4}$Matrix(y2Conj, Trans(Bar(y2Conj)), ye)(gen($i), gen($j))	0	0	1
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), ye)(gen($i), gen($j))	0	0	0
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), ye)(gen($i), gen($j))	1	1	1
$\frac{3}{4}$Matrix(y1Conj, Trans(Bar(y1Conj)), ye)(gen($i), gen($j))	1	0	0
$\frac{1}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), ye)(gen($i), gen($j))	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$-\frac{9}{4}$g2^{2}Matrix(ye)(gen($i), gen($j))	1
$-\frac{15}{4}$gY^{2}Matrix(ye)(gen($i), gen($j))	1
Tr[yd.Trans(Bar(yd))]Matrix(ye)(gen($i), gen($j))	3
Tr[ye.Trans(Bar(ye))]Matrix(ye)(gen($i), gen($j))	1
Tr[yu.Trans(Bar(yu))]Matrix(ye)(gen($i), gen($j))	3
$\frac{1}{2}$Matrix(y1, Trans(Bar(y1)), ye)(gen($i), gen($j))	0
$\frac{1}{2}$Matrix(y1Conj, Trans(Bar(y1Conj)), ye)(gen($i), gen($j))	0
$\frac{3}{4}$Matrix(y2, Trans(Bar(y2)), ye)(gen($i), gen($j))	1
$\frac{3}{2}$Matrix(ye, Trans(Bar(ye)), ye)(gen($i), gen($j))	1
$\frac{3}{4}$Matrix(y2Conj, Trans(Bar(y2Conj)), ye)(gen($i), gen($j))	0
$\frac{1}{2}$Matrix(y2, Trans(Bar(y2)), ye)(gen($i), gen($j))	0
$\frac{3}{4}$Matrix(y1, Trans(Bar(y1)), ye)(gen($i), gen($j))	1
$\frac{3}{4}$Matrix(y1Conj, Trans(Bar(y1Conj)), ye)(gen($i), gen($j))	0
$\frac{1}{2}$Matrix(y2Conj, Trans(Bar(y2Conj)), ye)(gen($i), gen($j))	0

$\beta^{(1)}_{Y_u}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$-\frac{9}{4}$g2^{2}Matrix(yu)(gen($i), gen($j))	1	1	1
g3^{2}Matrix(yu)(gen($i), gen($j))	-8	-8	-8
$-\frac{17}{12}$gY^{2}Matrix(yu)(gen($i), gen($j))	1	1	1
Tr[yd.Trans(Bar(yd))]Matrix(yu)(gen($i), gen($j))	3	3	3
Tr[ye.Trans(Bar(ye))]Matrix(yu)(gen($i), gen($j))	1	1	1
Tr[yu.Trans(Bar(yu))]Matrix(yu)(gen($i), gen($j))	3	3	3
$-\frac{3}{2}$Matrix(yd, Trans(Bar(yd)), yu)(gen($i), gen($j))	1	1	1
$\frac{3}{2}$Matrix(yu, Trans(Bar(yu)), yu)(gen($i), gen($j))	1	1	1

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$-\frac{9}{4}$g2^{2}Matrix(yu)(gen($i), gen($j))	1	1	1
g3^{2}Matrix(yu)(gen($i), gen($j))	-8	-8	-8
$-\frac{17}{12}$gY^{2}Matrix(yu)(gen($i), gen($j))	1	1	1
Tr[yd.Trans(Bar(yd))]Matrix(yu)(gen($i), gen($j))	3	3	3
Tr[ye.Trans(Bar(ye))]Matrix(yu)(gen($i), gen($j))	1	1	1

$\mathrm{Tr}[\mathrm{yu.Trans}(\mathrm{Bar}(\mathrm{yu}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	3	3	3
$-\frac{3}{2}\mathrm{Matrix}(\mathrm{yd},\mathrm{Trans}(\mathrm{Bar}(\mathrm{yd})),\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\frac{3}{2}\mathrm{Matrix}(\mathrm{yu},\mathrm{Trans}(\mathrm{Bar}(\mathrm{yu})),\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$-\frac{9}{4}\mathrm{g2}^2\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\mathrm{g3}^2\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	-8	-8	-8
$-\frac{17}{12}\mathrm{gY}^2\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\mathrm{Tr}[\mathrm{yd.Trans}(\mathrm{Bar}(\mathrm{yd}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	3	3	3
$\mathrm{Tr}[\mathrm{ye.Trans}(\mathrm{Bar}(\mathrm{ye}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\mathrm{Tr}[\mathrm{yu.Trans}(\mathrm{Bar}(\mathrm{yu}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	3	3	3
$-\frac{3}{2}\mathrm{Matrix}(\mathrm{yd},\mathrm{Trans}(\mathrm{Bar}(\mathrm{yd})),\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\frac{3}{2}\mathrm{Matrix}(\mathrm{yu},\mathrm{Trans}(\mathrm{Bar}(\mathrm{yu})),\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$-\frac{9}{4}\mathrm{g2}^2\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\mathrm{g3}^2\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	-8	-8	-8
$-\frac{17}{12}\mathrm{gY}^2\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\mathrm{Tr}[\mathrm{yd.Trans}(\mathrm{Bar}(\mathrm{yd}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	3	3	3
$\mathrm{Tr}[\mathrm{ye.Trans}(\mathrm{Bar}(\mathrm{ye}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\mathrm{Tr}[\mathrm{yu.Trans}(\mathrm{Bar}(\mathrm{yu}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	3	3	3
$-\frac{3}{2}\mathrm{Matrix}(\mathrm{yd},\mathrm{Trans}(\mathrm{Bar}(\mathrm{yd})),\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\frac{3}{2}\mathrm{Matrix}(\mathrm{yu},\mathrm{Trans}(\mathrm{Bar}(\mathrm{yu})),\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$-\frac{9}{4}\mathrm{g2}^2\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\mathrm{g3}^2\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	-8	-8	-8
$-\frac{17}{12}\mathrm{gY}^2\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\mathrm{Tr}[\mathrm{yd.Trans}(\mathrm{Bar}(\mathrm{yd}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	3	3	3
$\mathrm{Tr}[\mathrm{ye.Trans}(\mathrm{Bar}(\mathrm{ye}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\mathrm{Tr}[\mathrm{yu.Trans}(\mathrm{Bar}(\mathrm{yu}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	3	3	3
$-\frac{3}{2}\mathrm{Matrix}(\mathrm{yd},\mathrm{Trans}(\mathrm{Bar}(\mathrm{yd})),\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1
$\frac{3}{2}\mathrm{Matrix}(\mathrm{yu},\mathrm{Trans}(\mathrm{Bar}(\mathrm{yu})),\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1	1	1

$\mathcal{S}$	E ($\alpha=-4$)
$-\frac{9}{4}\mathrm{g2}^2\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1
$\mathrm{g3}^2\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	-8
$-\frac{17}{12}\mathrm{gY}^2\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1
$\mathrm{Tr}[\mathrm{yd.Trans}(\mathrm{Bar}(\mathrm{yd}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	3
$\mathrm{Tr}[\mathrm{ye.Trans}(\mathrm{Bar}(\mathrm{ye}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1
$\mathrm{Tr}[\mathrm{yu.Trans}(\mathrm{Bar}(\mathrm{yu}))]\mathrm{Matrix}(\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	3
$-\frac{3}{2}\mathrm{Matrix}(\mathrm{yd},\mathrm{Trans}(\mathrm{Bar}(\mathrm{yd})),\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1
$\frac{3}{2}\mathrm{Matrix}(\mathrm{yu},\mathrm{Trans}(\mathrm{Bar}(\mathrm{yu})),\mathrm{yu})(\mathrm{gen}(\$\mathrm{i}),\mathrm{gen}(\$\mathrm{j}))$	1